A SUBCONVEX BOUND IN THE INDEX ASPECT FOR THE DIRICHLET SERIES OF QUARTIC GAUSS SUMS

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ABSTRACT. Let $k \in \mathbb{Z}[i]$ be primary and squarefree, $K = \mathrm{N}(k)$, and let $\Lambda(k,s) = \sum_n (n/|n|) \sum_{x \bmod n} \overline{(x/n)_4} \exp(2\pi i \operatorname{Re}(k^3 x/n)) \mathrm{N}(n)^{-s}$, the sum over primary n prime to k. We prove that $|\Lambda(k, 1+it)| \ll_\varepsilon K^{13/18+\varepsilon}(1+|t|)^{15/2}$ for every $\varepsilon > 0$ and all real t. Up to a finite Euler factor and complex conjugation, $\Lambda(k,s)$ is the Dirichlet series of quartic Gauss sums at the index $-k^3/2$, for which the convexity bound is $K^{3/4+\varepsilon}$; the bound above saves $1/36$ in the exponent of K. The proof has two parts. First, from the functional equation of Eckhardt and Patterson for the quartic Kubota series twisted by ramified characters, we derive a Voronoi formula for the additive twists $m \mapsto \exp(2\pi i \operatorname{Tr}(bm/c))$ of the series at every odd modulus c, squarefree or not; at the primes dividing c exactly once its kernels are twisted Kloosterman sums, and at the higher prime powers they are bounded. Second, we run the amplified argument of Holowinsky and Nelson over $\mathbb{Z}[i]$, using an approximate functional equation that we derive from the analytic continuation and functional equation of the untwisted series at level $\Gamma_1(\lambda^4)$, in the form stated by David, Dunn, Hamieh and Lin after Diaconu.

*AI DISCLOSURE. I did not write this paper. This paper was drafted by Opus 5.5 (high) based upon research it conducted plus extensive reviews and feedback from Gemini 4 Argon (High) and GPT-6.1 Sol (High). My meager "contribution" boils down to: (a) establishing a protocol for systematic verification of LLM output via other LLMs and (b) generally pointing Opus towards an area where I saw a potential for the synthesis of existing work from real mathematicians, all of whom I have cited.

1. INTRODUCTION

1.1. **The result.** Let $\mathcal{O} = \mathbb{Z}[i]$, $\lambda = 1+i$, and call $c \in \mathcal{O}$ *primary* if $c \equiv 1 \pmod{\lambda^3}$. For odd c let $(\cdot/c)_4$ be the quartic residue symbol, and for $z \in \mathbb{C}$ put $\mathrm{e}(z) := \exp(2\pi i \operatorname{Tr} z)$ with $\operatorname{Tr} z = z + \bar{z}$. For a primary squarefree $k \in \mathcal{O}$

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with $K := \mathrm{N}(k)$ we consider

$$\Lambda(k,s) := \sum_{\substack{n \text{ primary} \\ (n,k)=1}} \frac{n}{|n|} \sum_{x \bmod n} \overline{\left(\frac{x}{n}\right)_4} \exp\bigl(2\pi i \operatorname{Re}(k^3 x/n)\bigr) \mathrm{N}(n)^{-s}. \tag{1.1}$$

Theorem A. *Let $k \in \mathcal{O}$ be primary and squarefree, and $K = \mathrm{N}(k)$. For every $\varepsilon > 0$,*

$$|\Lambda(k, 1+it)| \ll_\varepsilon K^{13/18+\varepsilon}(1+|t|)^{15/2} \qquad \text{for all real } t.$$

To place Theorem A, let $g_4(a,c) := \sum_{x \bmod c} (x/c)_4 \, \mathrm{e}(ax/c)$ for $a \in \lambda^{-2}\mathcal{O}$ and odd c, and for $v \in \{1, 1+\lambda^3\}$ let

$$\begin{aligned} \psi(a,s;v) &:= \sum_{c \equiv v \pmod 4} g_4(a,c) \frac{\bar{c}}{|c|} \mathrm{N}(c)^{-s}, \\ \psi_k(a,s;v) &:= \sum_{\substack{c \equiv v \pmod 4 \\ (c,k)=1}} g_4(a,c) \frac{\bar{c}}{|c|} \mathrm{N}(c)^{-s}, \end{aligned} \tag{1.2}$$

which converge absolutely for $\operatorname{Re} s > 3/2$. These are the Dirichlet series of quartic Gauss sums (Kubota's series) with angular character $\bar{c}/|c|$, in the normalisation in which the critical line is $\operatorname{Re} s = 1$. Since $\exp(2\pi i \operatorname{Re} w) = \mathrm{e}(w/2)$, conjugating (1.1) termwise gives (Section 6.2)

$$\Lambda(k,s) = \overline{\textstyle\sum_v \psi_k(\rho, \bar{s}; v)}, \qquad \rho := -k^3/2, \tag{1.3}$$

and by an algebraic identity of Castillo, de Faveri and Dunn [CDD, Lemma 6.1(i)], $\psi_k(\rho,s;v)$ differs from $\psi(\rho,s;v)$ by the finite Euler product $\Delta_k(s) = \prod_{p|k}(1 - (\bar{p}/|p|)^4 \mathrm{N}(p)^{3-4s})$. So Theorem A is a bound for Kubota's series in the aspect of the index $a = \rho$, along the indices $-k^3/2$ with k squarefree. The proof gives the same bound for each class v and for the series without the coprimality condition:

Corollary 1.1. *Let k and K be as in Theorem A and $\rho = -k^3/2$. For $v \in \{1, 1+\lambda^3\}$, every $\varepsilon > 0$ and all real t,*

$$\psi(\rho, 1+it; v) \ll_\varepsilon K^{13/18+\varepsilon}(1+|t|)^{15/2};$$

equivalently, since $\mathrm{N}(\rho) = K^3/4$, $\psi(\rho, 1+it; v) \ll_\varepsilon \mathrm{N}(\rho)^{13/54+\varepsilon}(1+|t|)^{15/2}$.

The proof of Corollary 1.1 is given at the end of Section 10 (Remark 10.4). Here $\mathrm{N}(\rho) = K^3/4$. For bounded t the convexity bound for these series is $K^{3/4+\varepsilon}$, that is, $\mathrm{N}(\rho)^{1/4}$ up to factors K^ε (see [DDHL, Proposition 4.3] and [CDD, (3.3)]; Section 10.1 below recovers $K^{3/4+\varepsilon}(1+|t|)^{3/2+\varepsilon}$ from the approximate functional equation). Theorem A improves the exponent of K from $3/4$ to $13/18 = 3/4 - 1/36$; in terms of the index, from $1/4$ to $13/54 = 1/4 - 1/108$.

1.2. **Context.** For the cubic analogue, Dunn and Radziwill [DR] prove Patterson's conjecture on the bias of cubic Gauss sums under the Generalized Riemann Hypothesis, and remark that one of two ways to bypass that hypothesis "would be to obtain 'subconvex' bounds in the r aspect for the Dirichlet series" of cubic Gauss sums $\tilde{g}(cr)$ [DR, §1.3]. In the quartic case, Castillo, de Faveri and Dunn [CDD] prove non-vanishing for a positive proportion of quartic Hecke L-functions over $\mathbb{Q}(i)$. Their argument uses only the convexity bound for the series $\tilde{\psi}(a, s, \xi)$, is arranged so that it does not need subconvex bounds for it, and they note that optimal pointwise bounds for $\tilde{\psi}(a, s, \xi)$ (and for the residues $\tau_4(a)$) are not available [CDD, §3.2]. Theorem A concerns the same series at the indices $a = -k^3/2$.

Bounds on average over the index are known. For the cubic series, David, de Faveri, Dunn and Stucky prove a Lindelöf-on-average bound for the second moment in the twist aspect [DdFDS, Lemma 6.6]: in their notation for the cubic series over $\mathbb{Z}[e^{2\pi i/3}]$, $\sum_{\mathrm{N}(m)\leq M}|\psi(hm, \sigma + it)|^2 \ll \mathrm{N}(h)^{1/2}M^{1+\varepsilon}(1 + |t|)^2$ for $1 < \sigma \leq 5/4$. De Faveri, Dunn and Hoffstein use a bound of the same type for the cubic series [FDH, (2.9)] and note that the quartic analogue is available through Rankin–Selberg theory [FDH, §2.2]. These are mean values over all indices; they give no pointwise bound for an individual index beyond convexity, and the indices $-k^3/2$ form a thin subset.

The structure of the proof follows Holowinsky and Nelson [HN], who prove $L(\pi \otimes \chi, \frac{1}{2}) \ll q^{3/4-\delta}$ for every $\delta < 1/36$, for a fixed cusp form π on $\mathrm{GL}_3(\mathbb{Z})$ and χ of prime conductor q: they write χ as the degenerate zero-frequency term of a Poisson-type formula, amplify, and then apply reciprocity, the GL_3 Voronoi formula, Cauchy–Schwarz and Weil's bound. Here $\chi = (\cdot/k)_4$ is a quartic character of squarefree modulus k, the coefficients are those of Kubota's series, and the role of the GL_3 Voronoi formula is played by Theorem B below. Our parameter choices are those of [HN], and the saving is the same, $1/36$.

The Voronoi formula is derived from the functional equation of Eckhardt and Patterson [EP, Theorem 1.7] for the series $\psi(e, \chi, \phi)$ attached to the S-adic fourfold cover of $\mathbb{Q}(i)$, which holds for quasicharacters χ that may be ramified at the finite places of S. The approximate functional equation for $\psi(\rho, s; v)$ itself uses the meromorphic continuation and the vector functional equation of the untwisted series at level $\Gamma_1(\lambda^4)$, due to Diaconu [Dia] (with the cusp data in part due to Suzuki) and stated in the form we use by David, Dunn, Hamieh and Lin [DDHL, Proposition 4.3 and the preceding discussion]; we list it as (D1)–(D4) in Section 2.3.

1.3. **Outline of the method.** *Peel, strata and characters (Section 3).* The coefficients $a(m)$ of Kubota's series vanish unless $m = cd^4$ with c squarefree, and satisfy $a(p^4x) = \xi_\omega(p)^4\mathrm{N}(p)a(x)$. To evaluate $A(c, b; X) = \sum_m a(m)\mathrm{e}(bm/c)V(\mathrm{N}(m)/X)$ at an odd modulus c, we remove the fourth powers of the primes of c exactly (the *peel*), split the part of m supported

on the primes of c into a squarefree *stratum* m_2, and expand the remaining additive character into multiplicative characters η modulo $c'' = c_D/m_2$. Each resulting series D_θ is identified with an Eckhardt–Patterson series $\psi(re_0, \chi_\theta, \phi_\theta)$ for $S = \{\infty, \lambda\} \cup \{p \mid c\}$ and a quasicharacter χ_θ ramified at the primes of c.

Local factors (Section 4). The functional equation of [EP, Theorem 1.7] is an operator $\tau(e)$ between finite-dimensional spaces $H(\chi) \to H(\check{\chi})$. We write its matrix in a unit-tuple basis as a product of local factors, one for each place of S, and compute these: at a prime $p \,\|\, c''$ with $\eta_p^4 \neq 1$ the sum over η_p is a twisted Kloosterman sum $q^{-1/2}K_{\chi_p}(l_p m; p)$ (Hasse–Davenport); at the exceptional components ($\eta_p^4 = 1$, or p in the stratum) the factors are rational in $\mathrm{N}(p)^{-\sigma}$ with poles only on $\operatorname{Re}\sigma = 1/4$; at $p^f \,\|\, c''$ with $f \geq 2$ an explicit evaluation of the η_p-sum through the p-adic logarithm bounds the factor by 2; at λ the factor is a Laurent polynomial divided by $1 - (2Z)^{-1}$. Expanding the rational factors into geometric series produces finitely many *pieces*, each with a scale Q_π and a kernel.

The Voronoi formula (Section 5). Inserting these on a line to the left and integrating term by term against the Mellin transform of V gives Theorem B: $A(c, b; X)$ is a sum over pieces of dual sums $\sum_m a_\pi^\circ(m)\widetilde{V}_\pi(\mathrm{N}(m)X_\pi/Q_\pi)\mathcal{K}_\pi(m; b)$, plus a polar term when the angular parameter vanishes. A Stirling bound for the archimedean factor (Lemma 5.1) controls the transforms.

Reduction (Section 6). Removing the coprimality to k by [CDD, Lemma 6.1(i)], writing $g_4(k^3 r, c) = (k/c)_4 g_4(r, c)$, and applying the approximate functional equation (Lemma 6.1, proved in Appendix A) reduces Theorem A for $|t| \leq K^{\varepsilon'}$ to the bound (HN-Q(i)) for smooth sums $\Sigma(a, N)$ of length N between $K^{3/2-1/18}$ and $K^{3/2+4\varepsilon'}$, twisted by the quartic character $\chi = (\cdot/k)_4$.

Amplification, Poisson and reciprocity (Section 7). As in [HN], we amplify over primes s, t of norms about $S = K^{1/9}$ and $T = K^{5/18}$, and write χ as the zero-frequency term of a Poisson formula over odd r of norm about $R = TN/(KS)$. This splits $\Sigma(a, N)$ into an F-term, an additive twist $\mathrm{e}(tm/(krs))$ of the coefficients, and an off-diagonal term $\mathcal{O}$, which is bounded directly (Proposition 7.1). Reciprocity turns the F-term into additive twists modulo rs, a modulus that is odd but need not be squarefree, which is why Theorem B is proved at every odd modulus.

The F-term (Section 8). We insert Theorem B at $c = rs$, organise the pieces by *patterns*, and apply Cauchy–Schwarz with (s, t) inside. After Poisson summation in m, the correlation sums factor over the primes of the modulus; at primes dividing it once they are complete sums of products of two Kloosterman sums, evaluated exactly, and at higher prime powers they are bounded pointwise, the loss being absorbed by an average over the squarefull parts of r.

Optimisation and the t-aspect (Sections 9 and 10). All terms are affine in $\log N/\log K$ on the range, and with the parameters of [HN] the diagonal terms are tight at $N = K^{3/2}$ (Section 9), which gives the exponent $3/4 - 1/36$. For the dependence on t, the same argument with the parameters unchanged costs a power of $1 + |t|$ that we compute exactly; the worst term costs $(1 + |t|)^{15/2}$, and for $1 + |t| \geq K^{1/36}$ the convexity bound suffices.

1.4. **Inputs and their status.** The proof uses the following results from the literature (listed with labels in Section 2.3).

- The functional equation of Eckhardt and Patterson [EP, Theorem 1.7], with their Theorems 1.3, 1.5, 1.9 and Definition 1.6 (I1). In the proof of Theorem 1.7 the computation of the Whittaker matrix τ^* [EP, pp. 259–260] is carried out by a formal argument, which the authors state can be made rigorous with generalized functions. We use the theorem as stated and do not re-prove that step. Sections 3–5 rest on it.
- The analytic properties (D1)–(D4) of the untwisted quartic series at level $\Gamma_1(\lambda^4)$ (I2): meromorphic continuation, entireness of the completed series for non-zero angular parameter, boundedness in vertical strips, the vector functional equation and the cusp relations. We take them as stated in [DDHL], where the continuation, the location of poles and the boundedness in vertical strips are derived from [Dia, Theorem 2.1], the vector functional equation from [Dia, (2.48)], and the cusp data from [Dia, (2.43)–(2.44)] and work of Suzuki. They are used only for the approximate functional equation (Lemma 6.1 and Appendix A).
- The algebraic identity [CDD, Lemma 6.1(i)] (I6), proved in that paper, used in Section 6.2 and in the residue bound of Lemma 5.2.
- Weil's bound for twisted Kloosterman sums modulo a prime (I3); the evaluation $\sum_{x \bmod k} \chi(x)\mathrm{e}(ux^{-1}/k) = \chi(u)\tau(\bar{\chi})$ for χ primitive modulo k (I4); and the bound $1/L(1 + i\tau, \xi^4) \ll \log(3 + |\tau|)$ for the Hecke character ξ^4 (I5). These are standard.

Further standard tools: quartic reciprocity with its supplements, the Hasse–Davenport product formula, Stirling's formula, the Phragmén–Lindelöf principle, Gauss's multiplication formula, Poisson summation on $\mathbb{Z}[i]$, the divisor bound in $\mathbb{Z}[i]$, and the Jacobi–Anger expansion. Appendix B records numerical checks of several local identities; no proof depends on them.

1.5. **Organisation.** Section 2 fixes notation, defines the sequences $a(m)$, lists the inputs and states the Voronoi formula (Theorem B). Sections 3–5 prove Theorem B. Section 6 reduces Theorem A to a bound for twisted sums. Sections 7–9 prove that bound for $|t| \leq K^{\varepsilon'}$, and Section 10 treats all t. Appendix A proves the approximate functional equation on the line $\mathrm{Re}\, s = 1$ with an explicit dependence on the height, and Appendix B describes the numerical checks.

2. Notation, the series, and the inputs

2.1. **Notation.** Throughout, $\mathcal{O} = \mathbb{Z}[i]$, $\lambda = 1 + i$ and N is the norm. The letters p, p' denote odd primes of $\mathcal{O}$, always taken primary, and $q_p := \mathrm{N}(p)$ (written q when p is clear). An element is *primary* if it is $\equiv 1 \pmod{\lambda^3}$; every odd c is a unit times a primary element. We write $\#c$ for the number of distinct prime divisors of c; then $C^{\#c} \ll \mathrm{N}(c)^{o(1)}$ for every fixed C. For $0 \neq c \in \mathcal{O}$ we write c_{sf} and c_{sq} for the squarefree and squarefull parts of c (coprime, $c = c_{\mathrm{sf}} c_{\mathrm{sq}}$), and $[\cdot,\cdot]$, $(\cdot,\cdot)$ for lcm and gcd.

We put $\mathrm{e}(z) := \exp(2\pi i \operatorname{Tr} z)$ with $\operatorname{Tr} z = z + \bar{z}$. The character e is trivial on $\lambda^{-2}\mathcal{O}$, so for odd c the function $\mathrm{e}_c(x) := \mathrm{e}(x/c)$ is a primitive additive character of $\mathcal{O}/c$. We write $(x/c)_4$ for the quartic residue symbol, $\chi_p := (\cdot/p)_4$ (of order 4 and conductor p), and

$$g_4(r,c) := \sum_{x \bmod c} \Big(\frac{x}{c}\Big)_4 \mathrm{e}\Big(\frac{rx}{c}\Big), \qquad \xi_\omega(c) := \Big(\frac{\bar{c}}{|c|}\Big)^\omega \quad (\omega \in \mathbb{Z}).$$

For a character ϑ modulo p^f let $G_f(\vartheta) := \sideset{}{^*}\sum_{x \bmod p^f} \vartheta(x)\mathrm{e}_{p^f}(x)$, where $\sideset{}{^*}\sum$ denotes a sum over units, and $G := G_1$. For a function Ψ on $(\mathcal{O}/p)^\times$ let $K_\Psi(w; p^f) := \sideset{}{^*}\sum_{x \bmod p^f} \Psi(x)\mathrm{e}_{p^f}(x + wx^{-1})$; for a character χ of $(\mathcal{O}/p)^\times$ this is a twisted Kloosterman sum.

Eckhardt and Patterson's variable is σ, with $\sigma = 0$ the centre; the unitary variable is $u := \sigma + 1/2$. A weight V is *inert with parameter* $Y \geq 1$ if V is smooth, supported in $[1,2]$, and $(y\,\mathrm{d}/\mathrm{d}y)^j V \ll_j Y^j$ for every $j \geq 0$. The expression $(\mathrm{N}(c)X)^{o(1)}$ means a quantity $\ll_\varepsilon (\mathrm{N}(c)X)^\varepsilon$ for every $\varepsilon > 0$. In Sections 6–10, $A \prec B$ means $A \ll_\varepsilon K^\varepsilon B$ for every $\varepsilon > 0$, the final ε being re-parametrised.

The *types* of a prime in a piece of the Voronoi formula are written $\mathsf{P}, \mathsf{A}, \mathsf{B}, \mathsf{D}, \mathsf{H}$; they are defined in (V5) and in Section 4.

2.2. The sequences.

Definition 2.1. Fix $j_0 \geq 0$. A *datum* is a triple (r, ω, v) with $r \in \{\pm 1, \pm i\} \cdot \lambda^j$ ($|j| \leq j_0$), $\omega \in \mathbb{Z}$ and $v \in \{1, 1 + \lambda^3\}$. For primary m put

$$z(m) := \sum_{\substack{m = cd^4,\ c \equiv v \pmod 4 \\ d \text{ primary}}} \xi_\omega(d)^4\, \mathrm{N}(d)^3\, g_4(r,c)\, \xi_\omega(c),$$

$$a(m) := z(m)\mathrm{N}(m)^{-1/2},$$

and $a(m) := 0$ for m not primary. The element r is a unit at every odd prime.

For $\operatorname{Re} u$ large, $\sum_m a(m)\mathrm{N}(m)^{-u} = \zeta_\lambda(4u - 1, \omega)\, \psi_v(u + \frac{1}{2}, r, \omega)$ in the notation of Section 2.3.

2.3. **The inputs.** The following results are used; the status of each is described in Section 1.4.

(I1) [EP]: Theorem 1.7, the functional equation [EP, p. 229]; with Theorem 1.3 (convergence, meromorphic continuation of finite order, poles; p. 228), Theorem 1.5 (Tate's local gamma factors; p. 229), Definition 1.6 (p. 229), and Theorem 1.9(4)–(5) (residue relations; p. 230). The proof of Theorem 1.7 [EP, §5, pp. 254–261] computes the Whittaker matrix τ^* on pp. 259–260 by a formal argument which the authors state can be made rigorous with generalized functions. Used in Sections 3–5.

(I2) The properties (D1)–(D4) below of the untwisted quartic series at level $\Gamma_1(\lambda^4)$, as stated in [DDHL], on the authority of [Dia, Theorem 2.1, (2.43)–(2.44), (2.48)] and work of Suzuki. Used only in Section 6 and Appendix A; Sections 3–5 do not use them (the analytic properties of the twisted series are derived from (I1) in Lemma 3.9).

(I3) Weil's bound $|K_\chi(w;p)| \le 2\mathrm{N}(p)^{1/2}$ for twisted Kloosterman sums modulo a prime (w a unit, χ any character of $(\mathcal{O}/p)^\times$). Used in Lemma 4.5 and Lemma 8.3.

(I4) For χ primitive modulo k: $\sum_{x \bmod k} \chi(x)\mathrm{e}_k(ux^{-1}) = \chi(u)\tau(\bar\chi)$ for every u (both sides vanish if $(u,k) \neq 1$). Used in Section 7.3.

(I5) For the Hecke character ξ^4, $\xi(c) = \bar c/|c|$: $1/L(1+i\tau, \xi^4) \ll \log(3+|\tau|)$. Used in Section 6.2.

(I6) [CDD, Lemma 6.1(i)]: let ξ be a Hecke character of the form ξ_ω, $0 \neq r \in \lambda^{-2}\mathcal{O}$, and α primary, squarefree and prime to r; let $\psi(r,s,\xi;v)$ denote the series (1.2) with ξ in place of $\bar c/|c|$, and ψ_α the same series restricted to $(c,\alpha)=1$. Then $\psi_\alpha(\alpha^3 r,s,\xi;v)\Delta_\alpha(s,\xi) = \psi(\alpha^3 r,s,\xi;v)$ with $\Delta_\alpha(s,\xi) := \prod_{p\mid\alpha}(1-\mathrm{N}(p)^{3-4s}\xi(p)^4)$. This is an algebraic identity, proved in [CDD] for $\mathrm{Re}\, s > 3/2$, and it persists under meromorphic continuation. For ξ trivial, $\Delta_p(5/4) = 1-\mathrm{N}(p)^{-2} \neq 0$. Used in Lemma 5.2(i) and Section 6.2.

The properties (D1)–(D4). For $0 \neq \nu \in \lambda^{-2}\mathcal{O}$, $\ell \in \mathbb{Z}$ and $\beta \in \{1, 1+\lambda^3\}$ let $\psi_\beta(s,\nu,\ell) := \sum_{c\equiv\beta \pmod 4} g_4(\nu,c)(\bar c/|c|)^\ell \mathrm{N}(c)^{-s}$, absolutely convergent for $\mathrm{Re}\, s > 3/2$; so $\psi(a,s;v) = \psi_v(s,a,1)$. Let $\kappa_1 = \infty, \dots, \kappa_{24}$ be the inequivalent essential cusps of $\Gamma_1(\lambda^4)$ for the Kubota multiplier, $\psi_{i1}(s,\nu,\ell)$ the corresponding Fourier–Whittaker coefficients of the Eisenstein series (normalised as in [DDHL]), and $\mathcal{V}$ the covolume of $\lambda^4\mathcal{O}$. Put

$$\zeta_\lambda(w,\ell) := \sum_{m\equiv 1 \pmod{\lambda^3}} \Big(\frac{\bar m}{|m|}\Big)^{4\ell} \mathrm{N}(m)^{-w},$$

$$Z_{i1}(s,\nu,\ell) := \zeta_\lambda(4s-3,\ell)\,\psi_{i1}(s,\nu,\ell),$$

$$\mathcal{G}(s,\ell) := \Gamma_{\mathbb{C}}\big(s+\tfrac{|\ell|}{2}-\tfrac34\big)\Gamma_{\mathbb{C}}\big(s+\tfrac{|\ell|}{2}-\tfrac12\big)\Gamma_{\mathbb{C}}\big(s+\tfrac{|\ell|}{2}-\tfrac14\big),$$

$$\Gamma_{\mathbb{C}}(s) := 2(2\pi)^{-s}\Gamma(s),$$

and $\widehat{\psi}_{i1} := \mathcal{G}(s,\ell)Z_{i1}$. We write $\zeta_\lambda(w) := \zeta_\lambda(w,1)$.

(D1) Each $\widehat{\psi}_{i1}(s,\nu,\ell)$ continues meromorphically to $\mathbb{C}$, and is entire if $\ell \neq 0$.
(D2) Each $\widehat{\psi}_{i1}(s,\nu,\ell)$ is bounded when $|\operatorname{Im} s|$ is large, in vertical strips of finite width.
(D3) (Vector functional equation.)

$$\widehat{\psi}_{i1}(s,\nu,\ell) = \mathrm{N}(\nu)^{1-s}(\bar{\nu}/|\nu|)^{\ell} \sum_{j=1}^{24} A_{ji}(2^{-s},\ell)\, \widehat{\psi}_{j1}(2-s,\nu,-\ell),$$

where each $A_{ji}(2^{-z},\ell)$ is a rational function of 2^{-z}, holomorphic for $\operatorname{Re} z \notin \{1, 5/4\}$.
(D4) (Cusp relations.) For the cusps $\kappa_{i_1} = 0$ and $\kappa_{i_2} = \lambda^4/(1+\lambda^3)$,

$$\mathcal{V}\psi_{i_1 1}(s,\nu,\ell) = (-1)^{\ell}\psi_1(s,\nu,\ell) \quad \text{and} \quad \mathcal{V}\psi_{i_2 1}(s,\nu,\ell) = (-1)^{\ell+1}\psi_{1+\lambda^3}(s,\nu,\ell).$$

For every i, $\mathcal{V}\psi_{i1}(s,\nu,\ell)$ is a finite linear combination of the series $\psi_\beta(s,(-1)^a\lambda^{2b}\nu,\ell)$, $\beta \in \{1, 1+\lambda^3\}$, over (a,b) in a finite set $S(\kappa_i,\nu) \subset \{0,1,2,3\}\times\mathbb{Z}_{\geq 0}$ that depends on ν only through its unit and its λ-order (with $b \leq \operatorname{ord}_\lambda(\nu)+5$). For the cusps $\kappa_i = \alpha_i/\gamma_i$ with $(\gamma_i,\lambda)=1$ one has $S(\kappa_i,\nu) = \{(0,0)\}$ and the coefficient is a unimodular constant; for the others the coefficients are unimodular constants times 2^{-bs} times Gauss sums $\Gamma_\beta(\nu, i^a\lambda^b)$, and these satisfy $|\Gamma_\beta(\nu,i^a\lambda^b)| \ll_{\operatorname{ord}_\lambda(\nu)} 1$ and depend on the odd part ν' of ν only through the class of ν' modulo 4 and a factor $\overline{(i^a\lambda^b/\nu')_4}$.

Besides these we use, as standard: quartic reciprocity with its supplements; the Hasse–Davenport product formula (Lemma 4.4); Stirling's formula; the Phragmén–Lindelöf principle; Gauss's multiplication formula; Poisson summation on $\mathbb{Z}[i]$; the divisor bound in $\mathbb{Z}[i]$; and the Jacobi–Anger expansion with $|J_\ell(x)| \leq (x/2)^{|\ell|}/|\ell|!$.

2.4. **The Voronoi formula.**

Theorem B (The Voronoi formula (V))**.** *Let (r,ω,v) be a datum and a as in Definition 2.1; let c be odd, $b \in \mathcal{O}$ with $(b,c)=1$, $X \geq 1$, V inert with parameter Y, and*

$$A(c,b;X) := \sum_m a(m)\, \mathrm{e}_c(bm)\, V(\mathrm{N}(m)/X).$$

Write $p^{e_p} \parallel c$. Then for every fixed $A > 0$,

$$\begin{aligned} A(c,b;X) = \sum_{\pi\in\Pi(c)} w_\pi \frac{X_\pi}{Q_\pi^{1/2}} \sum_m a_\pi^\circ(m)\, \widetilde{V}_\pi\Big(\frac{\mathrm{N}(m)X_\pi}{Q_\pi}\Big)\mathcal{K}_\pi(m;b) \\ + P(c,b;X) + O_A\big((\mathrm{N}(c)X)^{-A}\big), \end{aligned} \tag{2.1}$$

where all implied constants depend on ε, A, j_0 and polynomially on $(1+|\omega|)Y$, and:

(V1) (Pieces.) *Each* $\pi \in \Pi(c)$ *carries a* peel $D = D_\pi \mid c^\infty$ *with* $v_p(D) \le \lceil e_p/4 \rceil$, *a squarefree* stratum $m_2 \mid \mathrm{rad}(c_D)$, $c_D := c/(c, D^4)$, *a* type $T_p \in \{\mathsf{P}, \mathsf{A}, \mathsf{B}, \mathsf{D}, \mathsf{H}\}$ *at each* $p \mid c$ *(defined in* (V5)*), and finitely many integer indices (*2*-adic indices, i-indices, and Laurent indices* $n_p \ge 0$*). We put* $X_\pi := X/(\mathrm{N}(D)^4\mathrm{N}(m_2))$. *The number of pieces satisfies* $\#\Pi(c) \le (\mathrm{N}(c)X(1+|\omega|))^{o(1)}$.

(V2) (Scales and size.) $Q_\pi = 2^{O(1)} \prod_{p|c} q_p^{d_p}$ *with slopes* $d_p \le 3e_p$: $d_p = 3$ *for* P; $d_p \le 3$ *for* A; $d_p \in \{2, 3, -1\}$ *for* B; $d_p = 0$ *for* D; $d_p = 3e'$ *for* H *at* $p^{e'} \parallel c_D/m_2$; *each Laurent index lowers* d_p *by* $4n_p$. *Moreover* $|w_\pi| \le \mathrm{N}(D)$ *and* $|w_\pi| Q_\pi^{1/2} \sup_m |\mathcal{K}_\pi(m; b)| \le C^{\#c}\mathrm{N}(c)^{3/2}$.

(V3) (Dual sequences.) $a_\pi^\circ(m) = f_\pi(m) a^\natural(m)$, *where for primary* m

$$a^\natural(m) := \mathrm{N}(m)^{-1/2} \sum_{\substack{m = cd^4 \\ d \text{ primary}}} \xi_{-\omega}(d)^4 \mathrm{N}(d)^3 g_4(r, c) \xi_{-\omega}(c),$$

and f_π *is the indicator of one class modulo* λ^7 *times a constant of modulus* $\le C$. *The sequence* a_π° *does not depend on* b; *it depends on* c *only through the classes modulo* λ^7 *of the primes of* c *and the labels of* π. *Further,* $\sum_{\mathrm{N}(m) \le Z} |a_\pi^\circ(m)|^2 \ll Z^{1+\varepsilon}$.

(V4) (Transform.) $I_\pi(x) := x\widetilde{V}_\pi(x)$ *satisfies* $(x\,\mathrm{d}/\mathrm{d}x)^j I_\pi(x) \ll_{j,A} \min(x^{3/4-\varepsilon}, x^{-A})$. *The function* $\widetilde{V}_\pi$ *depends on* (c, b) *only through the labels of* π *and the classes modulo* λ^7 *of the primes of* c.

(V5) (Kernel.) $\mathcal{K}_\pi(m; b) = \theta_\pi(b) \prod_{p|c_D} k_{\pi,p}(m)$ *with* $|\theta_\pi(b)| = 1$, *where each* $k_{\pi,p}$ *is a function of* m *modulo* p^{e_p} *that contains the indicator* $\mathbf{1}_{(m,p)=1}$, *and at* $p \parallel c_D$:

- *(*P*)* $k = q^{-1/2} u_p K_{\chi_p}(l_p m; p) \mathbf{1}_{(m,p)=1}$ *with* $|u_p| = 1$, $l_p := -(4^4 B_p)^{-1} \bmod p$, B_p *as in* (V6); $|k| \le 2$;
- *(*A*)* $|k| \le Cq^{-1/2}$;
- *(*B*)* $(p \mid m_2)$ *the unramified part is independent of* b, *with* $|k| \le C$; *the ramified part has* $|k| \le Cq^{-1/2}$;
- *(*D*)* $(p \nmid c_D$, *i.e.* $4v_p(D) \ge e_p)$ *there is no factor and no condition on* (m, p);
- *(*H*)* $(p^{e'} \parallel c_D/m_2$ *with* $e' \ge 2)$ $|k| \le 2$.

At every p *with* $e_p \ge 2$, *whatever the type,* $|k_{\pi,p}| \le C$ *and* $k_{\pi,p}$ *has period* p^{e_p}.

(V6) (Frequencies.) *For* p *of type* P, $B_p = b \cdot \prod_{p' | c,\, p' \ne p} p'^{E_{p'}} \cdot u_\pi \cdot$ (*a unit modulo* p *fixed by the labels*), *where* u_π *is a monomial in* λ *and* i *fixed by the labels and* $E_{p'} \in \mathbb{Z}$ *depends only on the label at* p' *(Section 5.4:* $E = 2$ *for* P, $E = 4v_{p'}(D) - e_{p'}$ *for* D, $E = 2e'$ *for* H; *a Laurent index* n *lowers* E *by* $4n$*). All dependence of the pieces on characters of* $(\mathcal{O}/c)^\times$ *other than through* e_c *and the dual twist is through such* S*-unit shifts, common to all* $p \mid c$.

(V7) (Polar term.) $P(c,b;X) = 0$ *unless* $\omega = 0$*; if* $\omega = 0$*, then* $|P(c,b;X)| \ll C^{\#c}(\mathrm{N}(c)X)^{\varepsilon}X^{3/4}\mathrm{N}(c)^{-1/4}$.

Theorem B is proved in Sections 3–5 from (I1), (I3) and (I6); the labels are collected in Section 5.5.

3. Strata, character expansion, and functional equations

In Sections 3–5 a datum (r,ω,v) is fixed, a is the sequence of Definition 2.1, and c, b, X, V are as in Theorem B.

3.1. **Basic properties of the coefficients.**

Lemma 3.1. *(i) If* $(r,c)=1$*, then* $g_4(r,c)=0$ *unless* c *is squarefree, and* $|g_4(r,c)| = \mathrm{N}(c)^{1/2}$ *for* c *squarefree.*

(ii) $|a(m)| \le \sum_{d^4|m}\mathrm{N}(d) \le \tau(m)\mathrm{N}(m)^{1/4}$*, and* $\sum_{\mathrm{N}(m)\le Z}|a(m)|^2 \ll Z^{1+\varepsilon}$*. The same holds for the sequence* $a^\natural$ *of* (V3).

Proof. (i) If $p^2 \mid c$, then $x \mapsto x + (c/p)y$ fixes x modulo $\mathrm{rad}(c)$, hence fixes $(x/c)_4$, while $\mathrm{e}(r(c/p)y/c) = \mathrm{e}(ry/p)$ is a non-trivial character of y modulo p; summing over y gives 0. For squarefree c use the Chinese remainder theorem and $|G(\chi)| = q^{1/2}$. (ii) Only squarefree c contribute, so $|a(m)| \le \sum_{d^4|m}\mathrm{N}(d)^3\mathrm{N}(c)^{1/2}\mathrm{N}(m)^{-1/2} = \sum_{d^4|m}\mathrm{N}(d)$. The contribution of d to the second moment is $\ll \mathrm{N}(d)^2 Z/\mathrm{N}(d)^4$ up to divisor factors, and the sum over d converges. □

Lemma 3.2 (Twisted multiplicativity)**.** *For coprime primary* c_1, c_2*,*

$$g_4(r,c_1c_2) = (-1)^{C(c_1,c_2)}\Big(\frac{c_1}{c_2}\Big)_4^2 g_4(r,c_1)g_4(r,c_2),$$
$$C(a,b) := \frac{\mathrm{N}(a)-1}{4}\cdot\frac{\mathrm{N}(b)-1}{4}.$$

Hence for coprime m_1, m_2 *with* $m_2 = c_2d_2^4$ *(*c_2 *squarefree),*

$$a(m_1m_2) = \sum_{(v_1,v_2)} \varsigma(v_1,v_2)\,\zeta_{m_2}(m_1)\,a^{(v_1)}(m_1)\,a^{(v_2)}(m_2),$$

where the sum is over the pairs of classes with $v_1v_2 \equiv v \pmod 4$*,* $a^{(v')}$ *is the sequence of Definition 2.1 with class* v'*,* $\varsigma = \pm 1$ *depends only on the classes, and* $\zeta_{m_2}(m_1) := (m_1/c_2)_4^2 = \prod_{p|c_2}\chi_p^2(m_1)$.

Proof. The Chinese remainder theorem ($x = x_1c_2 + x_2c_1$) gives $g_4(r,c_1c_2) = (c_2/c_1)_4(c_1/c_2)_4\,g_4(r,c_1)g_4(r,c_2)$. Quartic reciprocity for primary elements gives $(c_2/c_1)_4 = (-1)^{C(c_1,c_2)}(c_1/c_2)_4$, and C depends on c_j only through $\mathrm{N}(c_j) \bmod 8$, that is, through the class of c_j modulo 4. For primary d, $d^4 \equiv 1 \pmod{\lambda^7}$ and $(d^4/c)_4 = 1$, so the d-parts carry no symbol. □

Lemma 3.3 (Fourth-power peel)**.** *For every odd prime* p *and every* x*,* $a(p^4x) = \xi_\omega(p)^4\mathrm{N}(p)\,a(x)$*. The same holds for* $a^{(v')}$*, and, with* $\xi_\omega(p)^4(p/|p|)^{4\ell}$ *in place of* $\xi_\omega(p)^4$*, for* $a(m)(m/|m|)^\ell$.

Proof. By Lemma 3.1(i) only squarefree c occur in z. If $d^4c = p^4x$ with c squarefree, then $4v_p(d) = 4 + v_p(x) - v_p(c) > 0$, so $p \mid d$, and $d \mapsto d/p$ is a bijection onto the pairs (d', c) with $d'^4c = x$, preserving c. Hence $z(p^4x) = \xi_\omega(p)^4\mathrm{N}(p)^3 z(x)$; divide by $\mathrm{N}(p^4x)^{1/2}$. □

3.2. **The peel.** On the support of a, $v_p(m) \equiv 0$ or $1 \pmod 4$. For $p^{e_p} \parallel c$ put $i_p(m) := \min(\lfloor v_p(m)/4 \rfloor, \lceil e_p/4 \rceil)$ and $D(m) := \prod_{p\mid c} p^{i_p(m)}$. Writing $m = D^4m'$, Lemma 3.3 gives $a(m) = u(D^4)a(m')$ with $u(D^4) := \xi_\omega(D)^4\mathrm{N}(D)$. Regrouping the m-sum by $D(m)$ and using $\mathrm{e}_c(bD^4m') = \mathrm{e}_{c_D}(b_Dm')$, we get

$$\begin{aligned} A(c,b;X) &= \sum_D u(D^4)\, A^{\mathrm{res}}\big(c_D, b_D; X/\mathrm{N}(D)^4\big), \\ c_D &:= c/(c, D^4), \quad b_D := bD^4/(c, D^4), \end{aligned} \tag{3.1}$$

where $A^{\mathrm{res}}(c_D, b_D; Z) := \sum_{m'} a(m')\mathrm{e}_{c_D}(b_Dm')V(\mathrm{N}(m')/Z)$, with m' restricted to $v_p(m') \le 1$ at every $p \mid c_D$ (exactly the p with $4i_p < e_p$) and unrestricted at the other primes of c. Here $(b_D, c_D) = 1$, $|u(D^4)| = \mathrm{N}(D)$, and the number of D is $\prod_p(\lceil e_p/4 \rceil + 1) \le \tau(c)^2$.

3.3. **Strata.** In A^{res} write $m' = m_1m_2$ with $(m_1, c_D) = 1$ and m_2 squarefree, $m_2 \mid \mathrm{rad}(c_D)$. Put $c'' := c_D/m_2$ and $X_\pi := X/(\mathrm{N}(D)^4\mathrm{N}(m_2))$. Then $\mathrm{e}_{c_D}(b_Dm_1m_2) = \mathrm{e}_{c''}(b_Dm_1)$, and by Lemma 3.2

$$\begin{aligned} A^{\mathrm{res}}\big(c_D, b_D; X/\mathrm{N}(D)^4\big) &= \sum_{m_2 \mid \mathrm{rad}(c_D)} \sum_{(v_1,v_2)} \varsigma\, a^{(v_2)}(m_2)\, A_{m_2}, \\ A_{m_2} &:= \sum_{(m_1,c_D)=1} a^{(v_1)}(m_1)\zeta_{m_2}(m_1)\mathrm{e}_{c''}(b_Dm_1)V\Big(\frac{\mathrm{N}(m_1)}{X_\pi}\Big), \end{aligned} \tag{3.2}$$

with $|a^{(v_2)}(m_2)| \le 1$ (m_2 squarefree) and $\zeta_{m_2} = \prod_{p\mid m_2}\chi_p^2$. The coefficient of a piece is $w := u(D^4)\varsigma a^{(v_2)}(m_2)$, so $|w| \le \mathrm{N}(D)$.

Remark 3.4. Without the peel, strata $m_2 = p^4$ (with $|a(p^4)| = \mathrm{N}(p)$) would carry coefficients of size $\mathrm{N}(p)$. In the Cauchy–Schwarz step of Section 8 the main term of a piece scales with $|w|^2$ times its scale, and these strata would be off by a factor of order $\mathrm{N}(p)^2$. The peel removes them exactly.

3.4. **Characters.** For $(m_1, c'') = 1$,

$$\begin{aligned} \mathrm{e}_{c''}(b_Dm_1) &= \varphi(c'')^{-1} \sum_{\eta \bmod c''} \eta(b_D)\, \tau_{c''}(\bar\eta)\, \eta(m_1), \\ \tau_{c''}(\bar\eta) &:= \sum_{x \bmod c''}^{*} \bar\eta(x)\mathrm{e}_{c''}(x), \end{aligned}$$

and by the Chinese remainder theorem $\tau_{c''}(\bar\eta) = \prod_{p^f \parallel c''} \bar\eta_p(c''/p^f)G_f(\bar\eta_p)$. For $f \ge 2$ only primitive η_p contribute (G_f of an imprimitive character

vanishes). Hence, by orthogonality,

$$A_{m_2} = \sum_{\eta} W(\eta) D_\theta,$$

$$W(\eta) := \prod_{p^f \| c''} \frac{\eta_p\big(b_D(c''/p^f)^{-1}\big) G_f(\bar{\eta}_p)}{\varphi(p^f)}, \tag{3.3}$$

$$D_\theta := \sum_{(m, c_D)=1} a^{(v_1)}(m)\theta(m) V\Big(\frac{\mathrm{N}(m)}{X_\pi}\Big),$$

with $\theta := \eta\zeta_{m_2}$, a character modulo the primes of c_D: $\theta_p = \eta_p$ at $p \mid c''$ and $\theta_p = \chi_p^2$ at $p \mid m_2$. For primitive η, $|W(\eta)| = \mathrm{N}(c'')^{-1/2} \prod_{p|c''}(1 - \mathrm{N}(p)^{-1})^{-1} \le 2^{\#c''}\mathrm{N}(c'')^{-1/2}$.

3.5. **The Eckhardt–Patterson setting.** Let $F := \mathbb{Q}(i)$, let S be a finite set of places of F containing ∞ and λ, $F_S := \prod_{v\in S} F_v$, $\mathfrak{o}^S$ the S-integers (a principal ideal domain), $(\cdot,\cdot)_S := \prod_{v\in S}(\cdot,\cdot)_v$ the quartic Hilbert symbol, $\widetilde{F}_S$ the cover of F_S by μ_4 with cocycle $(\cdot,\cdot)_S$, $s(x) := (x, 1)$, and $\epsilon\colon \mu_4 \to \mathbb{C}^\times$ injective [EP, p. 227]. Let $\mathbf{e}$ be an additive character of F_S with $\mathbf{e}(\mathfrak{o}^S) = 1$ and $\{x \in F : \mathbf{e}(x\mathfrak{o}^S) = 1\} = \mathfrak{o}^S$. The set Ω_ϵ [EP, p. 227] consists of the quasicharacters χ of $(F_S^{\times 4}(\mathfrak{o}^S)^\times)^\sim$ with $\chi = \epsilon$ on μ_4 and $\chi = 1$ on $s((\mathfrak{o}^S)^\times)$; no unramifiedness is imposed. The space $H(\chi)$ [EP, Definition 1.2, p. 228] consists of the functions ϕ on $\widetilde{F}_S$ with $\phi(hx) = (\chi(h)\|h\|_S)^{-1}\phi(x)$ for $h \in (F_S^{\times 4}(\mathfrak{o}^S)^\times)^\sim$; its dimension is $4^{|S|}$. With $g_S(\mathbf{e}, \epsilon, c) := \sum^*_{x \bmod c} \epsilon((x/c)_S)\mathbf{e}(x/c)$ [EP, Definition 1.1] and $L_S(\chi^4\|\cdot\|_S) := \prod_{v\notin S}(1 - (\chi^4(\pi_v)\|\pi_v\|_S)^{-1})^{-1}$, put

$$\psi(\mathbf{e}, \chi, \phi) := L_S(\chi^4\|\cdot\|_S) \sum_{c \in (\mathfrak{o}^S - 0)/(\mathfrak{o}^S)^\times} \tilde{g}_S(\mathbf{e}, \epsilon, s(c))\, \phi(s(c)).$$

By [EP, Theorem 1.3, p. 228] this converges for $\sigma(\chi^4) > 2$, continues meromorphically to Ω_ϵ with finite order, and in $\sigma(\chi^4) \ge 0$ has at most a simple pole, only at $\chi^4 = \|\cdot\|_S$.

Theorem 3.5 ([EP, Theorem 1.7, p. 229])**.** *Let $\chi \in \Omega_\epsilon$, $r \in F_S^\times$, $(r\mathbf{e})(x) := \mathbf{e}(rx)$, and let $\tau(\mathbf{e})\colon H(\chi) \to H(\check{\chi})$, where $\check{\chi} \in \Omega_\epsilon$ with $\chi\check{\chi} = 1$ on $s(F_S^{\times 4})$, be given by*

$$(\tau(\mathbf{e})\phi)(s(x')) := \sum_{x \in F_S^\times/F_S^{\times 4}} \frac{\Gamma_4(-(xx')^{-1}\mathbf{e}, \chi_S)}{\Gamma_1(\mathbf{e}, \chi^4)} \Big\|\frac{x}{x'}\Big\|_S \epsilon\big((x, -x')_S\big)\, \phi(s(x)).$$

Then $\psi(\mathbf{e}, \check{\chi}, \tau(\mathbf{e})\phi) = \psi(\mathbf{e}, \chi, \phi)$. Here $\Gamma_4(\mathbf{e}, \chi_S) := \prod_{v\in S}\Gamma_{4,v}(\mathbf{e}_v, \chi_v)$, where $\Gamma_{4,v}(\mathbf{e}, \varpi) := [F_v^\times : F_v^{\times 4}]^{-1}\sum_{\vartheta^4=1}\Gamma_{1,v}(\mathbf{e}, \varpi'\vartheta)$ for any extension ϖ' of ϖ to $F_v^\times$ [EP, Definition 1.6]*, and $\Gamma_{1,v}$ is Tate's gamma factor* [EP, Theorem 1.5]*: for ϖ ramified of conductor f, with d the exponent of the local*

different,

$$\Gamma_{1,v}(\mathbf{e},\varpi) = \varpi(\pi)^{-(d+f)}|\pi|^{(d+f)/2}|\pi|^{f/2}\sum_{x\in U/(1+\pi^f)}\varpi(x)\mathbf{e}(x/\pi^{d+f}),$$

and for ϖ unramified $\Gamma_{1,v}(\mathbf{e},\varpi) = \varpi(\pi)^{-d}(1-|\pi|\varpi(\pi)^{-1})/(1-\varpi(\pi))\cdot\mu(\mathfrak{o}_v)$.

This is [EP, Theorem 1.7]; its proof is [EP, §5, pp. 254–261], with the computation of τ^* on pp. 259–260 (see Section 1.4).

Convention 3.6. (a) Γ_n is the fixed-measure integral $\int \mathbf{e}(x)\chi(x)\,\mathrm{d}^\times x$ of [EP, p. 229], so that $\Gamma_{n,v}(a\mathbf{e},\varpi) = \varpi(a)^{-1}\Gamma_{n,v}(\mathbf{e},\varpi)$, with no factor $|a|^{1/2}$ (see Remark B.1).

(b) $\mathbf{e}_0 := \prod_{v\in S}\mathbf{e}_{0,v}$, where $(\mathbf{e}_{0,v})_v$ is the standard character of $\mathbb{A}_F/F$ with $\mathbf{e}_{0,\infty}(z) = \exp(2\pi i\operatorname{Tr} z)$. Since $\mathbf{e}_{0,v}$ is unramified at odd v, $\mathbf{e}_0$ satisfies both conditions above, and so does $r\mathbf{e}_0$ for every S-unit r. For $c\in\mathcal{O}$ prime to S and $x\in\mathcal{O}$, $\mathbf{e}_0(x/c) = \mathrm{e}_c(x)$. With the orientation of ϵ chosen so that $\epsilon((x/c)_S) = (x/c)_4$, we have $g_S(r\mathbf{e}_0,\epsilon,c) = g_4(r,c)$. For $\mathbf{e}_{0,\infty}$, $\Gamma_{1,\infty}(\mathbf{e}_0,|\cdot|^\sigma) = (2\pi)^{1-2\sigma}\Gamma(\sigma)/\Gamma(1-\sigma)$, the constant printed on [EP, p. 241]; for the character $\exp(-2\pi i\operatorname{Im} z)$ of [EP, §4] it is $\pi^{1-2\sigma}$. Only the shape of $\Gamma_{1,\infty}$ is used.

Lemma 3.7 (Global units)**.** *Let $S := \{\infty,\lambda\}\cup\{p\mid c_D\}$, let $\theta = \prod_p\theta_p$ be a character modulo the primes of c_D (θ_p of conductor p^{f_p}), and $\sigma\in\mathbb{C}$. Define χ_θ on fourth powers by: $\chi_\infty(z) := |z|_\infty^\sigma(z/|z|)^\omega$ ($|z|_\infty = |z|^2$); $\chi_p := \theta_p^{-1}$ on U_p^4 and $\chi_p(p^4) := \beta_p^4\mathrm{N}(p)^{-4\sigma}$ with $\beta_p := (p/|p|)^{-\omega}\prod_{p'\neq p}\theta_{p'}(p)$; $\chi_\lambda := 1$ on $U_\lambda^4 = 1+\lambda^7\mathcal{O}_\lambda$ and $\chi_\lambda(\lambda^4) := 2^{-4\sigma}(-1)^\omega\mu^4$ with $\mu := \prod_p\theta_p(\lambda)$. Then*

(i) the S-units modulo fourth powers inject into $F_S^\times/F_S^{\times 4}$;

(ii) $\chi_\theta\in\Omega_\epsilon$.

Proof. (i) $(\mathfrak{o}^S)^\times = \langle i\rangle\times\lambda^{\mathbb{Z}}\times\prod p^{\mathbb{Z}}$; λ and the p have independent valuation vectors, and i has order 4 in U_λ/U_λ^4 [EP, p. 249]. (ii) The cocycle is trivial on $F_S^{\times 4}$ and on $(\mathfrak{o}^S)^\times$ (product formula), so χ_θ is well defined on $(F_S^{\times 4}(\mathfrak{o}^S)^\times)^\sim$ if and only if $\chi(s(u^4)) = 1$ for $u = i,\lambda,p$. For i every component is 1. For λ: $\chi_\infty(\lambda^4) = \chi_\infty(-4) = 2^{4\sigma}(-1)^\omega$ and $\chi_p(\lambda^4) = \theta_p(\lambda)^{-4}$. For p: $\chi_\infty(p^4) = \mathrm{N}(p)^{4\sigma}(p/|p|)^{4\omega}$, $\chi_\lambda(p^4) = 1$, and $\chi_{p'}(p^4) = \theta_{p'}(p)^{-4}$. In each case the product is 1. □

Lemma 3.8 (Identification)**.** *With $u = \sigma+1/2$ and $\operatorname{Re} u > 1$,*

$$D_\theta(u) := \sum_{(m,c_D)=1} a(m)\theta(m)\mathrm{N}(m)^{-u} = \psi(r\mathbf{e}_0,\chi_\theta,\phi_\theta),$$

where $\phi_\theta\in H(\chi_\theta)$ is given on unit tuples y (archimedean component 1, finite components units, primary λ-component) by $\phi_\theta(s(y)) = \mathbf{1}_{y_\lambda\equiv v\ (4)}\prod_p\theta_p(y_p)$, and extended by equivariance. This holds for θ_p of every conductor.

Proof. The orbits of the S-units on $F_S^\times/F_S^{\times 4}$ ($256\cdot 16^{\#c_D}$ classes; the S-units modulo fourth powers number $16\cdot 4^{\#c_D}$) are represented exactly once

by the unit tuples with primary λ-component ($16 \cdot 4^{\#c_D}$ classes), so ϕ_θ is well defined; it is compatible with U_p^4 because $\chi_p = \theta_p^{-1}$ there. For c primary and prime to S write $c = h \cdot y$ with $h := (c, 1, \dots, 1) \in F_\infty^{\times 4}$ and $y := (1, c_\lambda, (c_p))$; then $(h, y)_S = 1$, so $\phi_\theta(s(c)) = \mathrm{N}(c)^{-\sigma-1}\xi_\omega(c)\mathbf{1}_{c\equiv_v}\theta(c)$. For $\pi \notin S$, $\|\pi\|_S = \mathrm{N}(\pi)$ and $\chi^4(\pi) = \mathrm{N}(\pi)^{4\sigma}(\pi/|\pi|)^{4\omega}\theta(\pi)^{-4}$, so $L_S = \sum_{d \text{ primary}, (d,S)=1} \theta(d)^4\xi_\omega(d)^4\mathrm{N}(d)^{-4\sigma-1}$. Each class of $\mathfrak{o}^S$ modulo S-units has exactly one primary generator prime to S, and $\tilde{g}_S(r\mathbf{e}_0, \epsilon, s(c)) = g_4(r, c)$. Multiplying out with $m = cd^4$ gives the claim. □

Lemma 3.9 (Analytic properties)**.** *D_θ continues to $\mathbb{C}$. It is holomorphic, except that when $\omega = 0$ and $\theta^4 = 1$ it may have a simple pole at $u = 3/4$ ($\sigma = 1/4$). The function $(u - 3/4)D_\theta(u)$ has polynomial growth in every vertical strip.*

Proof. For $\mathrm{Re}\,\sigma \geq 0$, i.e. $\sigma(\chi^4) \geq 0$, [EP, Theorem 1.3] applies, and $\chi_\theta^4 = \|\cdot\|_S$ if and only if $\omega = 0$, $\theta^4 = 1$ and $\sigma = 1/4$. For $\mathrm{Re}\,\sigma < 0$, Theorem 3.5 and Lemma 3.8 express D_θ as a finite combination of dual series $\psi(r\mathbf{e}_0, \check{\chi}, \cdot)$ with $\mathrm{Re}\,\check{\sigma} = -\mathrm{Re}\,\sigma > 0$, with coefficients the entries of $\tau(r\mathbf{e}_0)$. A dual series can have a pole only at $\check{\sigma} = 1/4$ under the same condition, where the archimedean factor vanishes (Section 5.1). The entries are $G_\infty(\sigma)$ (holomorphic for $\mathrm{Re}\,\sigma < 1/4$, Section 5.1) times finite-place factors whose only poles lie on $\mathrm{Re}\,\sigma = 1/4$ (Lemmas 4.3, 4.6 and 4.7, Section 4.4.2). Growth: D_θ is bounded on $\mathrm{Re}\,\sigma = 1/2 + \delta$ (absolute convergence, Lemma 3.1(ii)) and of polynomial growth on $\mathrm{Re}\,\sigma = -1/2 - \delta$ (Theorem 3.5, Stirling's formula, and absolute convergence of the dual series); [EP, Theorem 1.3] gives finite order, and the Phragmén–Lindelöf principle interpolates. □

4. Local factors

4.1. The unit-to-unit kernel.

Lemma 4.1 (Unit-tuple form of τ)**.** *Every class in $F_S^\times/F_S^{\times 4}$ is uniquely of the form $x = \upsilon \cdot y$ with $\upsilon = \lambda^{b_\lambda}\prod_{p|c_D} p^{b_p}$ (exponents in $\{0, \dots, 3\}$) and y a unit tuple. For $\phi \in H(\chi)$ put $F(x) := \phi(s(x))$. Then $F(\upsilon y) = \epsilon((\upsilon, y)_S)^\kappa F(y)$ with $\kappa = -1$; this follows from the multiplication $(x, \zeta)(x', \zeta') = (xx', \zeta\zeta'(x, x')_S^{-1})$ of* [EP, p. 227]*, which gives $s(\upsilon)s(y) = (1, (\upsilon, y)_S^{-1})s(\upsilon y)$, together with $\chi = \epsilon$ on μ_4 and $\chi(s(\upsilon)) = 1$. Write $F(y) = F_\lambda(y_\lambda)\prod_p \theta_p(y_p)$ on unit tuples; F_λ is a function on the 64 classes of U_λ/U_λ^4, equal to $\mathbf{1}_{\equiv_v}$ on the primary classes and extended by equivariance under the S-unit i (which multiplies it by $i^{\pm\omega}\theta(i)^{\mp 1}\epsilon((i, y)_\lambda)^{\pm\kappa}$, where $\theta(i) := \prod_p \theta_p(i)$). Let*

$$\mathbf{K}_v(x', x) := \frac{\Gamma_{4,v}(-(xx')^{-1}\mathbf{e}_0, \chi_v)}{\Gamma_{1,v}(\mathbf{e}_0, \chi_v^4)}\left|\frac{x}{x'}\right|_v \epsilon\big((x, -x')_v\big).$$

Then for m primary and prime to S,

$$
\begin{aligned}
(\tau\phi)(s(m)) &= \chi_\infty(-m)|m|_\infty^{-1} G_\infty(\sigma) \sum_{b_\lambda,(b_p)} \Lambda_\lambda(b_\lambda;m) \prod_p P_p(b_p;m),\\
G_\infty(\sigma) &:= \frac{\Gamma_{1,\infty}(\mathbf{e}_0,\chi_\infty)}{\Gamma_{1,\infty}(\mathbf{e}_0,\chi_\infty^4)},\\
P_p(b;m) &:= \mathrm{N}(p)^{b(\sigma+1)} (p/|p|)^{\omega b}\, \bar\theta_p(w_p)\, \epsilon((p,w_p)_p)^{-\kappa b}\, L_p(b;m),\\
w_p &:= \lambda^{b_\lambda} \prod_{p'\neq p} p'^{b_{p'}},\\
L_p(b;m) &:= \sum_{z\in U_p/U_p^4} \mathbf{K}_p(m,p^b z)\, \epsilon((p,z)_p)^{\kappa b}\, \theta_p(z),\\
\Lambda_\lambda(b_\lambda;m) &:= 2^{b_\lambda(\sigma+1)} (\lambda/|\lambda|)^{\omega b_\lambda} \sum_{y_\lambda}\\
&\qquad \mathbf{K}_\lambda(m,\lambda^{b_\lambda} w_\lambda y_\lambda)\, \epsilon\big((\lambda^{b_\lambda} w_\lambda, y_\lambda)_\lambda\big)^{\kappa} F_\lambda(y_\lambda),\\
w_\lambda &:= \prod_p p^{b_p}.
\end{aligned}
$$

The extension ϖ' of χ_p used in $\mathbf{K}_p$ has $\varpi'(p) = \beta_p \mathrm{N}(p)^{-\sigma}$.

Proof. Put $x' = m$ in Theorem 3.5 (for $r\mathbf{e}_0$; the index r multiplies each local factor by a character value). The factor Γ_4 factorises over S. At ∞, $F_\infty^\times = F_\infty^{\times 4}$ and $\Gamma_{1,\infty}(a\mathbf{e}_0,\chi) = \chi(a)^{-1}\Gamma_{1,\infty}(\mathbf{e}_0,\chi)$, so $\mathbf{K}_\infty(m,\upsilon) = \chi_\infty(-\upsilon m)|\upsilon/m|_\infty G_\infty(\sigma)$, and $\chi_\infty(\upsilon)|\upsilon|_\infty$ produces $\mathrm{N}(p)^{b(\sigma+1)}(p/|p|)^{\omega b}$ and $2^{b_\lambda(\sigma+1)}(\lambda/|\lambda|)^{\omega b_\lambda}$. Write $x = \upsilon y$ and $F(\upsilon y) = \epsilon((\upsilon,y)_S)^\kappa F(y)$. At p, $(\upsilon,y_p)_p = (p,y_p)_p^{b_p}$ (tame symbol of units). Substitute $y_p = w_p^{-1} z$: then $\theta_p(y_p) = \bar\theta_p(w_p)\theta_p(z)$ and $(p,y_p)^b = (p,w_p)^{-b}(p,z)^b$; the summand $\mathbf{K}_p \cdot F$ is a class function, so the z-sum over U_p/U_p^4 is well defined. The value of $\varpi'(p)$ is that of Lemma 3.7. □

Remark 4.2. The product formula of Lemma 4.1, with $\kappa = -1$, was compared numerically with the operator built directly from the definitions in Theorem 3.5; see Appendix B, item B3.

4.2. **Where the characters enter.** By Lemma 4.1, θ_p enters the dual weight only through: (o) the Gauss sums of θ_p inside $\mathbf{K}_p$ (Sections 4.3 and 4.5); (i) the values $\theta_p(m)$ inside L_p; (ii) values $\theta_p(\upsilon)$ at S-units υ: $\theta_p(p')$ through w_p and through $\beta_{p'}$ (inside $\mathbf{K}_{p'}$), $\theta_p(\lambda)$ through w_p and μ (inside Λ_λ), and $\theta_p(i)$ through $\theta(i)$. Each dependence of kind (ii) is a Laurent monomial in these values, or a rational function of $\beta_{p'}^4 \mathrm{N}(p')^{-4\sigma}$ or $(-1)^\omega \mu^4 2^{-4\sigma}$ that Section 4.4 expands into monomials. After the η-sum of Section 3.4, whose weight contains $\eta_p(b_D(c''/p^f)^{-1})$, a monomial $\prod_p \theta_p(\upsilon)^n$ multiplies the frequency by υ^n at every p simultaneously. The orientations (the sign

of ω in the angular factors and the orientation of ϵ) enter only unimodular constants and the labelling of the selected characters below; no bound depends on them.

4.3. **Generic local factors and the principal kernel.** Throughout this subsection $p \,\|\, c''$, $q := \mathrm{N}(p)$, ϖ' is the extension of χ_p with $\varpi'(p) = \beta_p q^{-\sigma}$, $W := \beta_p^4 q^{-4\sigma}$, and the characters of $F_p^\times$ trivial on fourth powers are $\vartheta_{j,n}(p^\beta z) := i^{n\beta}\chi_p(z)^j$. For $a = p^k u_a$, Convention 3.6(a) gives $\Gamma_{1,p}(a\mathbf{e}_0, \varpi'\vartheta_{j,n}) = (\varpi'\vartheta_{j,n})(a)^{-1}\Gamma_{1,p}(\mathbf{e}_0, \varpi'\vartheta_{j,n})$. We write $N_{p'}$ for the exponent of $\beta_{p'}$ in the factor at p' ($N = 4$ in Lemma 4.3; see Section 5.4).

Lemma 4.3 (Conductor p, $\theta_p^4 \neq 1$)**.** *$L_p(b; m) = 0$ unless $b \equiv 1$, and*

$$P_p(1; m) = q^{-3\sigma}\beta_p^4 (p/|p|)^\omega\, \bar\theta_p(w_p)\, \epsilon((p, w_p)_p)^{-\kappa} \cdot \frac{G(\bar\theta_p\bar\chi_p)}{G(\bar\theta_p^4)} \cdot (\bar\theta_p\chi_p^2)(-m),$$

up to the orientation $\chi_p \leftrightarrow \bar\chi_p$ fixed by κ and ϵ. The slope is 3, and $|G(\bar\theta\bar\chi)/G(\bar\theta^4)| = 1$.

Proof. Each $\varpi'\vartheta_{j,n}$ has conductor p ($\bar\theta_p\chi_p^j \neq 1$ because $\theta_p^4 \neq 1$), so [EP, Theorem 1.5] ($d = 0$, $f = 1$) gives $\Gamma_1(\mathbf{e}_0, \varpi'\vartheta_{j,n}) = (\varpi'(p)i^n)^{-1}q^{-1}G(\bar\theta_p\chi_p^j)$. Summing over n in [EP, Definition 1.6] forces $k \equiv 3 \pmod 4$:

$$\Gamma_{4,p}(a\mathbf{e}_0, \chi_p) = \tfrac14[k \equiv 3]\, \varpi'(p)^{-k-1}q^{-1}\theta_p(u_a)\sum_j \chi_p^{-j}(u_a)G(\bar\theta_p\chi_p^j),$$

and $\Gamma_1(\mathbf{e}_0, \chi_p^4) = W^{-1}q^{-1}G(\bar\theta_p^4)$. In $L_p(b; m)$ we have $a = -(p^b z m)^{-1}$, so $b \equiv 1$; $(p^b z, -m)_p = (p, -m)_p$ is a power of $\chi_p(-m)$, and the z-sum $\sum_z \epsilon((p,z))^\kappa\theta_p(z)\bar\theta_p(z)\chi_p^j(z)$ selects one j. With $k = -1$, $\varpi'(p)^0 W = W$, $|x/x'|_p = q^{-1}$ and $\mathrm{N}(p)^{\sigma+1}$ give $q^{-3\sigma}\beta_p^4$; the m-dependence is $\bar\theta_p(-m)\chi_p^{\pm 2}(-m) = (\bar\theta_p\chi_p^2)(-m)$. □

Lemma 4.4 (The η-sum at a prime)**.** *Let $\chi := \chi_p$, B and m units modulo p, and $y := -m/(4^4 B)$. Then*

$$\sum_{\eta^4 \neq 1} \eta(B)\frac{G(\bar\eta)G(\bar\eta\bar\chi)}{G(\bar\eta^4)}\bar\eta(-m)$$
$$= \frac{q\chi(-1)}{G(\chi)G(\chi^2)G(\chi^3)}\,\chi(y)^2\Big[(q-1)K_\chi(y; p) - \sum_{\nu^4=1}\bar\nu(y)G(\nu)G(\nu\chi)\Big].$$

Proof. Put $\varrho := \bar\eta$. Then $1/G(\varrho^4) = G(\bar\varrho^4)/q$, and the Hasse–Davenport product formula gives $\prod_{j=0}^3 G(\bar\varrho\chi^j) = \varrho^4(4)G(\bar\varrho^4)\prod_{j=1}^3 G(\chi^j)$. With $G(\varrho)G(\bar\varrho) = \varrho(-1)q$ and $G(\varrho\bar\chi)G(\bar\varrho\chi) = (\varrho\bar\chi)(-1)q$, the summand equals $\bar\varrho^4(4)\chi(-1)qG(\bar\varrho\chi^2)G(\bar\varrho\chi^3)\varrho(-m/B)/\prod_{j=1}^3 G(\chi^j)$. Put $\nu := \bar\varrho\chi^2$: then $\bar\varrho^4(4) = \nu^4(4)$, $\varrho(-m/B) = \bar\nu(-m/B)\chi^2(y)$, and $\nu^4 \neq 1$ if and only if $\eta^4 \neq 1$. Finally $\sum_{\text{all }\nu}\bar\nu(y)G(\nu)G(\nu\chi) = (q-1)K_\chi(y; p)$ (expand and sum over ν). □

The sign in the Hasse–Davenport formula used here was also checked numerically (Appendix B, item B5).

Lemma 4.5 (Principal kernel)**.** *Let $p \,\|\, c''$ and $p \nmid m_2$ (so $\theta_p = \eta_p$). The η_p-dependence of the global weight consists of: $\eta_p(b_D(c''/p)^{-1})G(\bar\eta_p)/\varphi(p)$ from Section 3.4; $G(\bar\eta_p\bar\chi_p)/G(\bar\eta_p^4)\cdot(\bar\eta_p\chi_p^2)(-m)\cdot\bar\theta_p(w_p)$ from Lemma 4.3; $\eta_p(p')^{N_{p'}}$ from the $\beta_{p'}$ of the other primes; and $\eta_p(\lambda^{n_\lambda}i^j)$ from Section 4.4.2. Hence the η_p with $\eta_p^4 \neq 1$ enter through Lemma 4.4 with*

$$B_p := b_D(c''/p)^{-1}w_p^{-1}\prod_{p'\neq p} p'^{N_{p'}}\lambda^{n_\lambda}i^j,$$

and their sum, divided by $\varphi(p)$, is $q^{-1/2}u_pK_{\chi_p}(l_pm;p)+R_p(m)$, where $l_p := -(4^4B_p)^{-1}$, $|u_p| = 1$ is independent of m, and R_p is a combination of four characters of m with $|R_p| \le 5q^{-1/2}$ (type A*). The factor $\chi_p(y)^2$ of Lemma 4.4 cancels against $\chi_p^2(-m)$ up to the constant $\bar\chi_p^2(4^4B_p)$, since $\chi_p^4 = 1$. By* (I3)*, $|q^{-1/2}K_{\chi_p}| \le 2$. If every prime of c'' is of type* P*, then $N = 4$ and $b_{p'} = 1$ for all p', so $B_p = b_D(c''/p)^2\lambda^{n_\lambda-b_\lambda}i^j$: the twists are the classical Chinese-remainder twists, and the product of the kernels over $p \mid c''$ is one twisted Kloosterman sum modulo c''.*

Proof. This follows from Lemmas 4.3 and 4.4 and (I3). For the bound on R_p: the factor in front of the bracket in Lemma 4.4 has modulus $q^{-1/2}$, and the four terms $\nu^4 = 1$ contribute at most $3q + q^{1/2}$; divided by $\varphi(p) = q-1$ this is at most $q^{-1/2}(3q+q^{1/2})/(q-1) \le 5q^{-1/2}$ for $q \ge 5$. □

4.4. **Exceptional components and the 2-adic block.**

Lemma 4.6 (Quartic θ_p at $p \,\|\, c_D$)**.** *Let $\theta_p = \chi_p^{j_0}$ and $R(W) := (1-q^{-1})/(1-q^{-1}W^{-1})$. Of the 16 extensions $\varpi'\vartheta_{j,n}$, the four with $j = j_0$ are unramified. For $a = p^ku_a$ the unramified part of $\Gamma_{4,p}(a\mathbf{e}_0)/\Gamma_1(\mathbf{e}_0,\chi_p^4)$ is $\frac14 W^{(\mathrm{r}-k)/4}R(W)$ for $\mathrm{r} := k \bmod 4 \in \{0,1,2\}$, and the monomial $\frac14 W^{(3-k)/4}$ for $k \equiv 3$; the ramified part ($j \neq j_0$) vanishes unless $k \equiv 3$, and then equals $\frac14\varpi'(p)^{-k-1}q^{-1}(-W+R(W))\theta_p(u_a)\sum_{j\neq j_0}\chi_p^{-j}(u_a)G(\bar\theta_p\chi_p^j)$. In $L_p(b;m)$ the unramified part is independent of z, so the z-sum $\sum_z \chi_p^{j_0}(z)\epsilon((p,z))^{\kappa b}$ is non-zero for exactly one $b \in \{0,1,2,3\}$; with $\kappa = -1$ and $\epsilon((p,z)_p) = \chi_p(z)$ (Convention 3.6(b)) it is $b \equiv j_0$. That b has slope $0,3,2,1$ for $b = 0,1,2,3$ (factors $R(W)$, W, $W\cdot R(W)$, $W\cdot R(W)$ respectively), i.e. slope $\equiv -j_0 \pmod 4$. The ramified part has $b = 1$, constant $\ll q^{-1/2}$, and slopes 3 (for $-W$) and -1 (for $R(W)$). All constants are $O(1)$ on $\operatorname{Re}\sigma \le 1/4-\delta$.*

Proof. With $z := \varpi'(p)i^n$ ($z^4 = W$), [EP, Theorem 1.5(i)] gives

$$\Gamma_1 = \frac{1-q^{-1}z^{-1}}{1-z} = \frac{(1-q^{-1})(1+z+z^2)+z^3-q^{-1}z^{-1}}{1-W},$$

and $\frac14\sum_{z^4=W} z^{-k}z^{n'} = W^{(n'-k)/4}[n' \equiv k \bmod 4]$; divide by $\Gamma_1(\mathbf{e}_0,\chi_p^4) = (1-q^{-1}W^{-1})/(1-W)$. The ramified part is the computation of Lemma 4.3,

with $(1-W)/(1-q^{-1}W^{-1}) = -W + R(W)$. The slopes follow from $\mathrm{N}(p)^{b(\sigma+1)}|x/x'|_p = q^{b\sigma}$ and $W = \beta_p^4 q^{-4\sigma}$. □

Type A. At $p \mid c''$ the additive weight of a quartic η_p is $\eta_p(B)G(\bar{\eta}_p)/\varphi(p)$, of modulus $\le q^{-1/2}$. So every quartic-η_p factor, and every remainder R_p of Lemma 4.5, has $|k| \le Cq^{-1/2}$ and slope ≤ 3.

4.4.1. *Type* B. At $p \mid m_2$, $\theta_p = \chi_p^2$ ($j_0 = 2$) and there is no additive weight at p. By Lemma 4.6 the unramified part has $b = 2$ (whatever κ), slope 2, constant $O(1)$, and does not depend on b_D; the ramified part has $|k| \le Cq^{-1/2}$ (slopes 3 and -1). The coefficient satisfies $|a(m_2)| \le 1$.

4.4.2. *The 2-adic block and the unit i.* By Lemma 4.1, Λ_λ depends on the characters only through $\mu = \prod_p \theta_p(\lambda)$ and $\theta(i)$ (a value in μ_4), on c only through the class of w_λ modulo λ^7, and on m only through its class modulo λ^7. Put $Z := \chi_\lambda(\lambda^4) = (-1)^\omega \mu^4 2^{-4\sigma}$ (Lemma 3.7); the sign $(-1)^\omega$ is a constant of modulus 1 and is absorbed into the coefficients below. By [EP, Definition 1.6] and [EP, Theorem 1.5] ($d = 2$ at λ; ramified conductors $f \le 7$, [EP, Table 4.1]), the unramified terms of $\Gamma_{4,\lambda}$ are rational with denominator $1-Z$, the ramified terms are monomials in μ and $2^{-\sigma}$ times Gauss sums, and $\Gamma_{1,\lambda}(\mathsf{e}_0, \chi_\lambda^4) = Z^{-2}(1-(2Z)^{-1})/(1-Z)$. Hence every entry of Λ_λ is a Laurent polynomial in $(\mu, 2^{-\sigma})$, of bounded degree with bounded coefficients, divided by $1-(2Z)^{-1}$: the only pole line is $\mathrm{Re}\,\sigma = 1/4$, and the bound is uniform in μ ($|\mu| = 1$). On $\mathrm{Re}\,\sigma < 1/4$, $1/(1-(2Z)^{-1}) = \sum_{k\ge 0}(2Z)^{-k}$ with ratio $2^{4\,\mathrm{Re}\,\sigma - 1}$. Each resulting monomial $c\,\mu^n 2^{\alpha\sigma} 2^{-k}$ becomes, after the η-sums, the frequency shift by λ^n at every p simultaneously (Section 4.2), a rescaling of the transform by a power of 2, and the weight 2^{-k}. The factor $\theta(i)^j$ is the shift by i^j.

4.4.3. *Monomials, pieces, truncation.* On $\mathrm{Re}\,\sigma < 1/4$ expand every $R(W_p)$ (types A, B) as $\sum_{n\ge 0}(1-q^{-1})q^{-n}W_p^{-n}$ (ratio $q^{4\,\mathrm{Re}\,\sigma-1}$), and the 2-adic denominators as in Section 4.4.2. A monomial $W_p^{-n} = \beta_p^{-4n} q^{4n\sigma}$ splits as follows: β_p^{-4n} contains $\prod_{p'\ne p}\theta_{p'}(p)^{-4n}$, a frequency shift at the other primes, and $q^{4n\sigma}$ rescales the transform. In the normalisation of Section 5.2 the n-th term is the piece with the same dual sequence and

$$Q_{\mathrm{eff}} = Qq^{-4n}, \qquad w_{\mathrm{eff}} = wq^{-n}$$
$$\big(I_n(x) = q^{-3n} I_0(q^{4n}x),\ \widetilde{V}_n(x) = q^n \widetilde{V}_0(q^{4n}x)\big),$$

and likewise with $(2,k)$ for the 2-adic index. A piece is thus labelled by $(D, m_2$, the type at each p, b_λ, the λ- and i-exponents, the Laurent indices). Its trivial size $|w_{\mathrm{eff}}|Q_{\mathrm{eff}}^{1/2}\sup|\mathcal{K}|$ gains $\prod_p q_p^{-3n_p} 2^{-3k}$. Keep the index vectors with $\sum_p n_p \log q_p + k\log 2 \le B\log(\mathrm{N}(c)X)$: there are at most $(\mathrm{N}(c)X)^{B\delta}\prod_p(1-q_p^{-\delta})^{-1} = (\mathrm{N}(c)X)^{B\delta+o(1)}$ of them for every $\delta > 0$, and the discarded pieces total $\ll (\mathrm{N}(c)X)^{-2B+o(1)} C^{\#c}\mathrm{N}(c)^{3/2}$ (trivial sizes, (V2)). With $B = B(A)$ and δ small this gives (V1) and the error term of (2.1).

These statements follow from Lemmas 4.1 and 4.6, [EP, Definition 1.6] and [EP, Theorem 1.5].

4.5. **Prime powers.** Let $p^f \parallel c''$ with $f \geq 2$, $q := \mathrm{N}(p) \equiv 1 \pmod 4$, $g := \lfloor f/2 \rfloor$, $h := f - g$, let ε_2 be the quadratic character of $(\mathcal{O}/p)^\times$ ($\varepsilon_2(-1) = 1$), and $\gamma_1 := \sum_{t \bmod p} \mathrm{e}_p(t^2)$. Division by 2 means multiplication by 2^{-1} modulo p^f. Only primitive η_p occur (Section 3.4). We use three elementary facts:

(F1) $\sum_{t \bmod p} \mathrm{e}_p(\alpha t^2 + \beta t) = \mathrm{e}_p(-\beta^2/(4\alpha))\varepsilon_2(\alpha)\gamma_1$ for α a unit, and $|\gamma_1| = q^{1/2}$;

(F2) for w a unit, $y^2 \equiv w \pmod{p^j}$ has at most 2 solutions ($y - y'$ and $y + y'$ are not both divisible by p);

(F3) a rational function with unit denominator at y_1 has a convergent Taylor expansion at y_1 with coefficients in $\mathcal{O}_{(p)}$.

Lemma 4.7 (Conductor p^f)**.** *For θ_p primitive modulo p^f all $\varpi'\vartheta_{j,n}$ have conductor p^f, $\Gamma_1(\mathbf{e}_0, \chi_p^4) = W^{-f}q^{-f}G_f(\bar\theta_p^4)$, and*

$$\frac{\Gamma_{4,p}(a\mathbf{e}_0)}{\Gamma_1(\mathbf{e}_0, \chi_p^4)} = \tfrac14[k \equiv -f]\,\varpi'(p)^{3f-k}\theta_p(u_a)\sum_j \chi_p^{-j}(u_a)\frac{G_f(\bar\theta_p\chi_p^j)}{G_f(\bar\theta_p^4)}.$$

So $L_p(b;m) \neq 0$ only for $b \equiv f \pmod 4$, the z-sum selects one class (Lemma 4.9(ii)), and

$$P_p(b;m) = q^{-3f\sigma}\beta_p^{3f+b}\cdot(\textit{unimodular cross factors})\cdot\frac{G_f(\bar\theta_p\chi_p^{c_1})}{G_f(\bar\theta_p^4)}\cdot(\bar\theta_p\chi_p^{c_2})(-m),$$

with integers c_1, c_2 depending only on b and κ. The slope is $3f$ and the constant is unimodular.

Proof. As for Lemma 4.3, with [EP, Theorem 1.5(ii)] at conductor f (θ_p^4 is primitive since 4 is a unit) and Lemma 4.9. □

Lemma 4.8 (Logarithm)**.** *Put $\lg(\mathsf{u}) := \mathsf{u}$ for f even and $\lg(\mathsf{u}) := \mathsf{u} - \mathsf{u}^2/2$ for f odd ($\mathsf{u} \in p^g\mathcal{O}$). Then $1 + \mathsf{u} \mapsto \lg(\mathsf{u})$ is an isomorphism $(1 + p^g\mathcal{O})/(1 + p^f\mathcal{O}) \to p^g\mathcal{O}/p^f\mathcal{O}$. Hence each character ϑ modulo p^f has a unique a_ϑ modulo p^h with $\vartheta(1 + \mathsf{u}) = \mathrm{e}_{p^f}(a_\vartheta \lg(\mathsf{u}))$; ϑ is primitive if and only if a_ϑ is a unit; $a_{\vartheta^n} = na_\vartheta$; $a_{\vartheta\vartheta'} = a_\vartheta$ for ϑ' modulo p^g; and the primitive ϑ with a given a_ϑ form one coset of the characters modulo p^g. Put $x_\vartheta := -a_\vartheta \bmod p$.*

Proof. We have $\lg((1+\mathsf{u})(1+\mathsf{v})-1) \equiv \lg(\mathsf{u})+\lg(\mathsf{v}) \pmod{p^f}$ (for f odd the terms $\mathsf{uv} - 2\mathsf{uv}/2$ cancel and the rest has valuation $\geq 3g \geq f$); lg is injective ($1 - \mathsf{u}/2$ is a unit); both groups have order q^h. The characters of $p^g\mathcal{O}/p^f\mathcal{O}$ are $\mathrm{e}_{p^f}(a\,\cdot)$, a modulo p^h, and $\lg(p^{f-1}t) \equiv p^{f-1}t$. □

Lemma 4.9 (Gauss sums)**.** *Let ϑ be primitive modulo p^f and y a unit with $y \equiv -a_\vartheta \pmod{p^h}$.*

(i) $G_f(\vartheta) = q^g\vartheta(y)\mathrm{e}_{p^f}(y)$ for f even, and $G_f(\vartheta) = q^g\vartheta(y)\mathrm{e}_{p^f}(y)\,\varepsilon_2(-a_\vartheta/2)\gamma_1$ for f odd, independently of the lift y; so $|G_f(\vartheta)| = q^{f/2}$.

(ii) $G_f(\vartheta\varphi) = \varphi(x_\vartheta)G_f(\vartheta)$ *for every character* φ *of conductor dividing* p.

Proof. Write $x = y'(1+\mathsf{u})$ with y' over representatives of $(\mathcal{O}/p^g)^\times$ (the class of $-a$ represented by y) and $\mathsf{u} = p^g t$, t modulo p^h. Then $G_f = \sum_{y'} \vartheta(y')\mathrm{e}_{p^f}(y') \sum_t \mathrm{e}_{p^f}(a\lg(\mathsf{u}) + y'\mathsf{u})$. For f even the t-sum is $q^g[y' \equiv -a \bmod p^g]$. For f odd, $\mathrm{e}_{p^f}(a\lg(\mathsf{u}) + y'\mathsf{u}) = \mathrm{e}_{p^{g+1}}((a+y')t)\mathrm{e}_p(-at^2/2)$; with $t = t_0 + pt_1$ the t_1-sum forces $y' \equiv -a \bmod p^g$, and for $y' = y$ the t_0-sum is $\varepsilon_2(-a/2)\gamma_1$ by (F1). Replacing y by $y(1+p^h t)$ multiplies $\vartheta(y)\mathrm{e}_{p^f}(y)$ by $\mathrm{e}_{p^f}((a+y)p^h t) = 1$. (ii) $a_{\vartheta\varphi} = a_\vartheta$. □

Lemma 4.10 (The η-sum at p^f)**.** *Let* B, m *be units modulo* p^f, Φ *any function on* $(\mathcal{O}/p)^\times$, $w := -m/(4^4 B)$, *and*

$$T(B,m;\Phi) := \sum_{\eta \text{ primitive mod } p^f} \eta(B)\Phi(x_{\bar\eta})\frac{G_f(\bar\eta)^2}{G_f(\bar\eta^4)}\bar\eta(-m).$$

Then, with $\rho_y := (w - y^2)/p^g$,

$$T = \varphi(p^f) \sum_{\substack{y \bmod p^g \\ y^2 \equiv w}} \Phi(y)\mathrm{e}_{p^f}\big(-(y + w/y)\big) \qquad (f \text{ even}),$$

$$T = \varphi(p^f)\varepsilon_2(2)\frac{\gamma_1^2}{q} \sum_{\substack{y \bmod p^g \\ y^2 \equiv w}} \Phi(y)\mathrm{e}_{p^f}\big(-(y + w/y)\big)\mathrm{e}_p\Big(\frac{\rho_y^2}{4y^3}\Big) \qquad (f \text{ odd}),$$

and $|T| \le 2\varphi(p^f)\max|\Phi|$ *for every* $f \ge 2$.

Proof. Put $\vartheta := \bar\eta$ and $y := y_\vartheta$. Lemma 4.9(i) with $a_{\vartheta^4} = 4a$, $y_{\vartheta^4} = 4y$ (and, for f odd, $\varepsilon_2(-a/2)^2 = 1$, $\varepsilon_2(-2a) = \varepsilon_2(2)\varepsilon_2(y)$) gives

$$\frac{G_f(\vartheta)^2}{G_f(\vartheta^4)} = q^g\vartheta(y)^{-2}\vartheta(4)^{-4}\mathrm{e}_{p^f}(-2y)\cdot\big[1,\ \text{resp. } \varepsilon_2(2)\varepsilon_2(y)\gamma_1\big].$$

So the summand is $q^g\Phi(y)\mathrm{e}_{p^f}(-2y)\vartheta(z)[\cdot]$ with $z := w/y^2$. Write $\vartheta = \vartheta_0\vartheta'$ with ϑ' modulo p^g (Lemma 4.8): $\sum_{\vartheta'}\vartheta'(z) = \varphi(p^g)[y^2 \equiv w \bmod p^g]$, and for $z = 1+\mathsf{v}$, $\mathsf{v} \in p^g\mathcal{O}$, $\vartheta_0(z) = \mathrm{e}_{p^f}(-y\lg(\mathsf{v}))$ with $y\mathsf{v} = w/y - y$. For f even, $\mathrm{e}_{p^f}(-2y - y\mathsf{v}) = \mathrm{e}_{p^f}(-(y + w/y))$ and $q^g\varphi(p^g) = \varphi(p^f)$. For f odd, y runs over units modulo p^{g+1} with $y^2 \equiv w \bmod p^g$ and the phase is $\mathrm{e}_{p^f}(\mathcal{V}(y))$, $\mathcal{V}(y) := -(y + w/y) + (w - y^2)^2/(2y^3)$. Fix a root y_1 modulo p^g and sum over the lifts $y_1 + p^g s$, s modulo p ($\Phi(y)\varepsilon_2(y)$ is constant on them). By (F3), with $w - y_1^2 = p^g\rho$, $\mathcal{V}(y_1 + p^g s) \equiv \mathcal{V}(y_1) + p^{2g}(s^2/y_1 - \rho s/y_1^2) \bmod p^f$ and $\mathcal{V}(y_1) \equiv -(y_1 + w/y_1) + p^{2g}\rho^2/(2y_1^3)$; (F1) gives $\mathrm{e}_{p^f}(-(y_1 + w/y_1))\mathrm{e}_p(\rho^2/(4y_1^3))\varepsilon_2(y_1)\gamma_1$. Multiply by $\varepsilon_2(2)\varepsilon_2(y_1)\gamma_1$ and $q^g\varphi(p^g) = \varphi(p^f)/q$. For the bound: there are at most 2 roots by (F2), and each term has modulus $\le \max|\Phi|$ ($|\gamma_1^2/q| = 1$). □

Remark 4.11. The same computation gives $|K_\Psi(w;p^f)| \le 2q^{f/2}$ for $f \ge 2$ and $|\Psi| \le 1$, and, for $\Phi = \psi$ of conductor dividing p and f odd, $T = \varepsilon_2(2)(\gamma_1/q^{1/2})\psi(-1)q^{-f/2}\varphi(p^f)K_{\psi\varepsilon_2}(w;p^f)$ with $\varepsilon_2(2)\gamma_1/q^{1/2} = \pm 1$. Neither statement is used below. Lemma 4.10 was checked numerically (Appendix B, item B6).

4.5.1. *Type* H. At $p^f \parallel c''$ ($f \ge 2$), $\theta_p = \eta_p\zeta_p$ with $\zeta_p \in \{1, \chi_p^2\}$ and η_p primitive. By Lemma 4.9(ii), $G_f(\bar\theta_p\chi_p^{c_1}) = \Phi(x_{\bar\eta_p})G_f(\bar\eta_p)$ with $\Phi := \bar\zeta_p\chi_p^{c_1}$, and $\bar\theta_p^4 = \bar\eta_p^4$; the factor $(\bar\zeta_p\chi_p^{c_2})(-m)$ and the values $\zeta_p(\upsilon)$ have modulus ≤ 1; the S-unit values $\eta_p(\upsilon)$ multiply B by υ. So the local factor is (a factor of modulus ≤ 1)$\cdot T(B_p, m; \Phi)/\varphi(p^f)$, and by Lemma 4.10, $|k_{\pi,p}| \le 2$, uniformly in the piece, b and m. With Lemmas 4.5 and 4.6 at residual exponent one this gives the last clause of (V5).

5. The transform and the Voronoi formula

5.1. **The archimedean factor.** By Tate's local formula for the character $|z|_\infty^\sigma(z/|z|)^\omega$, up to a factor c_∞^σ and a unimodular constant, where $c_\infty > 0$ does not depend on ω (in Tate's formula at a complex place the power of 2π does not depend on the angular character),

$$G_\infty(\sigma) = \frac{\Gamma(\sigma + |\omega|/2)\,\Gamma(1 - 4\sigma + 2|\omega|)}{\Gamma(1 - \sigma + |\omega|/2)\,\Gamma(4\sigma + 2|\omega|)}. \tag{5.1}$$

By Gauss's multiplication formula, $\Gamma(4\sigma + 2|\omega|)$ contains the factor $\Gamma(\sigma + |\omega|/2)$, so G_∞ is holomorphic for $\operatorname{Re}\sigma < 1/4 + |\omega|/2$. For $\omega = 0$, $G_\infty(\sigma) = c_\infty^\sigma \prod_{a\in\{1/4,1/2,3/4\}} \Gamma(a - \sigma)/\Gamma(a + \sigma)$ up to a unimodular constant (cf. [EP, p. 242]), with first pole at $\sigma = 1/4$ and a zero at $\sigma = -1/4$. Its growth on vertical lines is given by the following lemma.

Lemma 5.1 (Stirling bound for G_∞)**.** *Put $a = |\omega|/2$ and $x = \operatorname{Re}\sigma$. Then G_∞ is holomorphic for $\operatorname{Re}\sigma < 1/4 + a$, and its zeros there are simple, real, and exactly at $\sigma = -a - j/4$ ($j \ge 1$, $4 \nmid j$). For every $B \ge 1$ and $0 < \varepsilon \le 1/8$ there is C_B such that for all $a \ge 0$, $v = \operatorname{Im}\sigma \in \mathbb{R}$ and $-B \le x \le 1/4 - \varepsilon$,*

$$|G_\infty(\sigma)| \le C_B\,\varepsilon^{-1}(1 + |v| + |\omega|)^{-6x};$$

if moreover $|a + iv| \ge 4B + 4$, then $|G_\infty(\sigma)| = 4^{1-8x}|a + iv|^{-6x}(1 + O_B(1/|a + iv|))$. (The factor c_∞^σ is bounded above and below on the strip.)

Proof. Gauss's multiplication formula for $\Gamma(4\sigma + 4a)$ gives

$$G_\infty = (2\pi)^{3/2}4^{1/2-4\sigma-4a}\frac{\Gamma(1 - 4\sigma + 4a)}{\Gamma(1 - \sigma + a)\prod_{k=1}^3 \Gamma(\sigma + a + k/4)},$$

whence the holomorphy and the zeros. As $G_\infty(\bar\sigma) = \overline{G_\infty(\sigma)}$, we may take $v \ge 0$; put $\zeta = a + iv$. If $|\zeta| < 4B + 4$, the arguments lie in a compact set, $\operatorname{Re}(1 - 4\sigma + 4a) \ge 4\varepsilon$ gives $|\Gamma(1 - 4\sigma + 4a)| \ll_B \varepsilon^{-1}$, and the other factors

are entire. If $|\zeta| \geq 4B + 4$, then

$$|G_\infty| = \Big|\frac{\Gamma(\zeta + x)}{\Gamma(\zeta + 1 - x)}\Big| \cdot \Big|\frac{\Gamma(4\zeta + 1 - 4x)}{\Gamma(4\zeta + 4x)}\Big|.$$

For $\operatorname{Re}\zeta \geq 0$ and $|\alpha|, |\beta| \leq |\zeta|/4$ all points $\zeta + t$ on the segment from $\zeta + \beta$ to $\zeta + \alpha$ lie in $|\arg| \leq 3\pi/4$, where $\log\Gamma(w) = (w - \frac{1}{2})\log w - w + \frac{1}{2}\log 2\pi + O(1/|w|)$; integrating $\log w - 1/(2w)$ over t gives $\log|\Gamma(\zeta + \alpha)/\Gamma(\zeta + \beta)| = (\alpha - \beta)\log|\zeta| + O_B(1/|\zeta|)$. Hence $|G_\infty| = 4^{1-8x}|\zeta|^{-6x}(1 + O_B(1/|\zeta|))$, and $|\zeta| \leq 1 + v + |\omega| \leq 4|\zeta|$. □

A numerical check of Lemma 5.1 is described in Appendix B, item B7.

5.2. **The identity and the transform.** Write $(\mathcal{M}V)(u) := \int_0^\infty V(y)y^{u-1}\,\mathrm{d}y$ for the Mellin transform of V (entire and rapidly decaying on vertical lines). For each stratum sum and character θ,

$$\sum_{(m,c_D)=1} a(m)\theta(m)V(\mathrm{N}(m)/X_\pi) = \frac{1}{2\pi i}\int_{(3/2)} D_\theta(\sigma + \tfrac{1}{2})(\mathcal{M}V)(\sigma + \tfrac{1}{2})X_\pi^{\sigma+1/2}\,\mathrm{d}\sigma.$$

By Lemma 3.9 we move the line to $\operatorname{Re}\sigma = -1/2 - \delta$, picking up the residue at $\sigma = 1/4$ when $\omega = 0$ and $\theta^4 = 1$ (Lemma 5.2). On that line we insert Theorem 3.5 through Lemmas 3.8 and 4.1, expand into monomials (Sections 4.3–4.5, 4.4.2 and 4.4.3), and integrate term by term; the dual series and the expansions converge absolutely there. Together with the η-sums (Section 3.4, Lemmas 4.5 and 4.10) this gives the pieces of Theorem B, with

$$I_\pi(x) = x\widetilde{V}_\pi(x) = x^{1/2}\frac{1}{2\pi i}\int_{(-1/2-\delta)} \Phi_\pi(\sigma)(\mathcal{M}V)(\sigma + \tfrac{1}{2})x^\sigma\,\mathrm{d}\sigma,$$

where Φ_π is G_∞ times the constant of the piece's monomial (its powers of $\mathrm{N}(p)^{-\sigma}$ and $2^{-\sigma}$ are put into Q_π). The function Φ_π is holomorphic for $\operatorname{Re}\sigma < 1/4$. Shifting to $\operatorname{Re}\sigma = -A$ gives $I_\pi(x) \ll x^{1/2-A}$ for $x \geq 1$, and shifting to $\operatorname{Re}\sigma = 1/4 - \varepsilon$ gives $I_\pi(x) \ll x^{3/4-\varepsilon}$; the operator $(x\,\mathrm{d}/\mathrm{d}x)^j$ brings down $(\sigma + \frac{1}{2})^j$. This is (V4).

5.3. **Polar terms.**

Lemma 5.2 (Polar terms)**.** *Let $\omega = 0$ and $\theta^4 = 1$. (Then c'' is squarefree: a primitive character modulo p^f with $f \geq 2$ has order divisible by the residue characteristic.) Then $|\mathrm{Res}_{u=3/4} D_\theta| \ll C^{\#c''}\mathrm{N}(c'')^{1/4+\varepsilon}$, and* (V7) *holds.*

Proof. Write $\theta_p = \chi_p^{j_p}$. By quartic reciprocity and $g_4(r,c)(a/c)_4^{-1} = g_4(ra,c)$, the twist by χ_p^j is, up to class signs, the series at index rp^{4-j} restricted to $(m,p) = 1$. By Lemma 4.6 its unramified slope is $\equiv -j$, i.e. equal to the index exponent $4 - j$ modulo 4; this is consistent with the factor $\|rp^e\|_S^{-\sigma}$ in the functional equation of the index-rp^e series [EP, p. 242].

(i) Suppose some $\theta_p = \chi_p$ (index exponent 3, unramified slope 3). By (I6) with $\alpha = p$, $\psi_p(p^3 r)\Delta_p = \psi(p^3 r)$ with $\Delta_p \neq 0$ at the pole, so the residue is a multiple of $\rho_S(r_0 p^3, \eta)$ for indices r_0 prime to p, which vanishes by [EP, Theorem 1.9(5)] ($\rho_S(r_0 p^{n-1}, \eta) = 0$). So the residue is 0.

(ii) Otherwise every $p \mid c''$ has unramified slope $d_p \leq 2$ with constant $O(1)$, and a ramified part of slope 3 with constant $O(q^{-1/2})$ (Lemma 4.6). Apply Section 5.2 to D_θ with $V \geq 0$, $(\mathcal{M}V)(3/4) > 0$ and length Y:

$$\sum_m a(m)\theta(m)V(\mathrm{N}(m)/Y) = \mathrm{Res}\cdot Y^{3/4}(\mathcal{M}V)(3/4) + (\text{dual}).$$

The left side is $\ll Y^{1+\varepsilon}$ (Lemma 3.1); the dual side is

$$\begin{aligned}
&\ll (\mathrm{N}(c'')Y)^{\varepsilon} \sum_{\text{pieces}} |w| Q^{1/2} \sup|k| \\
&\leq (\mathrm{N}(c'')Y)^{\varepsilon} C^{\#c''} \prod_p \left(q^{d_p/2} + q^{3/2} q^{-1/2}\right) \\
&\leq (\mathrm{N}(c'')Y)^{\varepsilon} (2C)^{\#c''} \mathrm{N}(c''),
\end{aligned}$$

independently of Y. Take $Y = \mathrm{N}(c'')$. A pole of a dual series at $\sigma = -1/4$ is cancelled by the zero of G_∞ (Section 5.1).

For (V7): the polar part of $A(c, b; X)$ is the sum over D, m_2, the classes and the $4^{\#c''}$ quartic η of $w \cdot W(\eta) \cdot \mathrm{Res}\cdot X_\pi^{3/4}(\mathcal{M}V)(3/4)$, with $|W(\eta)| \leq 2^{\#c''}\mathrm{N}(c'')^{-1/2}$ (Section 3.4). Since $\mathrm{N}(c'') \geq \mathrm{N}(c)\mathrm{N}(D)^{-4}\mathrm{N}(m_2)^{-1}$, each term is at most $C^{\#c}(\mathrm{N}(c)X)^{\varepsilon}\mathrm{N}(D)X_\pi^{3/4}\mathrm{N}(c'')^{-1/4} \leq C^{\#c}(\mathrm{N}(c)X)^{\varepsilon}X^{3/4}\mathrm{N}(c)^{-1/4}$. □

5.4. **The exponents E.** In B_p of Lemma 4.5 (for general labels, with b_D and c'' from Section 3.2), a prime $p' \neq p$ of c contributes $p'^{E_{p'}}$ with

$$E_{p'} = v_{p'}(b_D) - v_{p'}(c'') - b_{p'} + N_{p'} \qquad (b_{p'} = N_{p'} := 0 \text{ if } p' \nmid c_D),$$

where $b_{p'}$ is the selected b at p' and $N_{p'}$ the exponent of $\beta_{p'}$ in the selected monomial. The values at primes with $e_{p'} = 1$ (Laurent index 0) are:

- P: 2; A (remainder of Lemma 4.5): 2; A (quartic, unramified, $b = 0, 1, 2, 3$): $-1, 2, 1, 0$; A (quartic, ramified, monomial $-W$, resp. $R(W)$): 2, resp. -2;
- B unramified: 2; B ramified ($-W$, resp. $R(W)$): 3, resp. -1;
- D: $4v_{p'}(D) - e_{p'}$ ($= 3$ when $e = 1$); H at $p'^f \parallel c''$: $2f$.

A Laurent index n lowers E by $4n$. The λ- and i-exponents form the monomial u_π of (V6). These values are read off from Lemmas 4.3, 4.6 and 4.7: $N = 4$ at generic primes and for the monomials W and $W \cdot R(W)$; $N = 0$ for $R(W)$; $N = 3f + b$ at p^f; the n-th Laurent term adds $-4n$.

5.5. **Proof of Theorem B.** Combine (3.1), (3.2), (3.3) and Section 5.2. The labels are justified as follows. (V1): Sections 3.2, 3.3 and 4.4.3. (V2): the slopes come from Lemmas 4.3, 4.6, 4.7 and Section 4.4.3; the size bound per prime is: P: $q^{3/2}\cdot 2$; A: $q^{3/2}\cdot Cq^{-1/2}$; B: $q\cdot C$ or $q^{3/2}\cdot Cq^{-1/2}$ (using $|a(m_2)|\le 1$); D: $\mathrm{N}(D)\le q^{(e+3)/4} < q^{3e/2}$; H: $q^{3e'/2}\cdot 2$; partial peel: $q^{i+3(e-4i)/2}\le q^{3e/2}$. (V3): Lemmas 3.8, 4.1, 3.1(ii) and Section 4.4.2. (V4): Section 5.2. (V5): Lemmas 4.5, 4.6, and Sections 4.4.1, 4.5.1. (V6): Section 4.2, Lemma 4.5, Section 5.4. (V7): Lemma 5.2. All of this rests on (I1) and (I3), and (V7) also on (I6) (Lemma 5.2(i)). □

6. Reduction to a twisted sum

6.1. **Standing hypotheses.** From now on k is primary and squarefree, $K := \mathrm{N}(k)$ is large, and $\chi := (\cdot/k)_4 = \prod_{p\mid k}\chi_p$, a primitive character modulo k of order 4. We put $\delta_0 := 1/36$. Until Section 10 we assume $|t|\le K^{\varepsilon'}$ with $\varepsilon' := \varepsilon/100$. In Sections 6–10 the letters s and t also denote primary primes (the amplifier variables of Section 7); the height is always written t in $1+it$ or $\mathcal{L} = 1+|t|$, and no confusion should arise.

6.2. **Coprimality.** Conjugating termwise and using $\exp(2\pi i\,\mathrm{Re}\,w) = \mathrm{e}(w/2)$, we get $\Lambda(k,s) = \overline{\sum_v \psi_k(\rho,\bar s;v)}$ with $\rho := -k^3/2$, where $\psi_k(\rho,s;v) = \sum_{c\equiv v\,(4),\,(c,k)=1} g_4(\rho,c)\xi_1(c)\mathrm{N}(c)^{-s}$ as in (1.2); write $\psi := \psi_1$ for the series without the coprimality condition. By (I6) with $\xi = \xi_1$ and $\alpha = k$, $\psi_k(\rho,s;v)\Delta_k(s) = \psi(\rho,s;v)$ with $\Delta_k := \prod_{p\mid k}(1-\xi_1(p)^4\mathrm{N}(p)^{3-4s})$. With $\zeta_\lambda(w) = \sum_{d\ \text{primary}}\xi_1(d)^4\mathrm{N}(d)^{-w}$ we have $\zeta_\lambda(4s-3)\Delta_k = \zeta_\lambda^{(k)}(4s-3)$ (the Euler factors at $p\mid k$ removed), so

$$\begin{aligned} Z(\rho,s;v) &:= \zeta_\lambda(4s-3)\psi(\rho,s;v) = \zeta_\lambda^{(k)}(4s-3)\psi_k(\rho,s;v)\\ &= \sum_{(m,k)=1} z_\rho(m)\mathrm{N}(m)^{-s}, \end{aligned} \tag{6.1}$$

$$z_\rho(m) := \sum_{\substack{cd^4=m,\ c\equiv v\\ (cd,k)=1}} \xi_1(d)^4\mathrm{N}(d)^3 g_4(\rho,c)\xi_1(c),$$

and

$$\begin{aligned} |\Lambda(k,1+it)| &\le \sum_v \frac{|Z(\rho,1-it;v)|}{|\zeta_\lambda^{(k)}(1-4it)|}\\ &\ll \log(3+|t|)\log\log(3K)\sum_v |Z(\rho,1-it;v)|, \end{aligned} \tag{6.2}$$

by (I5) and $\prod_{p\mid k}(1-\mathrm{N}(p)^{-1})^{-1}\ll \log\log 3K$.

6.3. **The twist.** For $(c,k)=1$ the substitution $x \mapsto xk^{-3}$ gives $g_4(k^3 r, c) = (k/c)_4 g_4(r,c)$, and quartic reciprocity gives $(k/c)_4 = (-1)^{C(k,c)}(c/k)_4$ with $C(k,c)$ depending on c only through its class modulo 4. As $(d^4/k)_4 = 1$,

$$z_\rho(m) = \varepsilon_v(k)\,\chi(m)\,z_{r_0}(m), \qquad r_0 := -1/2 = -i\lambda^{-2}, \quad \varepsilon_v(k) = \pm 1, \tag{6.3}$$

and $z_{r_0}\mathrm{N}^{-1/2}$ is the sequence of the datum $(r_0, 1, v)$ (Definition 2.1; the condition $(cd,k)=1$ is supplied by the factor $\chi(m)$).

6.4. **The approximate functional equation.** Put $\mathcal{E}(w) := e^{w^2}$.

Lemma 6.1 (Approximate functional equation)**.** *Let $s = 1 + i\tau$ with τ real, $\mathcal{L} := 1 + |\tau|$, and $X > 0$. Then*

$$Z(\rho, s; v) = \sum_m z_\rho(m)\mathrm{N}(m)^{-s} V_0\Big(\frac{\mathrm{N}(m)}{X}\Big) + \mathrm{N}(\rho)^{1-s}\frac{\bar\rho}{|\rho|}\sum_J \kappa_J \sum_m z_J^\sharp(m)\mathrm{N}(m)^{s-2} W_J\Big(\frac{\mathrm{N}(m)X}{\mathrm{N}(\rho)}; s\Big),$$

where:

- *J runs over a finite set, with $|J| \ll 1$ independent of k, and $|\kappa_J| \ll 1$ uniformly in k;*
- *$z_J^\sharp$ are the Dirichlet coefficients of $\zeta_\lambda(4z-3,-1)\psi_{v_J}(z,\rho_J,-1)$, i.e. of the series with angular parameter -1, class v_J and index $\rho_J = (-1)^a\lambda^{2b}\rho$ with $0 \le a,b \le 3$, so that $\sum_m z_J^\sharp(m)\mathrm{N}(m)^{s-2}$ is that series at $z = 2-s$;*
- *$V_0(y) := \frac{1}{2\pi i}\int_{(2)} \mathcal{E}(w) y^{-w}\,\mathrm{d}w/w$, and $W_J(y;s) := \frac{1}{2\pi i}\int_{(2)} \mathcal{E}(w) y^{-w}\gamma_J(s-w)\,\mathrm{d}w/w$ with γ_J given in* (A.2)*;*
- *for every $B > 0$ and $j \ge 0$, uniformly in k and τ,*

$$\Big(y\frac{\mathrm{d}}{\mathrm{d}y}\Big)^j W_J(y;s) \ll_{j,B} \Big(\frac{\mathcal{L}^6}{y}\Big)^B \qquad (y>0), \tag{U1}$$

and $(y\,\mathrm{d}/\mathrm{d}y)^j V_0(y) \ll_{j,B} \min(1, y^{-B})$.

In particular, on the line $\mathrm{Re}\,s = 1$ the bound (U1) *carries no fixed power of $\mathcal{L}$ beyond $\mathcal{L}^{6B}$, which is tied to the cut-off scale, and the logarithmic y-derivatives of W_J bring down no powers of $\mathcal{L}$ (at a general $\mathrm{Re}\,s = \sigma$ the same argument gives the exponent $6(1-\sigma+B)$). The weight W_J does depend on τ, through $\gamma_J(s-w)$; only its y-derivatives matter below, and powers of $\mathcal{L}$ in those come from the factors $\mathrm{N}(m)^{-s}$ and $\mathrm{N}(m)^{s-2}$ alone.*

The proof is given in Appendix A (Proposition A.4). It uses (D1)–(D4): Mellin inversion of $Z(\rho, s+w; v)\mathcal{E}(w)X^w/w$, the shift from $\mathrm{Re}\,w = 2$ to $\mathrm{Re}\,w = -2$, the functional equation on $\mathrm{Re}(s+w) = -1$, and Stirling's formula for the gamma ratio of (D3).

6.5. **Reduction to $\Sigma(a,N)$.** The dual series in Lemma 6.1 is again of Z-type, at the index $k^3\cdot r_J$ with $r_J=(-1)^a\lambda^{2b}r_0$ an S-unit. Its coefficients are defined as an unrestricted convolution; applying (I6) with $\xi=\xi_{-1}$, $\alpha=k$ and $r=r_J$ (which lies in $\lambda^{-2}\mathcal{O}$ and is prime to k) shows, as in Section 6.2, that they vanish unless $(m,k)=1$ and equal the coefficients of $\zeta_\lambda^{(k)}(4z-3,-1)\psi_k(k^3r_J,z,\xi_{-1};v_J)$. So Section 6.3 applies on both sides: $z_J^\sharp(m)=\pm\chi(m)z'(m)$ with $z'\mathrm{N}^{-1/2}$ the sequence of the datum $(r_J,-1,v_J)$. Take $X=\mathrm{N}(\rho)^{1/2}(1+|t|)^3$ in Lemma 6.1 at $s=1-it$. By (U1) and the decay of V_0, the terms with $\mathrm{N}(m)\ge K^{3/2+4\varepsilon'}$ contribute $O(K^{-100})$ (recall $\mathrm{N}(\rho)=K^3/4$ and $|t|\le K^{\varepsilon'}$). With a smooth dyadic partition of unity we obtain

$$\begin{aligned} Z(\rho,1-it;v) &\ll_\theta K^\theta \sum_{a\in\mathcal{A}_0}\sum_{\substack{N\le K^{3/2+4\varepsilon'}\\ \text{dyadic}}} N^{1/2}|\Sigma(a,N)|+K^{-100},\\ \Sigma(a,N)&:=\sum_m \frac{V(\mathrm{N}(m)/N)}{N}a(m)\chi(m), \end{aligned} \tag{6.4}$$

where $\mathcal{A}_0$ is a fixed finite set of data with $\omega=\pm1$ (with $j_0=4$), $\theta>0$ is arbitrary, and V is inert with parameter $K^{\varepsilon'}$. Here the phases $\mathrm{N}(m)^{\pm it}$ and the weights V_0, W_J are put into V: on a dyadic block $\mathrm{N}(m)\asymp N\le K^{3/2+4\varepsilon'}$, (U1) with $B>0$ small enough in terms of θ gives $(x\,\mathrm{d}/\mathrm{d}x)^jW_J(NxX/\mathrm{N}(\rho);s)\ll_{j,\theta}K^\theta$, so after the factor $x^{\pm it}$ the derivatives are $\ll_{j,\theta}K^\theta\mathcal{L}^j$; dividing the weight by K^θ makes it inert with parameter $K^{\varepsilon'}$ in the sense of Section 2.1, and the factor K^θ is kept outside. It is absorbed by the final re-parametrisation of ε and does not affect the saving.

For $N\le K^{3/2-2\delta_0}$, Cauchy–Schwarz and Lemma 3.1(ii) give $\Sigma(a,N)\prec 1\le N^{-1/2}K^{3/4-\delta_0}$. So, by (6.2) and (6.4), Theorem A for $|t|\le K^{\varepsilon'}$ follows from

$$\begin{aligned} &\Sigma(a,N)\prec N^{-1/2}K^{3/4-\delta_0}\quad\text{for } a\in\mathcal{A}_0,\\ &K^{3/2-2\delta_0}\le N\le K^{3/2+4\varepsilon'},\ V\text{ inert with parameter }K^{\varepsilon'}. \end{aligned} \tag{HN-Q(i))}$$

7. Amplification, Poisson summation, and reciprocity

In Sections 7–9 we prove (HN-Q(i)). We follow [HN]; every step used is proved below over $\mathbb{Z}[i]$.

7.1. **Parameters.** Following [HN, §3.6], put

$$S:=K^{1/9},\qquad T:=K^{5/18},\qquad R:=\frac{TN}{KS},\qquad H:=\frac{K}{R},$$

so that $R=K^{2/3}$ at $N=K^{3/2}$. On the range of N in (HN-Q(i)), $K^\varepsilon\ll R,S,T\ll K^{1-\varepsilon}$, $ST\le K^{-\varepsilon}R$ and $2S<T$ (Section 9).

7.2. **The amplifier.** Following [HN, §3.3], let $\mathcal{P}_S := \{s \text{ primary prime} : \mathrm{N}(s) \in [S, 2S],\ s \nmid k\}$, $\mathcal{P}_T$ likewise, and $\beta_s := \chi(s)/\#\mathcal{P}_S$ $(s \in \mathcal{P}_S)$, $\gamma_t := \bar\chi(t)/\#\mathcal{P}_T$ $(t \in \mathcal{P}_T)$. Then $\sum_s \beta_s\bar\chi(s) = \sum_t \gamma_t\chi(t) = 1$, $\beta_s \prec S^{-1}$, $\gamma_t \prec T^{-1}$, and

$$\Sigma(a, N) = \sum_{m,s,t} \frac{V(\mathrm{N}(m)/N)}{N}\, a(m)\, \beta_s\gamma_t\, \chi(tm/s).$$

Only the complete multiplicativity of χ is used. The s-sum is split into its 16 classes modulo λ^7.

7.3. **A formula for χ.** Following [HN, §3.2], fix a smooth W on $\mathbb{C}$, supported in $1 \le |z|^2 \le 2$, with $\int W = 1$. Put $\tau(\bar\chi) := \sideset{}{^*}\sum_{x \bmod k} \bar\chi(x)\mathrm{e}_k(x)$, $\varepsilon(\bar\chi) := \tau(\bar\chi)K^{-1/2}$, and $S_\chi(h, u; k) := \sideset{}{^*}\sum_{x \bmod k} \chi(x)\mathrm{e}_k(hx + ux^{-1})$. Poisson summation over the lattices $\mathcal{O}$ and $\lambda\mathcal{O}$ modulo k (the dual lattice for the trace pairing is $\lambda^{-2}\mathcal{O}$; only odd r are kept, through the factor $1 - \mathbf{1}_{\lambda|r}$) gives, for every $u \in \mathcal{O}$,

(7.1)

$$\chi(u) = K^{1/2}\sum_r \alpha_r \mathrm{e}_k(ur^{-1}) - \varepsilon(\bar\chi)^{-1}\sum_{h\ne0} \widehat{W}_*(h/H)\frac{S_\chi(h, u; k)}{K^{1/2}} + O(K^{-100}),$$

with $\alpha_r := 2\varepsilon(\bar\chi)^{-1}R^{-1}W(r/R^{1/2})\chi(r)\mathbf{1}_{\lambda\nmid r}$ (the factor 2 is the inverse density of odd r), h over $\lambda^{-2}\mathcal{O}$ with $\mathrm{N}(h) \le K^\varepsilon H$, and $\widehat{W}_*$ the combination of the two Fourier transforms (the second at frequency $h/2H$). The terms $h = 0$ give $\chi(u)$ by (I4), which is valid for every u since χ is primitive; nonzero multiples of k have $\mathrm{N}(h) \ge K \gg K^\varepsilon H$. The coefficient α_r is supported on odd r with $\mathrm{N}(r) \in [R, 2R]$ and $(r, k) = 1$, and $\alpha_r \ll R^{-1}$, $\sum_r|\alpha_r| \ll 1$.

7.4. **The splitting.** By (7.1) with $u = tm/s$, $\Sigma(a, N) = F - \varepsilon(\bar\chi)^{-1}\mathfrak{O}$, where

$$F := K^{1/2}\sum_{r,s,t} \alpha_r\beta_s\gamma_t \sum_m \frac{V(\mathrm{N}(m)/N)}{N} a(m)\, \mathrm{e}_k\Big(\frac{tm}{rs}\Big),$$

$$\mathfrak{O} := \sum_m \frac{V(\mathrm{N}(m)/N)}{N} a(m) \sum_{s,t} \beta_s\gamma_t \sum_{h\ne0} \widehat{W}_*(h/H)\frac{S_\chi(h, tm/s; k)}{K^{1/2}},$$

and (HN-Q(i)) follows from $F, \mathfrak{O} \prec N^{-1/2}K^{3/4-\delta_0}$ (the division tm/s is modulo k).

7.5. **Reciprocity.** Following [HN, §4.1]: on the support, $(k, rst) = 1$, and $\mathrm{e}_k(tm(rs)^{-1}) = \mathrm{e}(tm/(krs))\,\mathrm{e}_{rs}(-tmk^{-1})$, where k^{-1} is the inverse of k modulo rs. Indeed, if $\bar x$ denotes inverses, $\overline{rs}\cdot rs + \bar k\cdot k \equiv 1 \pmod{krs}$, so $\overline{rs}/k + \bar k/(rs) - 1/(krs) \in \mathcal{O}$, where e is trivial. By the choice of R, $|tm/(krs)| \ll (TN/(KRS))^{1/2} = 1$.

7.6. **Angular modes.** The weight $f(m) := V(\mathrm{N}(m)/N)\mathrm{e}(tm/(krs))$ is not a function of $\mathrm{N}(m)$. By the Jacobi–Anger expansion, $f(m) = \sum_{\ell\in\mathbb{Z}} f_\ell(|m|)(m/|m|)^\ell$ with

$$f_\ell(\varrho) = V(\varrho^2/N)\, i^\ell J_\ell\big(4\pi|t|\varrho/|krs|\big)\, e^{i\ell\arg(t/(krs))}.$$

Since $|J_\ell(x)| \le (x/2)^{|\ell|}/|\ell|!$ and $x \ll 1$, the modes with $|\ell| > K^\varepsilon$ contribute $O(K^{-100})$, and each $y \mapsto f_\ell((Ny)^{1/2})$ is inert with parameter K^ε, uniformly in (r,s,t,ℓ). Since $\xi_\omega(c)(c/|c|)^\ell = \xi_{\omega-\ell}(c)$ (and likewise for d^4), $a(m)(m/|m|)^\ell$ is the sequence of the datum $(r,\omega-\ell,v)$; its angular parameter may be 0. So

$$F = \sum_{|\ell|\le K^\varepsilon} K^{1/2} \sum_{r,s,t} \alpha_r\beta_s\gamma_t N^{-1} A_\ell(rs, -tk^{-1}; N) + O(K^{-90}),$$

with A_ℓ the sum of Theorem B for the datum $(r,\omega-\ell,v)$ and the weight f_ℓ. Write $F = F_1 + F_2$, where F_2 collects the terms with $(rs,t) \ne 1$; there $t \mid r$ (as $2S < T$), the modulus is rs/t and the frequency is $-k^{-1}$.

7.7. **The off-diagonal term.**

Proposition 7.1 (cf. [HN, §5])**.** $|\mathfrak{O}|^2 \prec H^2\big(1/(STH) + 1/K\big)$.

Proof. Apply Cauchy–Schwarz in m with Lemma 3.1(ii), then Poisson summation in m over the coset $1+\lambda^3\mathcal{O}$ modulo k. Since $N \ge K^{3/2-2\delta_0} \ge K^{1+\varepsilon}$ ($\delta_0 < 1/4$), only the zero frequency survives, and as in [HN]

$$|\mathfrak{O}|^2 \prec H^2(STH)^{-2} \sum_{s_i,t_i,h_i} K^{-1}|\Pi_0|,$$

$$\Pi_0 := \sum_{n \bmod k} \frac{S_\chi(h_1, t_1 n/s_1; k)\overline{S_\chi(h_2, t_2 n/s_2; k)}}{K}.$$

Opening the sums and summing over n forces $t_1/(s_1x) = t_2/(s_2y)$, i.e. $y = s_1t_2x/(s_2t_1)$; then $\chi(x/y)$ is constant and

$$|\Pi_0| = \Big|\sum_{x \bmod k}^{*} \mathrm{e}_k(\Delta_\mathfrak{O} x)\Big| \le \mathrm{N}((\Delta_\mathfrak{O}, k)), \qquad \Delta_\mathfrak{O} := s_2t_1h_1 - s_1t_2h_2,$$

for all h_1, h_2 (a Ramanujan sum modulo the squarefree k; only $(s_it_i,k)=1$ is used). Now $\mathrm{N}(\Delta_\mathfrak{O}) \ll K^\varepsilon STH < K^{1-\varepsilon}$ (as $ST \le K^{-\varepsilon}R$), so $k \mid \Delta_\mathfrak{O}$ if and only if $\Delta_\mathfrak{O} = 0$. The diagonal $\Delta_\mathfrak{O} = 0$ has $\prec STH$ solutions (divisor bound in $\mathbb{Z}[i]$). For $g \mid k$ the tuples with $g \mid \Delta_\mathfrak{O} \ne 0$ number $\prec (STH)^2/\mathrm{N}(g)$ (fix s_1, t_2, h_2 and $\Delta_\mathfrak{O}/g$; then s_2, t_1, h_1 by the divisor bound). So

$$|\mathfrak{O}|^2 \prec H^2(STH)^{-2}\Big[STH + \sum_{g\mid k} \frac{\mathrm{N}(g)}{K} \cdot \frac{(STH)^2}{\mathrm{N}(g)}\Big] \prec H^2\Big(\frac{1}{STH} + \frac{1}{K}\Big).$$

The sublattice $\lambda\mathcal{O}$ changes H by a factor 2. □

7.8. **Moduli restriction.**

Remark 7.2 (Why r is not restricted to squarefree values). Insert $\mathbf{1}_{r \text{ squarefree}} = \sum_{a^2 \mid r} \mu(a)$ in Section 7.3 and apply Poisson summation to $r = a^2 r'$. Each a gives dual frequencies up to $H\mathrm{N}(a)^2$ with weight $\mathrm{N}(a)^{-2}$, so in the estimate of Proposition 7.1 (absolute values after Cauchy–Schwarz; each off-diagonal tuple with $(\Delta_{\mathfrak{O}}, k) = 1$ contributes K^{-1}, a Ramanujan sum modulo a squarefree k with coprime argument having modulus 1) the ℓ^1-mass of the h-sums grows from H to $\asymp H \cdot \#\{a : \mathrm{N}(a)^2 \le 2R\}$, and the bound becomes $\asymp K^{-1}(HR^{1/2})^2 = H$, against the target $H/(ST)$: a loss $K^{7/18}$. Truncating a at $\mathrm{N}(a) \le A_1$ keeps the bound only for $A_1^2 \le R/(ST) = K^{5/18}$, and leaves in the F-term the r divisible by p^2 with $\mathrm{N}(p) > A_1$, of density $\gg A_1^{-1}$. This is a statement about this method (the Möbius sum is signed); it is why r is kept unrestricted. Only the prime λ is excluded (Section 7.3, at cost $O(1)$), so every modulus $c = rs$ is odd, and Theorem B is applied at every odd c, squarefree or not. (A unit factor of c changes $\mathrm{e}_c(bm)$ by a unit in b, which is absorbed in (V6).)

Lemma 7.3 (Squarefull averages)**.** *(i) The squarefull $m \in \mathcal{O}$ with $\mathrm{N}(m) \le Y$ number $\ll Y^{1/2}$.*

(ii) For $\Delta \in \mathcal{O} \smallsetminus \{0\}$, $R^{-1} \sum_{\mathrm{N}(r) \in [R,2R]} \mathrm{N}((\Delta, r_{\mathrm{sf}}))^{1/2} \mathrm{N}(r_{\mathrm{sq}})^{1/2} \ll \tau(\Delta) \log R$; in particular $R^{-1} \sum \mathrm{N}(r_{\mathrm{sq}})^{1/2} \ll \log R$.

Proof. (i) Write $m = a^2 b^3$. (ii) For squarefull n and $g \mid \Delta$, the r with $r_{\mathrm{sq}} = n$ and $g \mid r_{\mathrm{sf}}$ number $\ll R/(\mathrm{N}(n)\mathrm{N}(g))$; so the average is $\ll \sum_{g \mid \Delta} \mathrm{N}(g)^{-1/2} \sum_{n \text{ squarefull, } \mathrm{N}(n) \le 2R} \mathrm{N}(n)^{-1/2} \ll \tau(\Delta) \log R$ by (i) and partial summation. □

Use of Lemma 7.3. At $p^v \parallel M$ ($v \ge 2$) for a correlation modulus M, the local complete sum is bounded pointwise by $\sup|k^{(1)}| \sup|k^{(2)}| \le C^2$ (by (V5)), a loss $\mathrm{N}(p^v)^{1/2}$ against the squarefree-type bound $q^{-v/2}$; Lemma 7.3 absorbs the loss with a factor $\log R$. No p-adic stationary phase for correlations is needed. A pointwise bound $C \cdot \mathrm{N}(p^f)^{\delta}$ would instead cost $R^{2\delta}$; with no cancellation in the η-sum at p^f one has $\delta = 1/2$ and a loss $R^{1/2}$, so the square-root cancellation of Lemma 4.10 is needed.

8. The F-term and the Cauchy–Schwarz step

8.1. **Splitting.** Let $F_1 = F_{\mathrm{nc}} + F_{\mathrm{coll}}$, where F_{nc} collects the terms with $s \nmid r$ and F_{coll} those with $s \mid r$. We insert Theorem B at $c = rs$, $b = -tk^{-1}$, $X = N$. The polar terms and F_2 are treated in Section 8.8.

8.2. **Patterns.** A *pattern* Π at r consists of: the peel exponents $(v_p(D))_{p \mid r}$; the stratum part $m_2^{(r)} \mid \mathrm{rad}(r)$; the type at each $p \mid r$ with its Laurent index; the λ- and i-indices; the angular mode ℓ; the classes of r and of s modulo λ^7; and a label at s from a fixed finite list: P; A (each monomial, Laurent index 0); B unramified; B ramified (each monomial); D. In F_{nc},

Π is *realised* at (s,t) if a piece with these labels occurs in Theorem B at $c = rs$; otherwise its term is 0. The Laurent vectors at the primes of r are cut jointly (Section 4.4.3), so $\#\mathcal{P}(r) \le C^{\#r}\tau(r)^2 K^{O(\varepsilon)} \prec 1$. Both sides of every correlation in F_{nc} carry the same labels at every prime of r and at s.

Trivially bounded pieces. A piece with Laurent index $n_s \ge 1$ at s has trivial size $\le C\mathrm{N}(c)^{3/2}\mathrm{N}(s)^{-3n_s-1/2}$ (by (V2): w gains q^{-n}, $Q^{1/2}$ gains q^{-2n}, and the type A/B constants give $q^{-1/2}$ against P). These pieces contribute $\prec K^{1/2}(RS)^{3/2}N^{-1}S^{-7/2}$ to F ($n_s = 1$; less for $n_s \ge 2$): the exponent is $-2/9$ at $N = K^{3/2}$ and $-1/4$ at $N = K^{13/9}$, below the target (Section 9).

Lemma 8.1 (Local-unit factorisation)**.** *Fix a pattern Π at r, $s \nmid r$ with $(rs,t) = 1$, $c = rs$, $b = -tk^{-1}$, and a realisation π of Π. Then $\mathcal{K}_\pi(m;b) = \theta_\Pi(s,t)\prod_{p\mid c_D} k_{\Pi,p}(m)$ with $|\theta_\Pi(s,t)| = 1$, where:*

- *(a) at $p \,\|\, c_D$, $p \mid r$, of type P: $k_{\Pi,p}(m) = q^{-1/2}u_p K_{\chi_p}(l_p m;p)\mathbf{1}_{(m,p)=1}$ with $l_p = L_p s^{-e}t^{-1} \bmod p$, where L_p is a unit modulo p depending only on r, k, p and the labels of Π at the primes of r, and $e := E_s$ depends only on the label at s ($e = 2$ for P, $e = 3$ for D);*
- *(b) at $p = s$ of type P: $l_s = L_s t^{-1}$ with L_s depending only on r, k and the labels;*
- *(c) all other local factors satisfy the bounds of* (V5)*, with period p^{e_p};*
- *(d) the dual sequence a°_Π, the slopes, Q_π and X_π (up to bounded factors) are the same for every realisation of Π; $\widetilde{V}_\pi$ depends on (s,t) only through the weight f_ℓ, uniformly inert.*

Proof. By (V6) with $b = -tk^{-1}$, $B_p = -tk^{-1}s^{E_s}\prod_{p'\mid r, p'\neq p} p'^{E_{p'}}u_\pi \cdot (\text{unit})$ and $B_s = -tk^{-1}\prod_{p'\mid r} p'^{E_{p'}}u_\pi \cdot (\text{unit})$; only s and t vary within Π. Parts (c) and (d) are (V2)–(V5); the class of s is fixed by Π. The m-independent unimodular factors ($\theta_\pi(b)$, u_p, cross-factor constants) depend on (s,t) and are collected in $\theta_\Pi(s,t)$. □

8.3. **The sum F_{nc}.** With these conventions,

$$F_{\mathrm{nc}} = \sum_r \alpha_r \sum_{\Pi\in\mathcal{P}(r)} \sum_m \frac{a^\circ_\Pi(m)}{\mathrm{N}(m)^{1/2}} \sum_{s,t} \beta_s\gamma_t \Phi_\Pi(m;r,s,t) + O(F_{\mathrm{tail}}),$$

$$\Phi_\Pi(m;r,s,t) := \mathbf{1}_{s\nmid r} K^{1/2} w_\pi \frac{X_\pi}{N} Q_\pi^{-1/2}\mathrm{N}(m)^{1/2}\,\widetilde{V}_\pi\Big(\frac{\mathrm{N}(m)X_\pi}{Q_\pi}\Big)\mathcal{K}_\pi(m;-tk^{-1}),$$

where π is the realisation of Π at (s,t), $\mathcal{K}_\pi$ contains $\mathbf{1}_{(m,s)=1}$ exactly when the label at s is not D, and $F_{\mathrm{tail}} \prec K^{1/2}(RS)^{3/2}N^{-1}S^{-7/2} + K^{-90}$ collects the trivially bounded pieces and the truncation tails.

8.4. **Cauchy–Schwarz.** Following [HN, §4.4], truncate the m-sum at $\mathrm{N}(m) \le Z_\Pi := K^\varepsilon \max(Q_\pi/X_\pi)$ (by (V4)), and apply Cauchy–Schwarz

with (r,Π,m) outside and (s,t) inside:

$$|F_{\mathrm{nc}}|^2 \prec \Big[\sum_r |\alpha_r| \sum_\Pi \sum_{\mathrm{N}(m)\le Z_\Pi} \frac{|a_\Pi^\circ(m)|^2}{\mathrm{N}(m)}\Big] \cdot \Big[\sum_r |\alpha_r| \sum_\Pi \sum_m \Big|\sum_{s,t} \beta_s \gamma_t \Phi_\Pi(m;r,s,t)\Big|^2\Big] + |F_{\mathrm{tail}}|^2.$$

The first factor is $\prec 1$ (by (V3), $\sum|\alpha_r| \ll 1$ and $\#\mathcal{P}(r) \prec 1$). Opening the square, the second factor is $\sum_{s_i,t_i} \beta_{s_1}\bar\beta_{s_2}\gamma_{t_1}\bar\gamma_{t_2} M_\Pi$, with $M_\Pi := \sum_m \Phi_\Pi(m;r,s_1,t_1)\overline{\Phi_\Pi(m;r,s_2,t_2)} = (\text{unimodular}) \cdot \sum_m \mathcal{W}(m)\mathcal{P}(m)$, where

$$\mathcal{W}(m) := K w_1 \bar w_2 \frac{X_1X_2}{N^2}(Q_1Q_2)^{-1/2}\mathrm{N}(m)\widetilde{V}_1\Big(\frac{\mathrm{N}(m)X_1}{Q_1}\Big)\overline{\widetilde{V}_2\Big(\frac{\mathrm{N}(m)X_2}{Q_2}\Big)},$$
$$\mathcal{P}(m) := \kappa_1(m)\overline{\kappa_2(m)}, \quad \kappa_i := \prod_p k_{\Pi,p}^{(i)}.$$

The function $\mathcal{P}$ has period $M_0 \mid [rs_1, rs_2]$ (with the original exponents at the primes of r_{sq}), and $s_i \mid M_0$ unless the label at s is D. Further $Q_1/Q_2, X_1/X_2 \in [2^{-30}, 2^{30}]$; put $W_0 := |w_1 w_2|$.

Lemma 8.2 (One pattern after Poisson summation)**.** *Put* $\Sigma(\xi) := \mathrm{N}(M_0)^{-1}\sum_{x \bmod M_0} \mathcal{P}(x)\mathrm{e}(x\xi/M_0)$ *for* $\xi \in \lambda^{-2}\mathcal{O}$ *and* $\xi' := 2\xi \in \mathcal{O}$*, and suppose that* $|\Sigma(\xi)| \le \Lambda\,\mathrm{N}((\xi', M_0))^{1/2}\mathrm{N}(M_0)^{-1/2}$ *for* $\xi \ne 0$*. Then, with* $Q := \max Q_i$ *and* $X' := \max X_i$*,*

$$|M_\Pi| \prec KW_0QN^{-2}|\Sigma(0)| + KW_0(X'/N)N^{-1}\Lambda\,\tau(M_0)\mathrm{N}(M_0)^{1/2}.$$

Proof. Poisson summation over $\mathcal{O}$ on each class modulo M_0 (covolume 1, dual lattice $\lambda^{-2}\mathcal{O}$) gives $M_\Pi = \sum_{\xi\in\lambda^{-2}\mathcal{O}} \widehat{\mathcal{W}}(\xi/M_0)\Sigma(\xi)$. Split $\mathcal{W}$ dyadically in $\mathrm{N}(m) \sim Y$ and put $x := YX'/Q$ and $\theta_0 := 1/4 + \varepsilon$. By (V4), $\sup|\mathcal{W}_Y| \ll KW_0(X'/N^2)x\min(x^{-2\theta_0}, x^{-2A})$, and $\mathcal{W}_Y$ is inert at scale $Y^{1/2}$, so $\widehat{\mathcal{W}}_Y(0) \ll Y\sup|\mathcal{W}_Y|$ and $\widehat{\mathcal{W}}_Y(\xi/M_0) \ll_A Y\sup|\mathcal{W}_Y|K^{\varepsilon A}(1+\mathrm{N}(\xi)Y/\mathrm{N}(M_0))^{-A}$. For $\xi = 0$,

$$\sum_Y Y\sup|\mathcal{W}_Y| \ll KW_0QN^{-2}\sum_x x^2\min(x^{-2\theta_0}, x^{-2A}) \ll KW_0QN^{-2}.$$

For $\xi \ne 0$ only $\mathrm{N}(\xi') \le K^{3\varepsilon}\mathrm{N}(M_0)/Y$ matter, and

$$\sum_{0<\mathrm{N}(\xi')\le Z} \mathrm{N}((\xi', M_0))^{1/2} \ll \tau(M_0)Z;$$

the total is

$$\ll K^{3\varepsilon}\Lambda\,\tau(M_0)\mathrm{N}(M_0)^{1/2}KW_0(X'/N^2)\sum_x x^{1-2\theta_0}\min(1, x^{-2A}),$$

and the x-sum converges because $\theta_0 < 1/2$. □

For the principal pattern ($W_0 = 1$, $X' = N$, $Q \asymp (RS)^3$) the two terms are $K(RS)^3N^{-2}|\Sigma(0)|$ and $(K/N)\mathrm{N}(M_0)^{1/2}$, as in [HN]. For a Laurent index n at p, (V2) gives the gains q^{-6n} (first term) and q^{-2n} (second term), which are summable.

Lemma 8.3 (Local factors)**.** *By the Chinese remainder theorem (the twists in* (V5)–(V6) *are the classical ones),* $\Sigma(\xi) = \prod_{p^v \| M_0} \sigma_p(\xi)$ *with* $\sigma_p(\xi) := \mathrm{N}(p^v)^{-1} \sum_{x \bmod p^v} \mathcal{P}_p(x)\mathrm{e}(x\xi''/p^v)$*, where* ξ'' *is* ξ' *times a unit modulo* p^v*. Put* $h_p(D,\xi) := \mathrm{N}((D,\xi',p))^{1/2}q^{-1/2}$*, and let* Δ_e *be as in Lemma 8.4. Then:*

case	*bound for* $\sigma_p(0)$ *and* $\sigma_p(\xi)$
(L1) $p \,\|\, r$*,* $p \nmid s_1 s_2$*, type* P *on both sides*	$\le 3h_p(\Delta_e,0)$*, resp.* $\le 3h_p(\Delta_e,\xi)$
(L2) $p \,\|\, r$*, type* A *or* B*-ramified*	$\le C^2 q^{-1}$
(L3) $p \,\|\, r$*, type* B*-unramified*	$\le C^2$
(L4) $p = s_i$*,* $s_1 \ne s_2$ *(one side)*	P*:* $\le q^{-1}$ *(*$\xi'' \equiv 0$*),* $\le q^{-1/2} + q^{-1}$ *(*$\xi'' \not\equiv 0$*); A, B-ramified:* $\le Cq^{-1/2}$*; B-unramified:* $\le C$
(L5) $p = s_1 = s_2$	P*:* $\le 3h_s(t_2 - t_1, \xi)$*; A, B-ramified:* $\le C^2q^{-1}$*; B-unramified:* $\le C^2$
(L6) $p^v \,\|\, M_0$*,* $v \ge 2$	$\le C^2$ *(pointwise, by* (V5)*)*
type D *at* p *(both sides)*	$p \nmid M_0$*: no factor*

Hence the hypothesis of Lemma 8.2 holds with

$$\Lambda = (3C^2)^{\#M_0} \prod_{p \text{ in (L3), or B-unramified in (L4), (L5)}} q_p^{1/2} \cdot \mathrm{N}((M_0)_{\mathrm{sq}})^{1/2}.$$

Proof. (L1): with $k^{(i)}(x) = q^{-1/2}u_iK_\chi(l_ix)\mathbf{1}_{x\ne 0}$ (Lemma 8.1(a)), $\sigma_p(\xi) = q^{-2}\bar{u}_1u_2[\mathcal{C}(l_1,l_2,\xi'') - q]$, where $\mathcal{C}(l_1,l_2,h) := \sum_{n \bmod p} K_\chi(l_1n)\overline{K_\chi(l_2n)}\mathrm{e}_p(hn)$ and the $-q$ removes $n = 0$ ($|K_\chi(0)|^2 = |G(\chi)|^2 = q$). Expanding both sums and summing over n, $\mathcal{C} = q\sum_{x,y:\, l_1/x - l_2/y + h = 0} \chi(x/y)\mathrm{e}_p(x-y)$, so $\mathcal{C} = -q\chi(l_1/l_2)$ for $h = 0$, $l_1 \not\equiv l_2$; $\mathcal{C} = q(q-1)$ for $h = 0$, $l_1 \equiv l_2$; and for $h \not\equiv 0$ the substitution $u = x + l_1/h$ gives

$$\mathcal{C} = q\Big[\chi(h/l_2)\,\mathrm{e}_p\big(-(l_1+l_2)/h\big)K_\chi(l_1l_2/h^2;p) - \chi(l_1/l_2)\Big], \tag{8.1}$$

so $|\mathcal{C}| \le q(2q^{1/2}+1)$ by (I3). Since $l_1 \equiv l_2$ if and only if $p \mid \Delta_e$ (Lemma 8.4), this gives (L1) ($q \ge 5$). (L4), P: $(1/q)\sum_{x \bmod p} q^{-1/2}K_\chi(lx)\mathrm{e}_p(\xi x) = q^{-1/2}\chi(-l/\xi)\mathrm{e}_p(-l/\xi)$ for $\xi \not\equiv 0$, and $= 0$ for $\xi \equiv 0$; the indicator $\mathbf{1}_{(x,p)=1}$ changes this by $q^{-3/2}|G(\chi)| = q^{-1}$. (L5), P: by Lemma 8.1(b), $l^{(i)} = L_s t_i^{-1}$ with the same L_s, so $l_1 \equiv l_2$ if and only if $s \mid t_2 - t_1$, and the computation of (L1) applies. The other rows are the pointwise bounds of (V5). □

The identity (8.1) was also checked numerically (Appendix B, item B8).

Lemma 8.4 (The diagonal)**.** *Let $p \,\|\, r$ have type* P *on both sides, and put $\Delta_e := s_2^e t_2 - s_1^e t_1$ if $e \geq 0$ and $\Delta_e := s_1^{|e|} t_2 - s_2^{|e|} t_1$ if $e < 0$. Then $l_1 \equiv l_2 \pmod p$ if and only if $p \mid \Delta_e$. For $e \neq 0$, $\Delta_e = 0$ if and only if $(s_1, t_1) = (s_2, t_2)$, and $\Delta_0 = 0$ if and only if $t_1 = t_2$. Moreover $|\Delta_e| \leq K^{O(1)}$, so $\tau(\Delta_e) \prec 1$ when $\Delta_e \neq 0$. For $e = 2$, $\Delta_2 = (s_1, s_2)^2 \Delta_{\mathrm{HN}}/k$ with the quantity $\Delta_{\mathrm{HN}} := k(s_2^2 t_2 - s_1^2 t_1)/(s_1, s_2)^2$ of* [HN]*, and the local conditions at $p \nmid k s_1 s_2$ agree.*

Proof. By Lemma 8.1(a), $l_i = L_p s_i^{-e} t_i^{-1}$ with the same L_p and e, which gives the first statement. The s_i, t_i are primary primes, the norm ranges of s and t are disjoint ($2S < T$), and $\mathcal{O}$ is a unique factorisation domain; this gives the statements about $\Delta_e = 0$. The bound $|\Delta_e| \leq K^{O(1)}$ is clear, and the divisor bound gives $\tau(\Delta_e) \prec 1$. The identity for $e = 2$ is immediate from the definitions. □

8.5. **The main term.** Following [HN, §4.6]: for a pattern, $W_0(Q_\Pi/(RS)^3)|\Sigma(0)| = (r\text{-part}) \cdot (s\text{-part})$.

- *r-part* $= \prod_{p \| r} \rho_p h_p(\Delta_e, 0) \cdot \prod_{p^e \| r,\, e \geq 2} \rho_p q_p^{-e/2}$, with $\rho_p := |w_p|^2 q_p^{d_p - 3e_p} |\sigma_p(0)|/(h_p$, resp. $q_p^{-e/2})$. By the table and (V2): P: $\rho \leq 3$; A: $\rho \leq C^2 q^{d-3} q^{-1/2} q^{-6n} \leq C^2 q^{-1/2}$; B-ramified: $\leq C^2 q^{-1/2}$; B-unramified: $\leq q^{-1} C^2 q^{1/2}$; D: $\leq q^2 q^{-3} q^{1/2} = q^{-1/2}$. At $p^e \,\|\, r$ with $e \geq 2$: H: $\rho \leq 4q^{e/2}$; partial peel ($4i < e$): $\leq C^2 q^{2i - 12i + e/2}$; full peel: $\leq q^{2i - 3e + e/2} \leq q^{-5/2}$; stratum at the reduced exponent: $\leq C^2 q^{-3+e/2}$. So $\sum_{\text{types at } p} \rho_p \leq 3 + C' q_p^{-1/2}$ ($p \,\|\, r$) and $\leq C' e q_p^{e/2}$ ($p^e \,\|\, r$, $e \geq 2$). Since $\prod_{p \| r} h_p(\Delta_e, 0) \prod_{p^e \| r, e \geq 2} q_p^{-e/2} = \mathrm{N}((\Delta_e, r_{\mathrm{sf}}))^{1/2} \mathrm{N}(r)^{-1/2}$, for a fixed label at s we get $\sum_{\Pi \in \mathcal{P}(r)} (r\text{-part}) \prec \mathrm{N}(r_{\mathrm{sq}})^{1/2} \mathrm{N}((\Delta_e, r_{\mathrm{sf}}))^{1/2} \mathrm{N}(r)^{-1/2}$.
- *s-part* (against the factor S^3 in $(RS)^3$), with $\mathrm{HN}_s := S^{-1}$ if $s_1 \neq s_2$ and $\mathrm{HN}_s := h_s(t_2 - t_1, 0)$ if $s_1 = s_2$: P: $\leq 9\,\mathrm{HN}_s$; A (slope ≤ 3) and B-ramified: $\leq C^2 q^{-1} \leq C^2 \mathrm{HN}_s$; B-unramified ($|w_s| \leq 1$, slope 2): $\leq C^2 q^{-1} \leq C^2 \mathrm{HN}_s$; D ($|w_s| = \mathrm{N}(s)$, slope 0, no local factor): $\ll \mathrm{N}(s_1)\mathrm{N}(s_2) S^{-3} \ll \mathrm{HN}_s$. So the s-part is $\ll C^2 \mathrm{HN}_s$ for every label.
- *Averages.* $\alpha_r \ll R^{-1}$ on $\mathrm{N}(r) \in [R, 2R]$, $\beta_s \prec S^{-1}$ and $\gamma_t \prec T^{-1}$ on primes, so $\sum |\alpha_r \beta_{s_1} \beta_{s_2} \gamma_{t_1} \gamma_{t_2}| f \prec \mathbb{E} f$, where $\mathbb{E}$ is the uniform average over (r, s_1, s_2, t_1, t_2).
- $\Delta_e = 0$ *(B_0).* The r-part is $\prec 1$. For $e \neq 0$, $(s_1, t_1) = (s_2, t_2)$ has probability $\prec 1/(ST)$ and $\mathrm{HN}_s = 1$; for $e = 0$, $t_1 = t_2$ has probability $\prec 1/T$ and $\mathbb{E}[\mathrm{HN}_s] \prec S^{-1}$. So $B_0 \prec 1/(ST)$.
- $\Delta_e \neq 0$ *(B_1).* $\mathbb{E}_r[\mathrm{N}(r_{\mathrm{sq}})^{1/2} \mathrm{N}((\Delta_e, r_{\mathrm{sf}}))^{1/2}] \prec 1$ (Lemma 7.3(ii)), and $\mathbb{E}[\mathrm{HN}_s] \prec S^{-1}$. So $B_1 \prec R^{-1/2} S^{-1}$.

The main terms total $\prec K(RS)^3 N^{-2}(B_0 + B_1)$.

8.6. **The term $\xi \neq 0$ (B_2).** By Lemma 8.2 this term is $KN^{-1} W_0(X'/N) \Lambda\, \tau(M_0) \mathrm{N}(M_0)^{1/2}$. Per prime, against the factor $\mathrm{N}(r)^{1/2} \mathrm{N}([s_1, s_2])^{1/2}$ of [HN]: P, A: $\leq C$; B at $p \mid r$: $|w_p|^2 \cdot q^{-1}$ (from

$X') \cdot q^{1/2}$ (from Λ) $= q^{-1/2}$; D at $p \mid r$: $q^2 \cdot q^{-4} \cdot q^{-1/2}$ ($p \nmid M_0$) $= q^{-5/2}$; prime powers: the loss $\mathrm{N}(p^e)^{1/2}$ of (L6) (peels only lower it); at s: B-unramified $\le q^{-1}q^{1/2}$ per side, $\mathsf{D} \ll S^2S^{-4}S^{-1}$. So the sum over patterns is $\prec \mathbb{E}[\mathrm{N}(r_{\mathrm{sq}})^{1/2}\mathrm{N}(r)^{1/2}\mathrm{N}([s_1,s_2])^{1/2}] \prec R^{1/2}S$ (Lemma 7.3), that is,

$$B_2\text{-term} \prec \frac{K}{N}R^{1/2}S = K(RS)^3N^{-2} \cdot \frac{N/(R^2S)}{R^{1/2}S}.$$

8.7. **Collisions.** Put

$$F_{\mathrm{coll}} := K^{1/2}\sum_r \alpha_r \sum_{s\mid r} \beta_s \sum_{t:(rs,t)=1} \gamma_t N^{-1} A(rs, -tk^{-1}; N)$$

(for each ℓ). Fix (r,s) with $e := v_s(r) \ge 1$; then $e \le 6$, and r has at most 6 such s ($\mathrm{N}(r) \le 2R$, $\mathrm{N}(s) \ge S$). Label the pieces by patterns at the modulus rs (types at every prime of rs, s with exponent $e+1$; all outer), and apply Cauchy–Schwarz with (r, s, Π, m) outside and t inside:

$$|F_{\mathrm{coll}}|^2 \prec \Big[\sum_r |\alpha_r| \sum_{s\mid r} |\beta_s| \sum_{\Pi} \sum_{\mathrm{N}(m)\le Z_\Pi} \frac{|a^\circ_\Pi(m)|^2}{\mathrm{N}(m)}\Big] \cdot \Big[\sum_r |\alpha_r| \sum_{s\mid r} |\beta_s| \sum_{\Pi} \sum_m \Big|\sum_t \gamma_t \Phi_\Pi(m; r, s, t)\Big|^2\Big].$$

The first factor is $\prec S^{-1}$ (the density of $s \mid r$ is $\mathrm{N}(s)^{-1}$). In the second, both sides have modulus $c = rs$, the same pattern and the same indicators; t enters only through b, so at $p \,\|\, r$, $p \ne s$, of type P, $l_p^{(i)} = L_p t_i^{-1}$ (s now inside L_p), and (L1) holds with $\Delta = t_2 - t_1$; s^{e+1} and the other prime powers are (L6). Lemma 8.2 with $M_0 = rs$ and the ρ_p of Section 8.5 gives, per (r, s, t_1, t_2), a main term $\prec K(RS)^3N^{-2}\mathrm{N}((t_2 - t_1, c_{\mathrm{sf}}))^{1/2}\mathrm{N}(c)^{-1/2}\mathrm{N}(c_{\mathrm{sq}})^{1/2}$ and a $\xi \ne 0$ term $\prec (K/N)\mathrm{N}(c)^{1/2}\mathrm{N}(c_{\mathrm{sq}})^{1/2}$, with $\mathrm{N}(c_{\mathrm{sq}})^{1/2} = q^{(e+1)/2}\mathrm{N}(r'_{\mathrm{sq}})^{1/2}$, $r = s^e r'$. The r with $s^e \,\|\, r$ have relative weight $\ll q^{-e}$, and Lemma 7.3 handles r'. So $t_1 = t_2$ gives $K(RS)^3N^{-2}S^{-e}T^{-1}$, $t_1 \ne t_2$ gives $K(RS)^3N^{-2}S^{-e}S^{(e+1)/2}(RS)^{-1/2}$, and $\xi \ne 0$ gives $(K/N)S^{-e}(RS)^{1/2}S^{(e+1)/2}$. With $e \ge 1$ and the first factor,

$$|F_{\mathrm{coll}}|^2 \prec K(RS)^3N^{-2}\big(S^{-2}T^{-1} + R^{-1/2}S^{-3/2}\big) + (K/N)R^{1/2}S^{-1/2},$$

each term being at most the corresponding term of Sections 8.5–8.6 times $S^{-1/2}$.

8.8. **The terms with $(rs, t) \ne 1$ and the polar terms.** By (V2) every piece has trivial size $\le C^{\#c}\mathrm{N}(c)^{3/2}$, so, as in [HN, §4.3], $F_2 \prec N^{-1/2}K^{1/2}(KRS/(TN))^{3/2}$ ($c = rs/t$, $N \prec K^{3/2}$), i.e. $|F_2|^2 \prec (K/N)(KRS/(TN))^3$. This form is the trivial bound $|F_2|^2 \prec K(RS/T)^3N^{-2}$ times K^3/N^2, a factor ≥ 1 only for $N \prec K^{3/2}$; the direct bound $K(RS/T)^3N^{-2}$ holds for every N, coincides with the form above at $N = K^{3/2}$, and has slack there (its excess over the

target is $-4/9$, Section 10.5). By (V7), for the modes with $\omega-\ell=0$, $F_{\mathrm{polar}}\prec K^{1/2}N^{-1}N^{3/4}(RS)^{-1/4}$: the exponent is $-5/72$ at $N=K^{3/2}$ and $-1/24$ at $N=K^{13/9}$. Any residue bound $\mathrm{N}(c)^{\varrho}$ with $\varrho<17/56$ would suffice; (V7) gives $\varrho=1/4$. (For $(rs,t)\neq 1$ the polar terms are smaller.)

9. Combination and optimisation

From Sections 7 and 8, using $|F_1|^2\le 2|F_{\mathrm{nc}}|^2+2|F_{\mathrm{coll}}|^2$,

$$\begin{aligned}|F|^2 \prec \frac{K}{N}\Big(\frac{KRS}{TN}\Big)^3 + K(RS)^3N^{-2}\Big(\frac{1}{ST}+\frac{1+N/(R^2S)}{R^{1/2}S}\Big)\\ +|F_{\mathrm{coll}}|^2+|F_{\mathrm{polar}}|^2+|F_{\mathrm{tail}}|^2,\end{aligned}\tag{9.1}$$

$$|\mathfrak{O}|^2\prec H^2\Big(\frac{1}{STH}+\frac{1}{K}\Big),$$

which is the bound of [HN] for the F-term, plus the collision, polar and trivially bounded terms. The target is $|F|^2,|\mathfrak{O}|^2\prec N^{-1}K^{3/2-2\delta_0}$. With $S=K^{1/9}$, $T=K^{5/18}$ and $R=TN/(KS)$, every term is a power of K whose exponent is affine in $\nu:=\log N/\log K$ on $[13/9,3/2]$: indeed $N/(R^2S)=K^{14/9-\nu}$ with $14/9-\nu\ge 1/18>0$ there, so $1+N/(R^2S)\asymp N/(R^2S)$. Hence it suffices to check the endpoints. Table 1 lists the exponents of K; the targets are $-1/18$ (for squares) at $N=K^{3/2}$ and 0 at $N=K^{13/9}$.

term	$N=K^{3/2}$	$N=K^{13/9}$
$\mathfrak{O}$, diagonal $H/(ST)$	$-1/18$	0
$\mathfrak{O}$, off-diagonal H^2/K	$-1/3$	$-2/9$
F, first term (F_2)	$-1/2$	$-4/9$
F, B_0	$-1/18$	$-1/9$
F, B_1+B_2	$-1/18$	$-1/36$
F_{coll} (three terms)	$-1/6$, $-1/6$, $-2/9$	$-2/9$, $-7/36$, $-7/36$
$\lvert F_{\mathrm{polar}}\rvert$ (target $-1/36$, resp. 0)	$-5/72$	$-1/24$
$\lvert F_{\mathrm{tail}}\rvert$, $n_s=1$ (target $-1/36$, resp. 0)	$-2/9$	$-1/4$
$\lvert F_{\mathrm{tail}}\rvert$, $n_s=2$	$-5/9$	$-7/12$

Table 1. Exponents of K of the terms in (9.1) and Proposition 7.1.

Moreover the reciprocity condition $KRS/(TN)=1$ holds; the exponent of $R/(ST)$ is $5/18$, resp. $2/9$, which is positive; and the exponents of R, S, T lie in $(0,1)$. The optimisation is tight at $N=K^{3/2}$: the terms B_0, B_1+B_2 and the diagonal of $\mathfrak{O}$ all equal the target there, as in [HN]. For $K^{3/2}<N\le K^{3/2+4\varepsilon'}$ every exponent changes by $O(\varepsilon')$, which the notation $\prec$ absorbs ($\varepsilon'=\varepsilon/100$). Hence

$$\Sigma(a,N)\prec N^{-1/2}K^{3/4-1/36}$$

on the whole range of (HN-Q(i)), and by Section 6, $|\Lambda(k,1+it)|\prec K^{13/18}$ for $|t|\le K^{\varepsilon'}$, where $13/18=3/4-1/36$; in terms of $\mathrm{N}(\rho)\asymp K^3$ this is the exponent $13/54$. At $N=K^{3/2}$ the Voronoi modulus is $\mathrm{N}(c)=RS=K^{7/9}$ and the dual length is $\mathrm{N}(c)^3/N=K^{5/6}$.

Remark 9.1. The exponents in Table 1, the conditions above, and the same quantities at the intermediate point $N = K^{53/36}$ were also computed in exact rational arithmetic; see Appendix B, item B10.

10. The t-aspect

In this section we prove Theorem A with the exponent $15/2$ of $1 + |t|$ for all real t. We write $\mathcal{L} := 1 + |t|$ (the letter T is the amplifier length of Section 7.1, and t in Sections 7–8 denotes an amplifier prime), $\mathcal{L} = K^u$, and u, p, η are real numbers; x is the dual variable of (V4). Factors $(K\mathcal{L})^{O(\varepsilon)}$ are not displayed.

10.1. **Large height, $\mathcal{L} \geq K^{\delta_0}$.** Lemma 6.1 and Section 6.5 with $X = \mathrm{N}(\rho)^{1/2}\mathcal{L}^3$ (by (U1) both sides of the approximate functional equation have effective length $K^{3/2}\mathcal{L}^3$, Remark A.5(i)) give

$$|\Lambda(k, 1 + it)| \ll (K\mathcal{L})^{\varepsilon} \sum_{a \in \mathcal{A}_0} \sum_{N \leq K^{3/2}\mathcal{L}^3} N^{1/2} |\Sigma(a, N)|,$$

and Cauchy–Schwarz with Lemma 3.1(ii) gives $\Sigma(a, N) \prec 1$ as in Section 6.5. So $|\Lambda| \ll (K\mathcal{L})^{\varepsilon} K^{3/4}\mathcal{L}^{3/2}$, the convexity bound. As $K^{3/4} = K^{13/18}K^{\delta_0} \leq K^{13/18}\mathcal{L}$, we get $|\Lambda| \ll K^{13/18+\varepsilon}\mathcal{L}^{5/2+\varepsilon}$.

10.2. **Small height, $\mathcal{L} \leq K^{\delta_0}$.** We keep the parameters of Section 7.1 unchanged: S and T are fixed powers of K, and $R = TN/(KS)$ depends on N and on nothing else. The weight V of $\Sigma(a, N)$ now contains the phase y^{-it} and is inert with parameter $\mathcal{L}K^{\varepsilon}$; by (U1) the factor y^{-it} is its only source of powers of $\mathcal{L}$. Write $N = K^{3/2+\eta}$; then $\eta \leq 3u + \varepsilon$, i.e. $K^{\eta} = N/K^{3/2} \leq K^{\varepsilon}\mathcal{L}^3$, so a loss $K^{c\eta}$ ($c \geq 0$) is a loss $\mathcal{L}^{3c}$. For $N \leq K^{3/2-2\delta_0}$ nothing changes ($\Sigma(a, N) \prec 1$). For $K^{3/2-2\delta_0} \leq N \leq K^{3/2}\mathcal{L}^{3+\varepsilon}$ it suffices to prove

$$\text{(HN-Q(ii))} \qquad N^{1/2}|\Sigma(a, N)| \prec K^{3/4-\delta_0}\mathcal{L}^{15/2},$$

for then, since $\mathcal{L} \leq K^{\delta_0}$ gives $(K\mathcal{L})^{\varepsilon} \leq K^{2\varepsilon}$, the $O(\log K)$ dyadic N and the $O(1)$ classes cost $K^{o(1)}$, and $|\Lambda| \prec K^{3/4-\delta_0}\mathcal{L}^{15/2} = K^{13/18}\mathcal{L}^{15/2}$. Every loss below is polynomial in $\mathcal{L}$ (a power $\mathcal{L}^{\mathsf{p}}$, or a shift $\eta > 0$); no saving available only for small $\mathcal{L}$ is used.

In Theorem B the weight V enters only through $\widetilde{V}_\pi$ (by (V4)) and the polar term P (by (V7)); w_π, Q_π, $\mathcal{K}_\pi$ and a_π° do not depend on V (Sections 3–4). So the constants "polynomial in $(1 + |\omega|)Y$" of Theorem B are those of the transform, re-derived for the tall weight in Lemma 10.1, and those of P, whose $\mathcal{L}$-cost is the polar row of Section 10.5. The one further constant is that of the truncation remainder $O((\mathrm{N}(c)X)^{-A})$ (Section 4.4.3), which inherits the transform bound through the trivial size of the discarded pieces; by Lemma 10.2(b) that costs at most $\mathcal{L}^{9/2}$, which is absorbed by the cutoff $C\log(K\mathcal{L})$ of Section 10.3 with C large, at cost $K^{o(1)}$ in the count of kept indices. No other constant of Theorem B carries a power of $\mathcal{L}$.

Lemma 10.1 (The transform for a tall weight)**.** *Let* $\mathsf{F}(y) := f_\ell((Ny)^{1/2})$ *(Section 7.6), supported in* $[1,2]$*, with* $(y\,\mathrm{d}/\mathrm{d}y)^j\mathsf{F} \ll_j \mathcal{L}^j K^\varepsilon$ *uniformly in the amplifier variables; let* $\omega'' := \omega - \ell$*, and let* I *be the function of* (V4) *built from* F*. Then*

(i) $I(x) \ll K^{O(\varepsilon)} x^{3/4-\varepsilon}$ *for all* $x > 0$*;*
(ii) $(x\,\mathrm{d}/\mathrm{d}x)^j I \ll_j K^{O(\varepsilon)}\mathcal{L}^j x^{3/4-\varepsilon}$*;*
(iii) $I(x) \ll_B K^{-B}x^{1/2}\mathcal{L}$ *for* $x \geq \mathcal{L}^6 K^\varepsilon$*.*

In particular the dual sums of Theorem B are cut at $\mathrm{N}(m) \leq \mathcal{L}^6 K^\varepsilon Q_\pi/X_\pi$*.*

Proof. The Mellin transform $\widetilde{\mathsf{F}}(z) := \int \mathsf{F}(y) y^{z-1}\,\mathrm{d}y$ is entire and, by j integrations by parts, $\widetilde{\mathsf{F}}(z) \ll_j K^{O(\varepsilon)} 2^{|\mathrm{Re}\, z|}\min(1, (\mathcal{L}/|z|)^j)$. By Section 5.2, $I(x) = x^{1/2}\frac{1}{2\pi i}\int \Phi(\sigma)\widetilde{\mathsf{F}}(\sigma+\frac12)x^\sigma\,\mathrm{d}\sigma$ with Φ equal to G_∞ times a constant, and $|\Phi(\sigma+iv)| \ll \varepsilon^{-1}(1+|v|)^{-6\sigma}(1+|\omega''|)^{6B} \ll (1+|v|)^{-6\sigma}K^{O(B\varepsilon)}$ by Lemma 5.1 ($|\omega''| \leq K^\varepsilon + 1$). On $\mathrm{Re}\,\sigma = 1/4-\varepsilon$ the integrand is $\ll (1+|v|)^{-3/2+6\varepsilon}K^{O(\varepsilon)}$, which is integrable: this gives (i), with no power of $\mathcal{L}$. For $j=0$, (ii) is (i). For $j \geq 1$ the factor σ^j coming from $(x\,\mathrm{d}/\mathrm{d}x)^j$ gives $\int(1+|v|)^{j-3/2+6\varepsilon}\min(1,(\mathcal{L}/|v|)^{j+2})\,\mathrm{d}v \ll \mathcal{L}^{j-1/2+6\varepsilon}$, which is (ii). On $\mathrm{Re}\,\sigma = -B$ (no pole is crossed, Section 5.2), $|\Phi| \ll (1+|v|)^{6B}K^{O(B\varepsilon)}$, and $j = 6B+2$ integrations by parts give $I(x) \ll x^{1/2}\mathcal{L}(2K^{C\varepsilon}\mathcal{L}^6/x)^B$; for $x \geq \mathcal{L}^6K^{\varepsilon_1}$ ($\varepsilon_1 > C\varepsilon$ arbitrarily small) and B large this is $\ll K^{-B'}x^{1/2}\mathcal{L}$ for any B'. □

Lemma 10.2 (The $\mathcal{L}$-losses)**.** *With the weights of Lemma 10.1:*

(a) Lemma 8.2 holds with its first term multiplied by $\mathcal{L}^9$ *and its second term by* $\mathcal{L}^5$*;*
(b) a piece of Theorem B has trivial size $\ll \mathcal{L}^{9/2}Q_\pi^{1/2}\max|\mathcal{K}_\pi|$*, so the trivial bounds of Section 8.2 (pieces with* $n_s \geq 1$*) and of Section 8.8 (*$(rs,t) \neq 1$ *and its polar terms) hold with an extra factor* $\mathcal{L}^{9/2}$*.*

Proof. (a) $\mathcal{W}_Y$ is inert at scale $Y^{1/2}/\mathcal{L}$ (by Lemma 10.1(ii), up to K^ε), and $\sup|\mathcal{W}_Y| \ll KW_0(X'/N^2)|I(x)|^2/x$ with $x = YX'/Q$; by Lemma 10.1(iii) the x-sums of the proof of Lemma 8.2 are cut at $x \leq \mathcal{L}^6K^\varepsilon$. For $\xi = 0$: $\sum_Y Y\sup|\mathcal{W}_Y| \ll KW_0QN^{-2}\sum_{x\leq\mathcal{L}^6}x^{3/2-2\varepsilon} \ll \mathcal{L}^9K^{O(\varepsilon)}KW_0QN^{-2}$. For $\xi \neq 0$ the surviving ξ' have $\mathrm{N}(\xi') \leq \mathcal{L}^2K^{3\varepsilon}\mathrm{N}(M_0)/Y$ (the inertness scale), which replaces $K^{3\varepsilon}$ by $\mathcal{L}^2K^{3\varepsilon}$, and $\sum_{x\leq\mathcal{L}^6}x^{1/2-2\varepsilon} \ll \mathcal{L}^3$; together $\mathcal{L}^5$. (b) We have $(X_\pi/Q_\pi^{1/2})\sum_{\mathrm{N}(m)\leq\mathcal{L}^6Q_\pi/X_\pi}|a_\pi^\circ(m)||\widetilde{V}_\pi|\max|\mathcal{K}_\pi|$ with $|\widetilde{V}_\pi| = |I|/x \ll x^{-1/4}$, and (V3) gives $(X_\pi/Q_\pi^{1/2})(Q_\pi/X_\pi)\sum_{x\leq\mathcal{L}^6}x^{3/4}\max|\mathcal{K}_\pi| \ll \mathcal{L}^{9/2}Q_\pi^{1/2}\max|\mathcal{K}_\pi|$. □

10.3. **Where $\mathcal{L}$ enters.** The following steps cost nothing: the reduction of Section 6.2 (a factor $\log(3+4|t|)$ from (I5)); the amplifier, the formula (7.1) and the reciprocity of Section 7.5 (the phase there is $\mathrm{e}(tm/(krs))$ with t the amplifier prime, so it carries no $\mathcal{L}$); the angular modes (Section 7.6); Proposition 7.1 (after Cauchy–Schwarz the weight enters through $|y^{-1/2-it}V_0(y)|^2$,

which has no phase, and the Poisson step needs $N \geq K^{1+\varepsilon}\mathcal{L}^2$, which holds for $N \geq K^{3/2-2\delta_0}$ and $\mathcal{L} \leq K^{\delta_0}$); the finite-place structure of Theorem B (the moduli of the local factors depend on $\operatorname{Re}\sigma$ only; the Laurent indices are truncated at $C\log(K\mathcal{L})$, with tails $O((K\mathcal{L})^{-10})$); the patterns and Lemmas 8.1, 8.3 and 8.4; the first factor of Section 8.4 ($m \leq Z_\Pi = \mathcal{L}^6 K^\varepsilon \max(Q_\pi/X_\pi)$, a cost $(K\mathcal{L})^{o(1)}$); the polar term (the residue is independent of t and enters as $\operatorname{Res}\cdot X_\pi^{3/4}\widetilde{\mathsf{F}}(3/4)$, with $|\widetilde{\mathsf{F}}(3/4)| \ll K^\varepsilon$); and the optimisation (only through N). The steps that cost something are:

step	how $\mathcal{L}$ enters	cost
approximate functional equation (Lemma 6.1, Section 6.5)	lengths $N \leq K^{3/2}\mathcal{L}^3K^\varepsilon$; weight V inert with parameter $\mathcal{L}K^\varepsilon$	shift $\eta \leq 3u+\varepsilon$
transform (Lemma 10.1)	$I(x) \ll x^{3/4-\varepsilon}$ for all x; dual length $\mathcal{L}^6 Q_\pi/X_\pi$	none in size; cutoff $x \leq \mathcal{L}^6$
Lemma 8.2, $\xi = 0$ (B_0, B_1, F_{coll}, first term)	$\sum_{x\leq\mathcal{L}^6} x^{3/2}$	$\mathcal{L}^9$
Lemma 8.2, $\xi \neq 0$ (B_2)	$\mathcal{L}^2$ (truncation) $\cdot \sum_{x\leq\mathcal{L}^6} x^{1/2} = \mathcal{L}^2\cdot\mathcal{L}^3$	$\mathcal{L}^5$
trivial bounds (Sections 8.2, 8.8; polar at $(rs,t)\neq 1$)	$\sum_{x\leq\mathcal{L}^6} x^{3/4}$	$\mathcal{L}^{9/2}$
first term of F_2 for $N > K^{3/2}$	direct bound of Section 8.8	none (slack)
Section 10.1 (convexity)		$\mathcal{L}^{3/2}$

10.4. **The worst term.** The main diagonal B_0 (Section 8.5) is tight at $N = K^{3/2}$, and its exponent has slope 2 in η. For it $|F|^2$ loses $\mathcal{L}^9$ (dual length $\mathcal{L}^6$, zero frequency) and $K^{2\eta}$, so $|F|$ loses $\mathcal{L}^{9/2}K^\eta \leq \mathcal{L}^{9/2}\mathcal{L}^3 = \mathcal{L}^{15/2}$.

10.5. **Exponents.** Keep the exponents (of K) $1/9$ for S and $5/18$ for T, let $\nu = 3/2+\eta$ be that of N, and $\nu-1-1/9+5/18$ that of R. For a squared term of Section 9, let $E(\eta)$ be its exponent minus the target exponent $3/2-2\delta_0-\nu$ (so $E(0)$ is its excess over the target at $N = K^{3/2}$), let p be the power of $\mathcal{L}$ that it carries (Lemma 10.2), put $L(u) := \max(0, E(3u) + \mathsf{p}u)/2$, and $A_{\mathrm{loss}} := \sup_{0<u\leq\delta_0} L(u)/u$. For a term that is not squared (the polar terms and the s-Laurent pieces), E and p refer to the term itself and $L(u) := \max(0, E(3u) + \mathsf{p}u)$. Then $N^{1/2}|\Sigma(a,N)|/K^{3/4-\delta_0} \prec \mathcal{L}^{A_{\mathrm{loss}}}$. The first term of F_2 is taken in its direct form (Section 8.8). At $\eta = 3\delta_0 = 1/12$ one has $R = K^{3/4}$, and $R < K$, $R > ST$, $KRS/(TN) = 1$, $N > K^{1+2\delta_0}$ (needed in Proposition 7.1), and $S, T, R > 1$. Table 2 gives, for each term, p, $E(0)$ and $\sup_u L(u)/u$, and Table 3 gives $E(\eta)$ for the leading squared terms at five values of η.

So the terms that are tight at $N = K^{3/2}$ (the diagonal of $\mathfrak{O}$, B_0, B_1+B_2) are paid for by the passage to $N > K^{3/2}$ (B_0 has slope 2) and by the powers

term	p	$E(0)$	$\sup_u L(u)/u$
squared terms			
$\mathfrak{O}$, diagonal	0	0	0
$\mathfrak{O}$, off-diagonal	0	$-5/18$	0
F_2, first term (direct form)	9	$-4/9$	0
F, B_0	9	0	15/2
F, $B_1 + B_2$	9	0	23/4
F_{coll}, diagonal	9	$-1/9$	11/2
F_{coll}, off-diagonal	9	$-1/9$	19/4
F_{coll}, $\xi \neq 0$	5	$-1/6$	1/4
terms that are not squared			
polar	0	$-1/24$	0
polar, with the conservative factor $\mathcal{L}^{9/2}$	9/2	$-1/24$	3
polar, $(rs,t) \neq 1$	9/2	$-1/4$	0
s-Laurent piece, $n_s = 1$	9/2	$-7/36$	1/2
s-Laurent piece, $n_s = 2$	9/2	$-19/36$	0

TABLE 2. The $\mathcal{L}$-exponents. The supremum over the squared terms is 15/2, over the other terms 3; so $A_{\mathrm{loss}} = 15/2$. The term $F_{B_1+B_2}$ is charged $\mathcal{L}^9$ for both B_1 and B_2.

term	$\eta = 0$	1/72	1/36	1/18	1/12
$\mathfrak{O}$, diagonal	0	0	0	0	0
F_2, first term (direct form)	$-4/9$	$-5/12$	$-7/18$	$-1/3$	$-5/18$
F, B_0	0	1/36	1/18	1/9	1/6
F, $B_1 + B_2$	0	1/144	1/72	1/36	5/72
F_{coll}, diagonal	$-1/9$	$-1/12$	$-1/18$	0	1/18
F_{coll}, off-diagonal	$-1/9$	$-13/144$	$-5/72$	$-1/36$	1/72
F_{coll}, $\xi \neq 0$	$-1/6$	$-23/144$	$-11/72$	$-5/36$	$-1/8$

TABLE 3. $E(\eta)$ for the leading squared terms (no power of $\mathcal{L}$ included).

of $\mathcal{L}$; at $\eta = 1/12$ the collision terms become positive (1/18 and 1/72) and are paid for by $\mathcal{L}^{11/2}$ and $\mathcal{L}^{19/4}$. For $N \leq K^{3/2}$ ($\eta \leq 0$) the exponents are those of Section 9, $E(\eta) \leq E(0) \leq 0$, and only $\mathcal{L}^{\mathsf{p}/2} \leq \mathcal{L}^{9/2}$ is lost. The regime of Section 10.1 gives the exponent $3/2 + 1/(36\delta_0) = 5/2$. The values in Tables 2 and 3 were computed in exact rational arithmetic (Appendix B, item B11).

10.6. **Conclusion.** By Section 10.5, (HN-Q(ii)) holds, so $|\Lambda(k, 1 + it)| \prec K^{13/18}\mathcal{L}^{15/2}$ for $\mathcal{L} \leq K^{\delta_0}$ (Section 10.2), and by Section 10.1, $|\Lambda| \ll K^{13/18+\varepsilon}\mathcal{L}^{5/2+\varepsilon}$ for $\mathcal{L} \geq K^{\delta_0}$. Both are $\ll_\varepsilon K^{13/18+\varepsilon}\mathcal{L}^{15/2}$. This proves Theorem A. □

Remark 10.3 (A weaker form). A weaker form of Theorem A, with an exponent of $1 + |t|$ depending on ε, follows without the $\mathcal{L}$-bookkeeping: run

Sections 6–9 for $|t| \le K^{\varepsilon'}$, $\varepsilon' = \varepsilon/100$ (the lengths, weights and every use of inertness then cost $K^{O(\varepsilon')}$), and for $|t| > K^{\varepsilon'}$ use the trivial bound from Lemma 6.1, which gives $K^{3/4+\varepsilon}\mathcal{L}^{A_*}$ with an absolute A_*, together with $K^{3/4} \le K^{13/18}\mathcal{L}^{1/(36\varepsilon')}$.

Remark 10.4 (Proof of Corollary 1.1). The argument above bounds each $Z(\rho, 1-it; v)$ separately: by (6.4) (and its analogue in Section 10.1) and the bounds for $\Sigma(a, N)$, $|Z(\rho, 1-it; v)| \prec K^{13/18}\mathcal{L}^{15/2}$ for $\mathcal{L} \le K^{\delta_0}$ and $|Z(\rho, 1-it; v)| \ll (K\mathcal{L})^{\varepsilon}K^{13/18}\mathcal{L}^{5/2}$ for $\mathcal{L} \ge K^{\delta_0}$. Since $\psi(\rho, s; v) = Z(\rho, s; v)/\zeta_\lambda(4s-3)$ and $1/|\zeta_\lambda(1-4it)| \ll \log(3+|t|)$ by (I5) (the Euler factor at λ is bounded above and below on $\mathrm{Re} = 1$), and $\log(3+|t|) \prec 1$ when $\mathcal{L} \le K^{\delta_0}$, we get $\psi(\rho, 1-it; v) \ll_\varepsilon K^{13/18+\varepsilon}\mathcal{L}^{15/2}$ for all real t; replacing t by $-t$ gives Corollary 1.1. Finally $\mathrm{N}(\rho) = K^3/4$, so $K^{13/18} = (4\mathrm{N}(\rho))^{13/54} \asymp \mathrm{N}(\rho)^{13/54}$.

Appendix A. The approximate functional equation

In this appendix we prove Lemma 6.1 from (D1)–(D4). Throughout, k is primary and squarefree, $\rho = -k^3/2$, $v \in \{1, 1+\lambda^3\}$, $s = 1 + i\tau$ with $\tau \in \mathbb{R}$, and $\mathcal{L} = 1 + |\tau|$. All implied constants are independent of k and τ unless stated otherwise. We use the notation of Section 2.3 with angular parameter $\ell = \pm 1$; since $\mathcal{G}(z, \ell)$ depends only on $|\ell|$, we write $\mathcal{G}(z) := \mathcal{G}(z, \pm 1) = \Gamma_{\mathbb{C}}(z - \frac{1}{4})\Gamma_{\mathbb{C}}(z)\Gamma_{\mathbb{C}}(z + \frac{1}{4})$.

A.1. **The series and its functional equation.** By definition, $Z(\rho, u; v) = \zeta_\lambda(4u-3, 1)\psi_v(u, \rho, 1)$. By the cusp relations in (D4), $\psi_1(u, \rho, 1) = -\mathcal{V}\psi_{i_1 1}(u, \rho, 1)$ and $\psi_{1+\lambda^3}(u, \rho, 1) = \mathcal{V}\psi_{i_2 1}(u, \rho, 1)$, so

$$Z(\rho, u; v) = \varepsilon_v \, \mathcal{V} \, Z_{i_v 1}(u, \rho, 1),$$
$$\varepsilon_1 = -1, \ \varepsilon_{1+\lambda^3} = 1, \tag{A.1}$$
$$i_v := i_1 \text{ or } i_2 \text{ according as } v = 1 \text{ or } 1 + \lambda^3.$$

Lemma A.1. *(a) For* $\mathrm{Re}\, u > 3/2$, $Z(\rho, u; v) = \sum_m z_\rho(m)\mathrm{N}(m)^{-u}$ *with absolute convergence;* $|z_\rho(m)| \le \tau(m)\mathrm{N}(m)$, *so* $|Z(\rho, u; v)| \ll 1$ *on* $\mathrm{Re}\, u = 3$, *uniformly in* k.

(b) $Z(\rho, u; v)$ *is entire in* u.

(c) There are T_0, M *(depending on* k*) and an absolute* c_1 *such that* $|Z(\rho, u; v)| \le Me^{3\pi|\mathrm{Im}\, u|/2}(1 + |\mathrm{Im}\, u|)^{c_1}$ *for* $-1 \le \mathrm{Re}\, u \le 3$ *and* $|\mathrm{Im}\, u| \ge T_0$.

Proof. (a) The series $\psi_v(u, \rho, 1)$ converges absolutely for $\mathrm{Re}\, u > 3/2$ and $\zeta_\lambda(4u-3, 1)$ for $\mathrm{Re}\, u > 1$, so their Dirichlet convolution converges absolutely for $\mathrm{Re}\, u > 3/2$; by the identity (6.1), which rests on (I6) (Section 6.2), its coefficients are the $z_\rho(m)$ defined there. (The analytic argument below uses only (D1)–(D4); the bound for the coefficients holds equally for the unrestricted convolution.) Trivially $|g_4(\rho, c)| \le \mathrm{N}(c)$, so $|z_\rho(m)| \le \sum_{cd^4=m} \mathrm{N}(d)^3\mathrm{N}(c) \le \tau(m)\mathrm{N}(m)$, and on $\mathrm{Re}\, u = 3$, $|Z| \le \sum_m \tau(m)\mathrm{N}(m)^{-2} \ll 1$. (b) By (D1) with $\ell = 1 \ne 0$,

$\widehat{\psi}_{i_v1}(u,\rho,1) = \mathcal{G}(u)Z_{i_v1}(u,\rho,1)$ is entire. The function $1/\mathcal{G}$ is entire, since $1/\Gamma$ is. By (A.1), $Z(\rho,u;v)$ is entire. (c) By (D2) there are T_0, M_0 with $|\widehat{\psi}_{i_v1}(u,\rho,1)| \le M_0$ for $-1 \le \operatorname{Re} u \le 3$ and $|\operatorname{Im} u| \ge T_0$. By Stirling's formula, uniformly for $\operatorname{Re} u$ in a compact set and $|\operatorname{Im} u| \ge 1$, $1/|\Gamma_{\mathbb{C}}(u+c)| = (2\pi)^{\operatorname{Re} u+c}/(2|\Gamma(u+c)|) \ll (1+|\operatorname{Im} u|)^{1/2-\operatorname{Re} u-c}e^{\pi|\operatorname{Im} u|/2}$. Multiply the three factors of $1/\mathcal{G}(u)$ and use (A.1). □

Let J be the set of quadruples (j,a,b,β) with $1 \le j \le 24$, $(a,b) \in S(\kappa_j,\rho)$ and $\beta \in \{1, 1+\lambda^3\}$. We have $\rho = (-i)\lambda^{-2}k^3$ with k^3 primary, so ρ has unit part $-i$ and λ-order -2 for every k; by (D4) the sets $S(\kappa_j,\rho)$ are therefore independent of k, and $b \le 3$. Hence J is a fixed finite set. For $\mathsf{J} = (j,a,b,\beta) \in J$ put $\rho_{\mathsf{J}} := (-1)^a\lambda^{2b}\rho$, $v_{\mathsf{J}} := \beta$, $b_{\mathsf{J}} := b$,

$$R(z) := \frac{\mathcal{G}(2-z)}{\mathcal{G}(z)}, \qquad \gamma_{\mathsf{J}}(z) := R(z)\, A_{j\,i_v}(2^{-z},1)\, 2^{-b_{\mathsf{J}}(2-z)}, \tag{A.2}$$

and $Z^{\sharp}_{\mathsf{J}}(z) := \zeta_\lambda(4z-3,-1)\psi_{v_{\mathsf{J}}}(z,\rho_{\mathsf{J}},-1) = \sum_m z^{\sharp}_{\mathsf{J}}(m)\mathrm{N}(m)^{-z}$ (absolutely convergent for $\operatorname{Re} z > 3/2$, with $|z^{\sharp}_{\mathsf{J}}(m)| \le \tau(m)\mathrm{N}(m)$ as in Lemma A.1(a)).

Lemma A.2. *There are constants κ_{J} with $|\kappa_{\mathsf{J}}| \ll 1$ uniformly in k such that, for $\operatorname{Re} u = -1$,*

$$Z(\rho,u;v) = \mathrm{N}(\rho)^{1-u}\frac{\bar\rho}{|\rho|}\sum_{\mathsf{J}\in J}\kappa_{\mathsf{J}}\,\gamma_{\mathsf{J}}(u)\,Z^{\sharp}_{\mathsf{J}}(2-u),$$

where $Z^{\sharp}_{\mathsf{J}}(2-u)$ is given by its absolutely convergent Dirichlet series.

Proof. On $\operatorname{Re} u = -1$ the functions $A_{ji}(2^{-u},1)$ are holomorphic by (D3), since $-1 \notin \{1, 5/4\}$. Apply (D3) with $\nu = \rho$, $\ell = 1$, $i = i_v$, and divide by $\mathcal{G}(u)$, using $\mathcal{G}(2-u,-1) = \mathcal{G}(2-u)$:

$$Z_{i_v1}(u,\rho,1) = \mathrm{N}(\rho)^{1-u}\frac{\bar\rho}{|\rho|}\,R(u)\sum_{j=1}^{24}A_{ji_v}(2^{-u},1)\,Z_{j1}(2-u,\rho,-1).$$

By the second part of (D4), applied with $\ell = -1$ and multiplied by $\zeta_\lambda(4z-3,-1)$,

$$\mathcal{V}Z_{j1}(z,\rho,-1) = \sum_{(a,b)\in S(\kappa_j,\rho)}\sum_{\beta} c_{j,a,b,\beta}\,2^{-bz} \\ \zeta_\lambda(4z-3,-1)\psi_\beta(z,(-1)^a\lambda^{2b}\rho,-1),$$

where each $c_{j,a,b,\beta}$ is a unimodular constant times a Gauss sum $\Gamma_{\beta'}(\rho,i^a\lambda^b)$; by (D4) these have modulus $\ll_{\mathrm{ord}_\lambda(\rho)} 1$, and $\mathrm{ord}_\lambda(\rho) = -2$ for every k. Put $z = 2-u$, multiply by ε_v and use (A.1); the claim follows with $\kappa_{\mathsf{J}} := \varepsilon_v c_{j,a,b,\beta}$. At $\operatorname{Re} u = -1$ we have $\operatorname{Re}(2-u) = 3$, where $Z^{\sharp}_{\mathsf{J}}$ converges absolutely. □

Lemma A.3. *Each γ_{J} is holomorphic on $\operatorname{Re} z < 1$. For every compact interval $I \subset (-\infty,1)$,*

$$|\gamma_{\mathsf{J}}(x+iy)| \ll_I (1+|y|)^{6(1-x)} \qquad (x \in I,\ y \in \mathbb{R}).$$

Proof. The rational factor. $A_{ji}(2^{-z},1) = \mathcal{R}(2^{-z})$ for a rational function $\mathcal{R}$. A pole $q_0 \neq 0$ of $\mathcal{R}$ produces poles of $z \mapsto \mathcal{R}(2^{-z})$ on the line $\operatorname{Re} z = -\log_2|q_0|$; since $A_{ji}(2^{-z},1)$ is holomorphic for $\operatorname{Re} z \notin \{1, 5/4\}$, every such pole has $|q_0| \in \{2^{-1}, 2^{-5/4}\}$. For $x \in I$ the point $q = 2^{-z}$ lies in the compact annulus $\{2^{-\max I} \leq |q| \leq 2^{-\min I}\}$, which avoids the circles $|q| = 2^{-1}$ and $|q| = 2^{-5/4}$ (as $\max I < 1$) and the point 0. On it $\mathcal{R}$ is continuous, hence bounded: $|A_{ji}(2^{-z},1)| \ll_I 1$ for $\operatorname{Re} z \in I$, and $A_{ji}(2^{-z},1)$ is holomorphic on $\operatorname{Re} z < 1$.

The exponential factor. $|2^{-b(2-z)}| = 2^{-b(2-x)} \leq 1$ for $x \leq 2$, and it is entire.

The gamma ratio. We have $R(z) = \prod_{c\in\{-1/4,0,1/4\}} \Gamma_{\mathbb{C}}(2-z+c)/\Gamma_{\mathbb{C}}(z+c)$ and $\Gamma_{\mathbb{C}}(2-z+c)/\Gamma_{\mathbb{C}}(z+c) = (2\pi)^{2z-2}\Gamma(2-z+c)/\Gamma(z+c)$. The function $\Gamma(2-z+c)$ has its poles at $z = 2+c+n$ $(n \geq 0)$, all with $\operatorname{Re} z \geq 7/4$, and $1/\Gamma(z+c)$ is entire; so R is holomorphic on $\operatorname{Re} z < 7/4$, and $|(2\pi)^{2z-2}| = (2\pi)^{2x-2} \ll_I 1$. By Stirling's formula, uniformly for α in a compact set and $|y| \geq 1$, $|\Gamma(\alpha \pm iy)| = \sqrt{2\pi}|y|^{\alpha-1/2}e^{-\pi|y|/2}(1+O(1/|y|))$. Applying this with $\alpha = 2-x+c$ and $\alpha = x+c$, there is $y_0 = y_0(I)$ with

$$\Big|\frac{\Gamma(2-z+c)}{\Gamma(z+c)}\Big| \asymp_I |y|^{(2-x+c)-(x+c)} = |y|^{2-2x} \qquad (|y| \geq y_0).$$

For $|y| \leq y_0$ the ratio is continuous on the compact set $I \times [-y_0, y_0]$ (it has no poles there), hence bounded. Multiplying the three factors, $|R(x+iy)| \ll_I (1+|y|)^{6-6x}$. Combining the three factors gives the lemma. □

A.2. **Proof of Lemma 6.1.**

Proposition A.4. *Lemma 6.1 holds, with J, κ_J, ρ_J, v_J and γ_J as above.*

Proof. For $X > 0$ and $c \in \mathbb{R}$ put

$$\mathcal{I}(c) := \frac{1}{2\pi i}\int_{(c)} Z(\rho, s+w; v)\,\mathcal{E}(w)X^w\frac{\mathrm{d}w}{w}.$$

Recall $|\mathcal{E}(c+iy)| = e^{c^2-y^2}$.

Step 1: Mellin inversion. On $\operatorname{Re} w = 2$ we have $\operatorname{Re}(s+w) = 3$. By Lemma A.1(a), $\sum_m |z_\rho(m)|\mathrm{N}(m)^{-3}\int|\mathcal{E}(2+iy)|X^2\,\mathrm{d}y/|2+iy| < \infty$, so we may interchange sum and integral:

$$\mathcal{I}(2) = \sum_m z_\rho(m)\mathrm{N}(m)^{-s}V_0(\mathrm{N}(m)/X), \qquad V_0(y) = \frac{1}{2\pi i}\int_{(2)}\mathcal{E}(w)y^{-w}\frac{\mathrm{d}w}{w}.$$

Step 2: the contour shift. By Lemma A.1(b) the integrand is holomorphic in $-2 \leq \operatorname{Re} w \leq 2$ except for the simple pole at $w = 0$, where the residue is $Z(\rho, s; v)$ since $\mathcal{E}(0) = 1$. On the horizontal segments $\operatorname{Im} w = \pm T$, $-2 \leq \operatorname{Re} w \leq 2$, with $T \geq T_0 + |\tau|$, Lemma A.1(c) gives

$$\Big|Z(\rho, s+w; v)\mathcal{E}(w)\frac{X^w}{w}\Big| \leq Me^{3\pi(T+|\tau|)/2}(1+T+|\tau|)^{c_1}\,e^{4-T^2}\,\frac{X^2+X^{-2}}{T}$$
$$\longrightarrow 0 \qquad (T \to \infty).$$

The integral $\mathcal{I}(-2)$ converges absolutely, by Step 3 below. Hence $Z(\rho,s;v)=\mathcal{I}(2)-\mathcal{I}(-2)$.

Step 3: the functional equation. On $\operatorname{Re} w=-2$ we have $\operatorname{Re}(s+w)=-1$, and Lemma A.2 gives

$$Z(\rho,s+w;v)=\mathrm{N}(\rho)^{1-s}\frac{\bar\rho}{|\rho|}\sum_{\mathfrak{J}}\kappa_{\mathfrak{J}}\,\mathrm{N}(\rho)^{-w}\gamma_{\mathfrak{J}}(s+w)\sum_m z_{\mathfrak{J}}^\sharp(m)\mathrm{N}(m)^{s+w-2}.$$

By Lemma A.3 with $I=\{-1\}$, $|\gamma_{\mathfrak{J}}(s+w)|\ll(1+|\tau+y|)^{12}$ for $w=-2+iy$, and $\sum_m|z_{\mathfrak{J}}^\sharp(m)|\mathrm{N}(m)^{-3}<\infty$; so the double sum–integral converges absolutely, and

$$\mathcal{I}(-2)=\mathrm{N}(\rho)^{1-s}\frac{\bar\rho}{|\rho|}\sum_{\mathfrak{J}}\kappa_{\mathfrak{J}}\sum_m z_{\mathfrak{J}}^\sharp(m)\mathrm{N}(m)^{s-2}\\ \frac{1}{2\pi i}\int_{(-2)}\mathcal{E}(w)\Big(\frac{X\mathrm{N}(m)}{\mathrm{N}(\rho)}\Big)^w\gamma_{\mathfrak{J}}(s+w)\frac{\mathrm{d}w}{w}.$$

Substitute $w\mapsto -w$. Since $\mathcal{E}$ is even, $\mathrm{d}w/w$ is invariant and the orientation of the line is reversed, so $\frac{1}{2\pi i}\int_{(-2)}f(w)\,\mathrm{d}w/w=-\frac{1}{2\pi i}\int_{(2)}f(-w)\,\mathrm{d}w/w$ for the integrand f above. Thus

$$-\mathcal{I}(-2)=\mathrm{N}(\rho)^{1-s}\frac{\bar\rho}{|\rho|}\sum_{\mathfrak{J}}\kappa_{\mathfrak{J}}\sum_m z_{\mathfrak{J}}^\sharp(m)\mathrm{N}(m)^{s-2}\,W_{\mathfrak{J}}\Big(\frac{\mathrm{N}(m)X}{\mathrm{N}(\rho)};s\Big),$$
$$W_{\mathfrak{J}}(y;s)=\frac{1}{2\pi i}\int_{(2)}\mathcal{E}(w)y^{-w}\gamma_{\mathfrak{J}}(s-w)\frac{\mathrm{d}w}{w},$$

and $Z(\rho,s;v)=\mathcal{I}(2)-\mathcal{I}(-2)$ is the identity of Lemma 6.1. The constants $|J|$ and $|\kappa_{\mathfrak{J}}|$ are bounded independently of k by Lemma A.2.

Step 4: the bound (U1). Fix $s=1+i\tau$. For $\operatorname{Re}w>0$ we have $\operatorname{Re}(s-w)<1$, so by Lemma A.3 the integrand $\mathcal{E}(w)y^{-w}\gamma_{\mathfrak{J}}(s-w)/w$ is holomorphic on $\operatorname{Re}w>0$, and on every strip $c'\le\operatorname{Re}w\le c''$ with $0<c'<c''$ it is bounded by $\ll_{c',c''}e^{c''^2-(\operatorname{Im}w)^2}(1+|\tau|+|\operatorname{Im}w|)^{6c''}y^{-\operatorname{Re}w}$. The horizontal integrals therefore vanish, and for every $B>0$

$$W_{\mathfrak{J}}(y;s)=\frac{1}{2\pi i}\int_{(B)}\mathcal{E}(w)y^{-w}\gamma_{\mathfrak{J}}(s-w)\frac{\mathrm{d}w}{w}.$$

For $j\ge0$, differentiation under the integral sign (justified by the absolute and locally uniform convergence of the differentiated integral) gives

$$\Big(y\frac{\mathrm{d}}{\mathrm{d}y}\Big)^jW_{\mathfrak{J}}(y;s)=\frac{1}{2\pi i}\int_{(B)}\mathcal{E}(w)(-w)^jy^{-w}\gamma_{\mathfrak{J}}(s-w)\frac{\mathrm{d}w}{w}.$$

On $w=B+iy'$ we have $s-w=(1-B)+i(\tau-y')$, and Lemma A.3 with $I=\{1-B\}$ gives $|\gamma_{\mathfrak{J}}(s-w)|\ll_B(1+|\tau-y'|)^{6B}\le\mathcal{L}^{6B}(1+|y'|)^{6B}$. Hence

$$\Big|\Big(y\frac{\mathrm{d}}{\mathrm{d}y}\Big)^jW_{\mathfrak{J}}(y;s)\Big|\ll_{j,B}y^{-B}\mathcal{L}^{6B}\int_{\mathbb{R}}e^{B^2-y'^2}|B+iy'|^{j-1}(1+|y'|)^{6B}\,\mathrm{d}y'\\ \ll_{j,B}\Big(\frac{\mathcal{L}^6}{y}\Big)^B,$$

which is (U1). The constants depend only on j, B, the fixed rational functions $A_{ji_v}(\cdot,1)$ and $b_\jmath \le 3$, so they are uniform in k and τ.

Step 5: the weight V_0. For $y \ge 1$ move the line of V_0 to $\operatorname{Re} w = B$, which gives $V_0(y) \ll_B y^{-B}$; for $y \le 1$ move it to $\operatorname{Re} w = -1$, passing the pole at $w = 0$ with residue 1, which gives $V_0(y) = 1 + O(y)$. For $j \ge 1$ the integrand of $(y\,\mathrm{d}/\mathrm{d}y)^j V_0(y) = \frac{1}{2\pi i}\int_{(2)} \mathcal{E}(w)(-w)^j y^{-w}\,\mathrm{d}w/w$ is entire, and moving the line to $\operatorname{Re} w = \pm B$ gives $\ll_{j,B} \min(y^B, y^{-B})$. In all cases $(y\,\mathrm{d}/\mathrm{d}y)^j V_0(y) \ll_{j,B} \min(1, y^{-B})$.

Finally, the dual sum in Lemma 6.1 converges absolutely: $|z_\jmath^\sharp(m)|\mathrm{N}(m)^{-1} \le \tau(m)$, and $W_\jmath(\mathrm{N}(m)X/\mathrm{N}(\rho); s) \ll (\mathcal{L}^6\mathrm{N}(\rho)/(\mathrm{N}(m)X))^2$ by (U1) with $B = 2$. □

Remark A.5. (i) Taking $B = \varepsilon$ in (U1) gives $W_\jmath(y; s) \ll_\varepsilon \mathcal{L}^{6\varepsilon} y^{-\varepsilon}$ for $y \le 1$, and taking B large shows that $W_\jmath(y; s)$ is negligible for $y \ge \mathcal{L}^6 K^\varepsilon$. With $X = \mathrm{N}(\rho)^{1/2}\mathcal{L}^3$ both sums in Lemma 6.1 therefore have effective length $\mathrm{N}(\rho)^{1/2}\mathcal{L}^3 K^{o(1)}$. (ii) The proof uses no division by ζ_λ and no information about Z to the right of $\operatorname{Re} u = 3$ or to the left of $\operatorname{Re} u = -1$, and the W-contour is never moved to the left of $\operatorname{Re} w = 0$; this is why the possible poles of the coefficients $A_{ji}(2^{-z}, 1)$ on $\operatorname{Re} z = 1$ play no role on the line $\operatorname{Re} s = 1$.

Appendix B. Numerical checks

The following computations were made as checks on the local identities and on the bookkeeping of exponents. Items B1–B9 are checks only, and no proof in this paper depends on them. Items B10 and B11 recompute, in exact arithmetic, the exponents stated in Tables 1, 2 and 3; those values are part of the proofs in Sections 9 and 10, and each is an elementary computation from the formulas given there. Items B1–B8 are floating-point evaluations of the finite sums involved (characters, Gauss sums, Hilbert symbols and the local factors of Theorem 3.5 computed from their definitions), with the stated maximal discrepancies; items B10 and B11 use exact rational arithmetic.

(B1) *Convention 3.6.* With the normalisation of Convention 3.6(a) the summand of $\tau(\mathbf{e})$ is a class function on $F_S^\times/F_S^{\times 4}$ to relative accuracy $6.2 \cdot 10^{-15}$, and $\tau\phi \in H(\check\chi)$ to $3.6 \cdot 10^{-15}$; with an extra factor $|a|^{1/2}$ the summand is not a class function (relative defect about 1) and the level-one functional equation fails. For the character $\exp(-2\pi i \operatorname{Im} z)$ of [EP, §4] the archimedean constant $\pi^{1-2\sigma}$ was confirmed in the same way.

(B2) *Lemmas 3.7 and 3.8.* At $S = \{\infty, \lambda, p\}$ with $\mathrm{N}(p) = 9, 13$: $\prod_v \chi_v(\upsilon^4) = 1$ for $\upsilon = i, \lambda, p$ to $2.3 \cdot 10^{-16}$, and the S-units act freely; for θ_p of conductor p, the associate invariance of $g(c)\phi(s(c))$ and the class invariance hold to $6.2 \cdot 10^{-15}$.

(B3) *Lemma 4.1.* The product formula, with $\kappa = -1$, was compared with the operator built directly from the definitions in Theorem 3.5 (a sum over all classes of $F_S^\times/F_S^{\times 4}$, evaluated place by place), on all

of $H(\chi)$. At one odd place ($S = \{\infty, \lambda, p\}$) and at two odd places ($S = \{\infty, \lambda, 3, 3+2i\}$ and $\{\infty, \lambda, 3, 1+4i\}$; ten pairs of characters, ramified at both places, at one, or at neither; three values of σ; all 1024 reachable residue patterns of m) the two agree to relative discrepancy at most $5.8 \cdot 10^{-14}$. Dropping $\bar{\theta}_p(w_p)$, flipping κ, replacing w_p by w_p^{-1}, or dropping any one Hilbert-symbol factor gives disagreement of order 1. Not covered: $\omega \neq 0$; m with positive valuation at a place of S; three or more odd places.

(B4) *Lemma 4.3.* The kernel $\mathbf{K}_p$ at $\mathrm{N}(p) = 9, 13$ and three values of σ: the form agrees to 10^{-15} and the constant to 6 digits.

(B5) *Lemma 4.4.* For $\mathrm{N}(p) = 9, 13, 17, 29, 37$ and all (B, m) the identity holds to $4 \cdot 10^{-14}$, with Hasse–Davenport sign $+1$; the variants with $y = -m/B$, with $y = -4^4 m/B$, and with $\chi \mapsto \bar{\chi}$ fail.

(B6) *Lemma 4.10 and Remark 4.11.* The bound and the identity of Remark 4.11 were checked for $q = 5$ ($f \leq 9$), $q = 13, 17$ ($f \leq 5$), $q = 29, 37, 41$ ($f = 3$), and the inert prime 3 ($q = 9$, $f \leq 6$); the deviations are at most $1.1 \cdot 10^{-9}$.

(B7) *Lemma 5.1.* With 30-digit arithmetic, the ratio $|G_\infty(\sigma)|/(1 + |v| + |\omega|)^{-6 \operatorname{Re}\sigma}$ was evaluated (through the Gauss product form) for $\operatorname{Re}\sigma \in \{0.2, 0.1, 0, -0.1, -0.2, -0.25, -0.3, -0.5, -1, -1.5, -5\}$, $|\omega| \in \{0, 1, 2, 5, 20, 100, 1000\}$ and 50 values of v from 0 to 10^5; it is bounded on each line, uniformly over the grid, and tends to $4^{1-8\operatorname{Re}\sigma}$ for large $|v|$. This is a finite grid; the uniformity rests on the proof.

(B8) *The identity* (8.1). For $\mathrm{N}(p) = 13, 17, 29, 37, 41, 53$ the relative error is at most $6.7 \cdot 10^{-15}$, and $\max|\mathcal{C}|/q^{3/2} \leq 2.01$ for $h \neq 0$.

(B9) *Theorem 3.5 for ramified* χ (not used; the theorem is cited from [EP]). Both sides of the functional equation, with the range of the Gauss-sum moduli truncated at $M \leq 4 \cdot 10^5$ and compared on windows where both truncated sums are converged, agree to $1.3 \cdot 10^{-4}$ at level one and, for θ ramified at $p = 3$ or $3 + 2i$, to $9 \cdot 10^{-4}$ in the best windows (median $6 \cdot 10^{-3}$ to $9 \cdot 10^{-3}$ over all converged windows); negative controls fail at order 1, and $\tau(\check{\chi})\tau(\chi)$ is the identity to $3.3 \cdot 10^{-15}$. The precision is limited by the truncation.

(B10) *Section 9.* Every exponent of Table 1, the conditions stated after it, the polar and s-Laurent terms and the collision terms were computed with exact fractions at $N = K^{13/9}$, $K^{53/36}$ and $K^{3/2}$, together with $\theta = 13/18$, $\mathrm{N}(c) = K^{7/9}$ and the dual length $K^{5/6}$; all checks pass.

(B11) *Section 10.5.* The quantities $E(0)$ and $\sup_u L(u)/u$ of Table 2, the values of Table 3, the conditions at $\eta = 1/12$, and the exponent $5/2$ of the convexity regime were computed with exact fractions; the result is $A_{\mathrm{loss}} = 15/2$.

Remark B.1. Item B1 is the reason for the normalisation in Convention 3.6(a): the proofs use only the stated property

$\Gamma_{n,v}(a\mathbf{e}, \varpi) = \varpi(a)^{-1}\Gamma_{n,v}(\mathbf{e}, \varpi)$ of the fixed-measure integral of [EP, p. 229].

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